

Project Name:-

High-Voltage Circuit Design and Power Electronics Simulation for EV Applications
Using MATLAB & PCB Design Tools

by

Ankit Kumar

1st March Batch

PROJECT DESCRIPTION

Design a basic rectifier and filter Stages.

Project Objective :-

The main objective of project is to design Rectifier Circuit using Proteus and showing its DC output using various filter stages. The project aims to achieve a constant ripple free DC output.

Important Pre-requisite :-

Before starting the project it is crucial to have a good understanding of basic electronics concepts, circuit design principle and components. Also, knowledge of how to use Proteus simulation software is essential, as it will be platform for designing, simulating and analyzing the output of rectifier when using different filter stages.

Require Components

<u>Component List</u>			
<u>Sl No.</u>	<u>Component Name</u>	<u>Function</u>	<u>Library name</u>
1.	VSINE	AC Voltage Source	-
2.	AC Voltmeter	For Measurement	-
3.	Transformer	Voltage Conversion	2P2S
4.	Bridge Rectifier	Convert AC to DC	2W01G
5.	Capacitor	Filter	CAP
6.	DC Voltmeter	For Measurement	-
7.	Resistor	Load Resistor	-
8.	Oscilloscope	Measurement graph	-
9.	Inductor	Filter	-

REQUIRED INPUT PARAMETERS AND CALCULATIONS :-

Input Parameter :-

In the context of Rectifier circuit project using Proteus, the input parameter generally refer to the characteristics of the AC input voltage waveform that the rectifier circuit will be designed to handle. Here are the key input parameters.

1.Source Voltage :-

In the Proteus software as a source voltage we will take the value of the peak voltage.

$$\text{Peak voltage} = \text{Nominal voltage} \times \text{sq root}(2)$$

Peak voltage is the maximum instantaneous voltage reached by AC cycle. Nominal voltage is average voltage of AC supply.

$$\begin{aligned}
 \text{Peak voltage} &= \text{Nominal voltage} \times \text{sq root}(2) \\
 &= 70.71\text{V} \times \text{sq root}(2) \\
 &= 100\text{V}
 \end{aligned}$$

$$\text{Frequency} = 50\text{Hz}$$

Here I used VSINE component in Proteus to represent input voltage. I set amplitude to 100V and frequency to 50Hz.

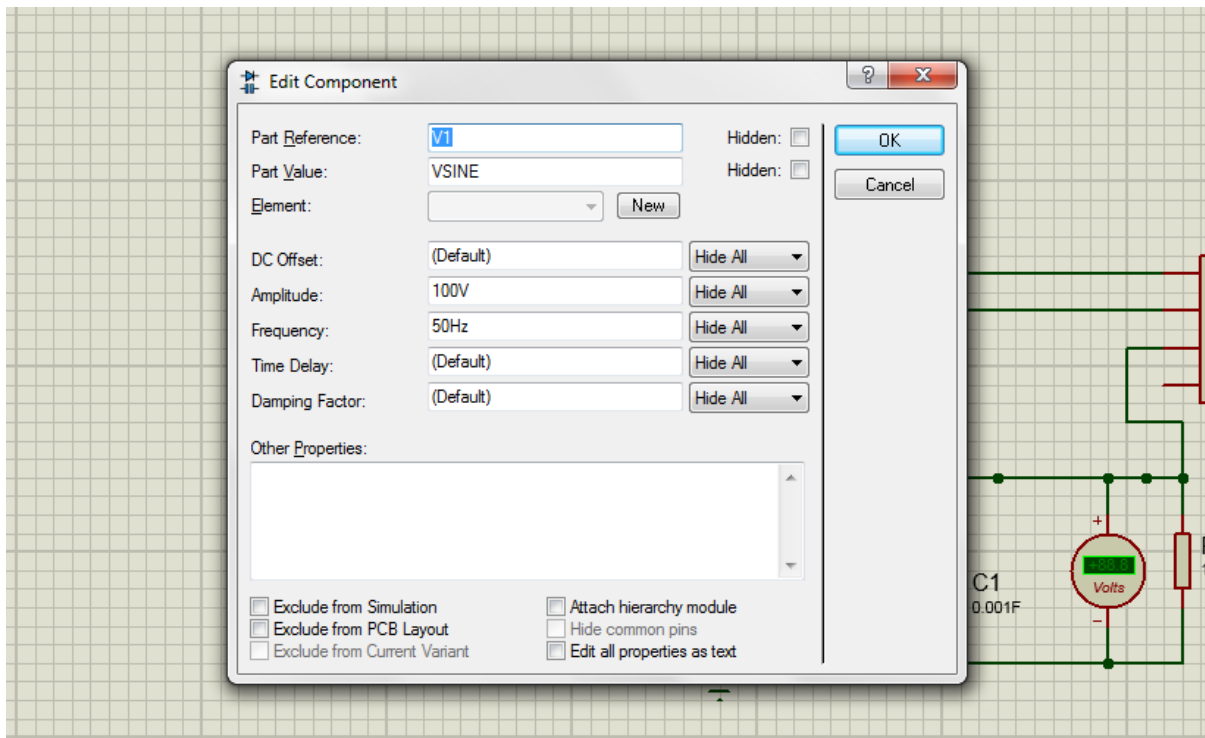


Fig 1. Input Sinewave Parameter

2.AC voltmeter :-

I used an AC Voltmeter to measure the output voltage of VSINE source. To get the accurate reading I connected voltmeter in parallel with VSINE source. This allows voltmeter directly to read voltage of source without affecting the circuit functionality. Voltmeter are always connected in parallel to the source as it has high internal resistance.

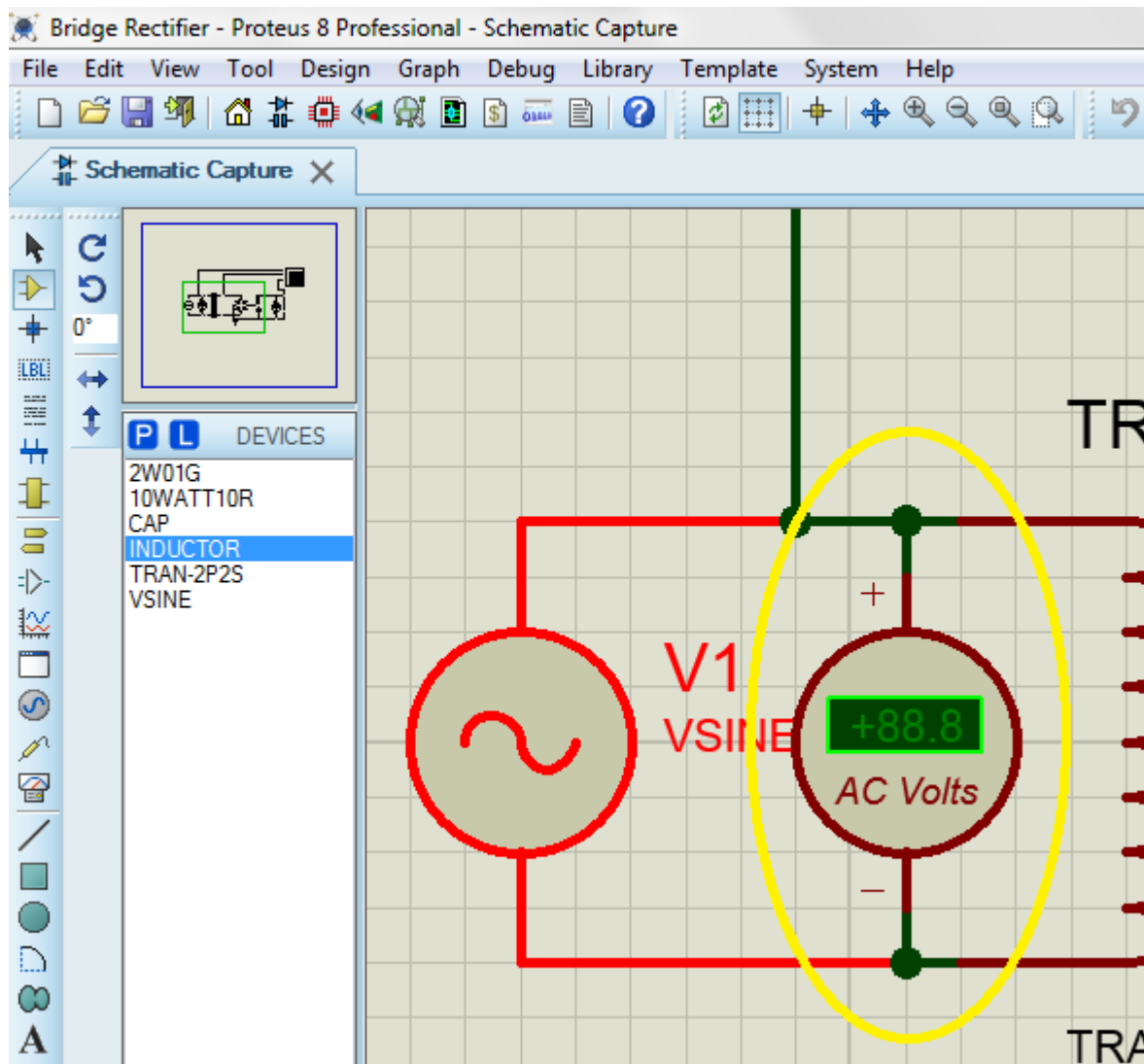


Fig 2. Voltmeter

3. Transformer :-

Here we have used step down transformer to step-down the sine voltage to required level. It step down transformer incoming voltage from the supply mains and isolates the output circuit from line supply.

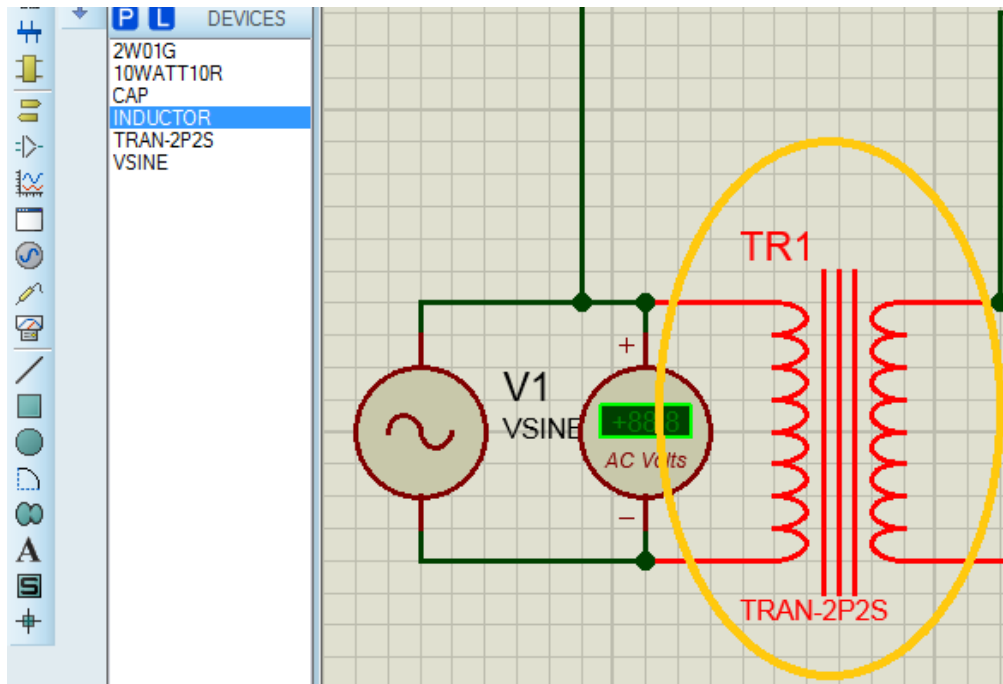


Fig 3. Transformer

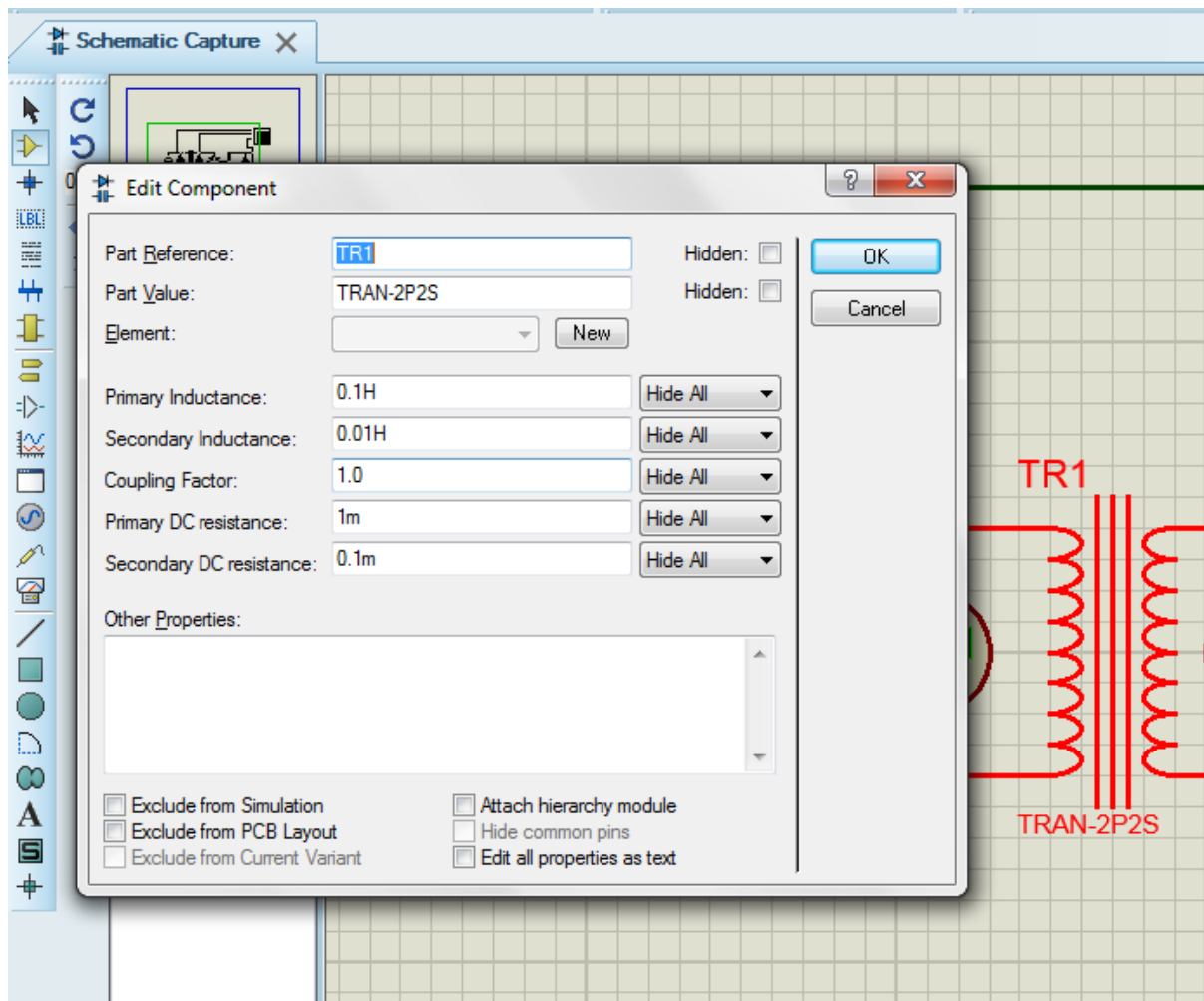


Fig 4. Input parameter for transormer

4. RECTIFIER

Rectifier are electronic Device which help to convert AC to DC . Basically there are two type of rectifier –

1. Half wave rectifier
2. Full Wave rectifier- Bridge rectifier , Center tapped rectifier

Here I have used full wave bridge rectifier which is used to convert AC to DC. It has total 4 terminals. 2 terminals are connected to voltage source and remaining 2 is connected to load Resistor.

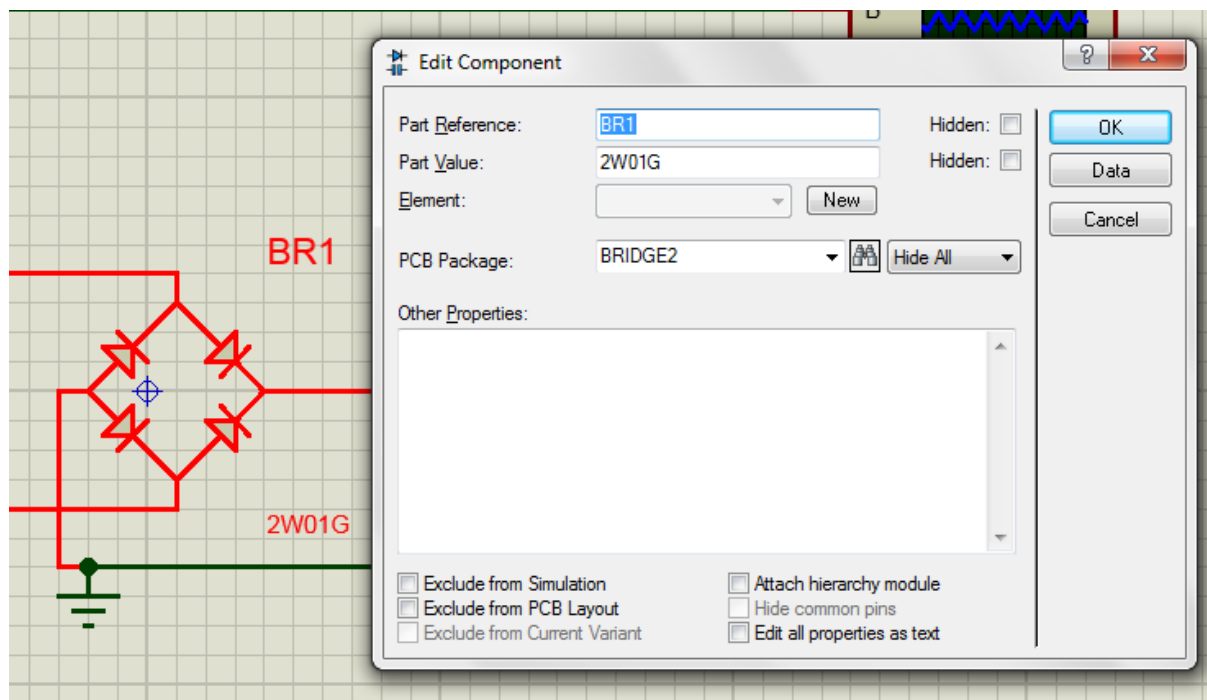


Fig 5. Full Wave Bridge Rectifier

5. Filter

After conversion of AC to DC there are ripples present in the wave to get pure form of DC we use filter which help to remove AC part from DC .

There are basically 4 type of filter –

- i. Series inductor filter
- ii. Shunt capacitor filter
- iii. LC filter

iv. Pi-filter or capacitive input filter

Basically we use inductor and capacitor to remove the ripple and get pure AC.

INDUCTOR allow DC and block AC.

CAPACITOR allow AC and block DC.

i. Series inductor filter

Here I have used inductor in series with the load to get pure DC .

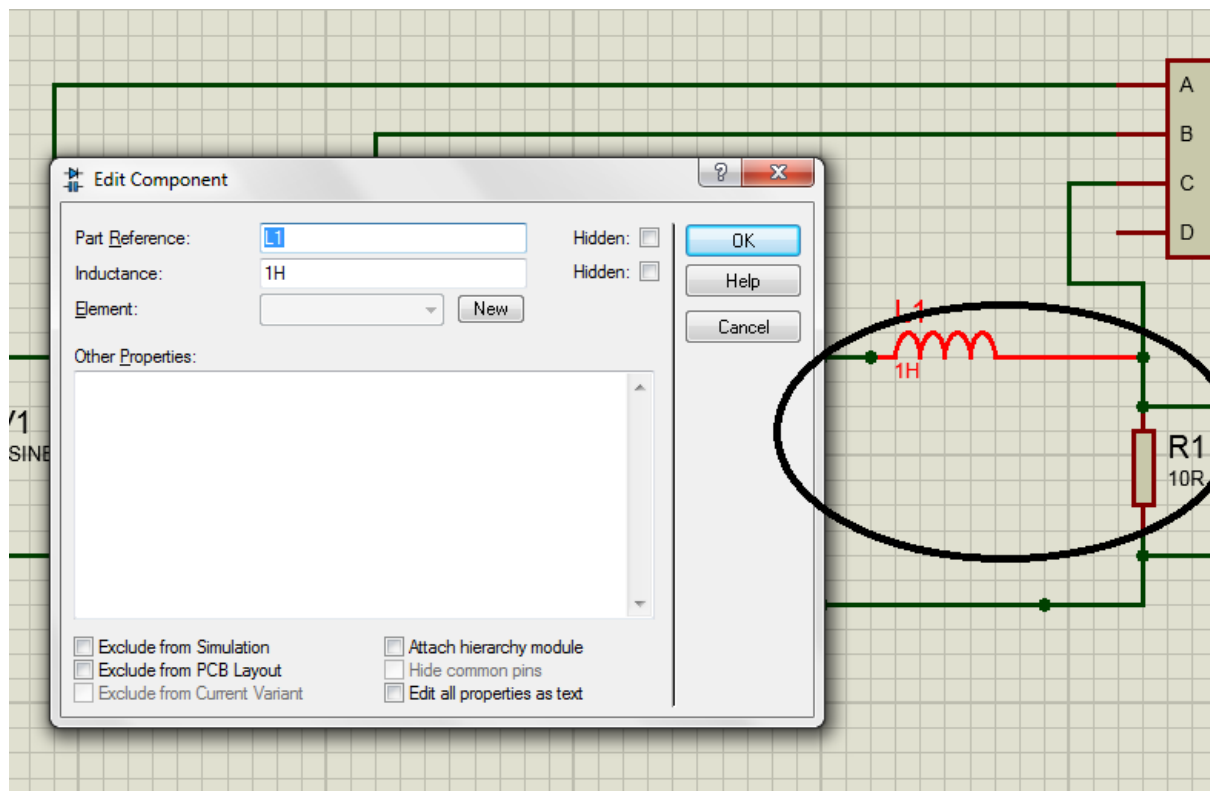


Fig 6. Series inductor Filter

ii. Shunt capacitor filter

Here I have used capacitor in parallel with the load to get Pure DC.

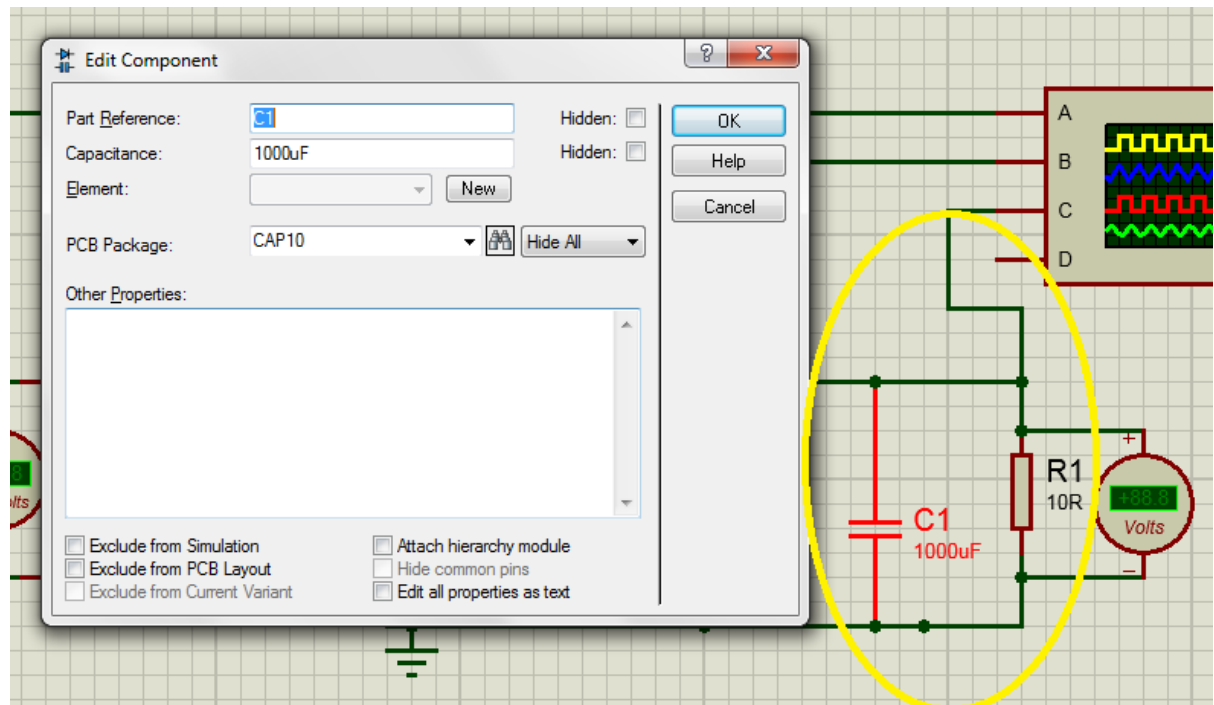


Fig 7. Shunt capacitor Filter

iii. LC filter

Here I have used capacitor and inductor both, inductor in series and capacitor parallel to load.

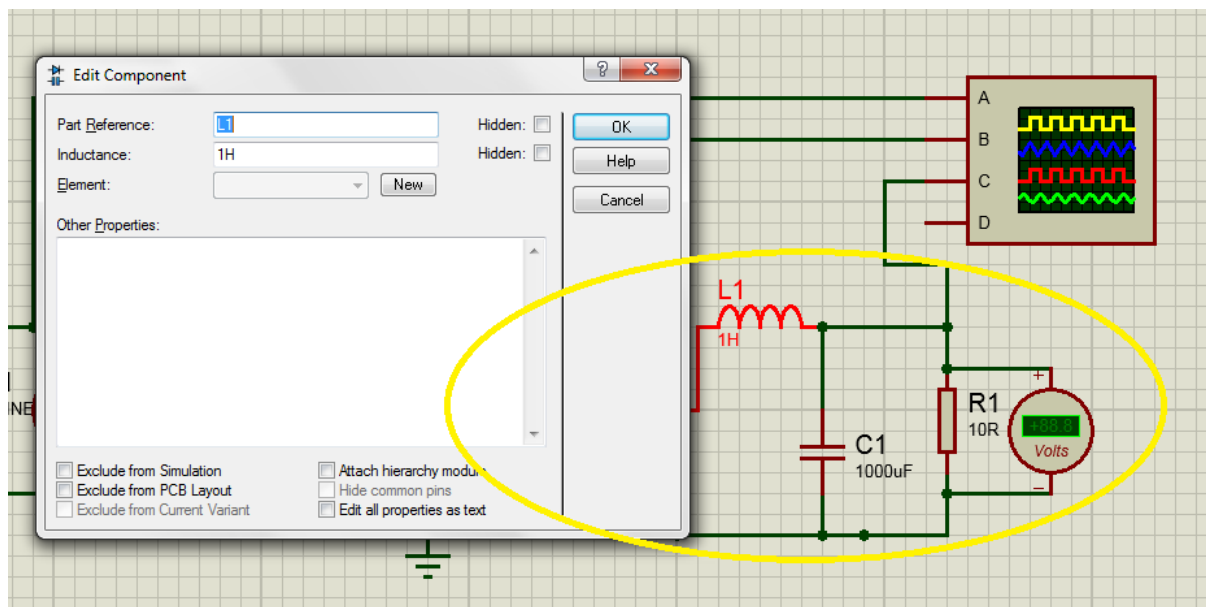


Fig 7(i). LC Filter

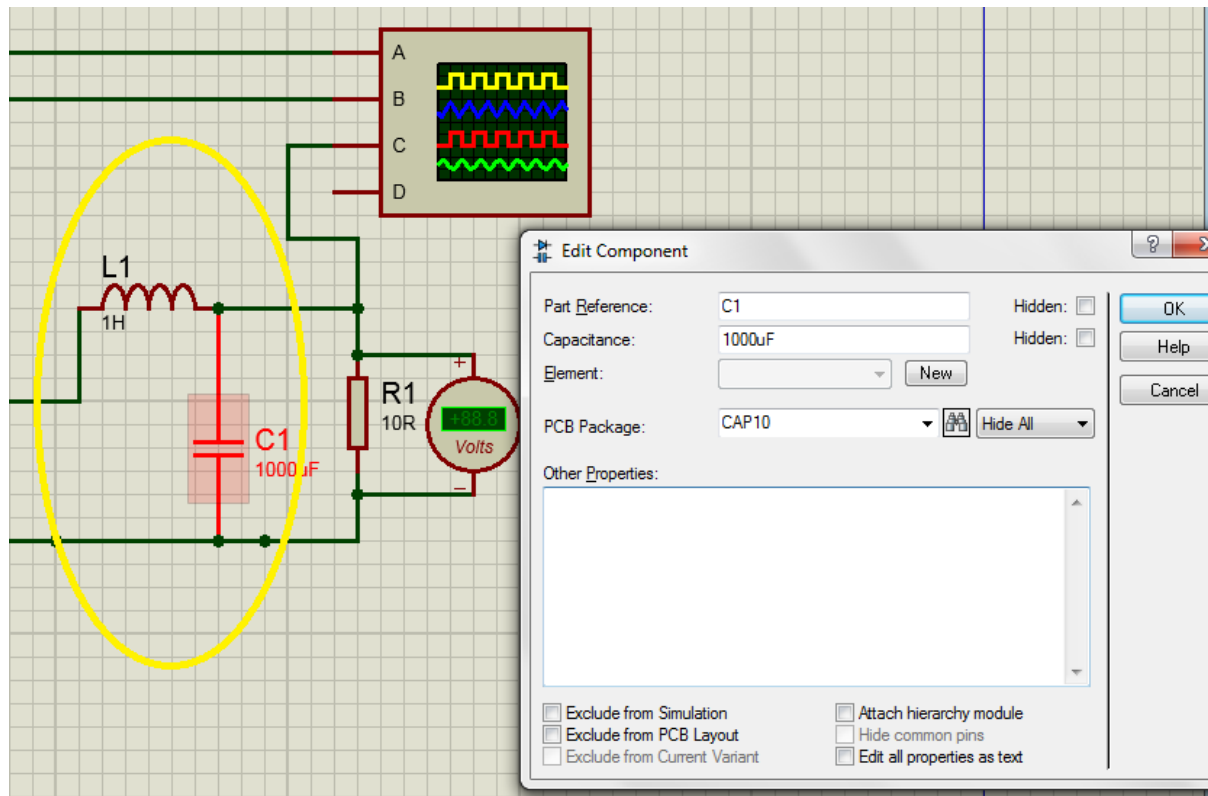


Fig 7(ii). LC Filter

iv. Pi- filter

Here I have used 2 capacitors and inductor in pi shape so it is known as Pi filter. The major part of filtering is done by 10uF capacitor and remaining ripples are removed by combining inductor and capacitor.

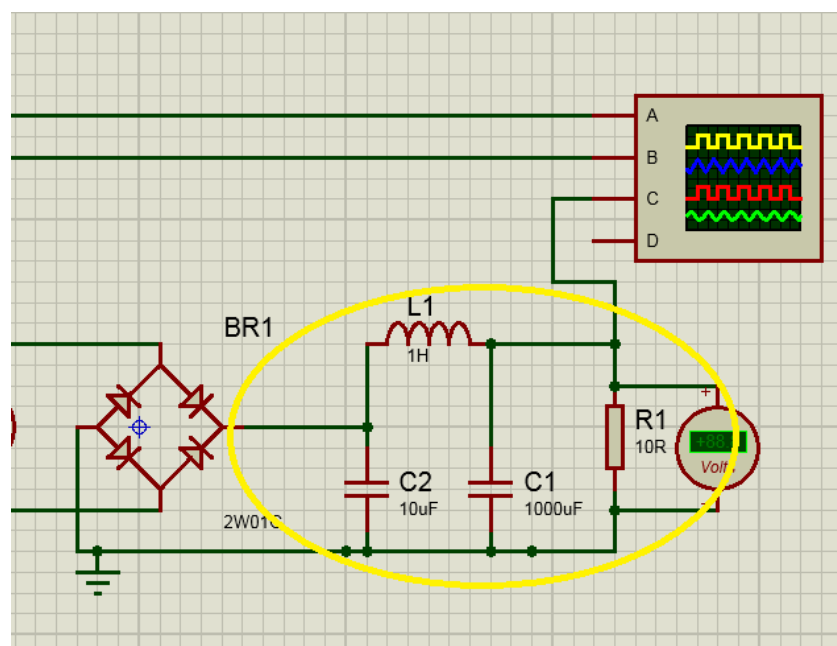


Fig 8. Pi Filter

6. DC Voltmeter

Here I used DC voltmeter to measure the output of DC voltage across load resistor, I connected DC Voltmeter in parallel to load resistor. This allows voltmeter to measure the voltage of circuit without affecting the circuit as voltmeter has high internal resistance.

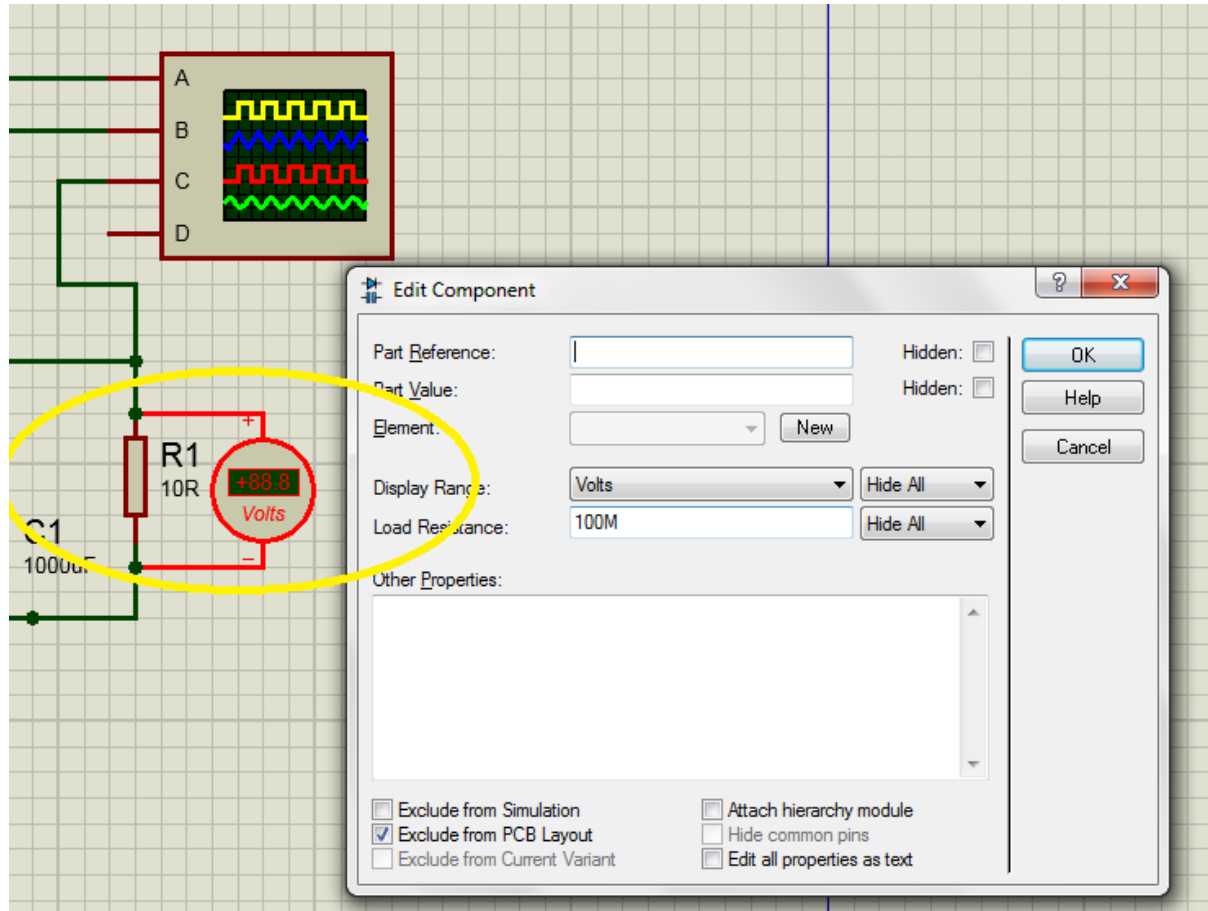


Fig 9. DC Voltmeter

6. Oscilloscope

An Oscilloscope is used to visualize and analyses voltage signals by displaying their waveforms over time. It helps in measuring AC and DC signals, Frequency, amplitude.

I have used oscilloscope in my circuit with following connections:-

- **A Terminal** - Connected to the primary winding of input AC Signal.
- **B Terminal** - Connected to the secondary winding of transformer.
- **C Terminal** – Connected to load resistor.
- **D Terminal** - Not connected.

Observation and Expected Signal:

I have connected the oscilloscope to observe the circuit behaviour.

AC Input Signal – It is displaying sine wave since it is an alternating current.

DC Output Signal – It is displaying different type of output wave with different filters.

Circuit Diagram of Series inductor filter :-

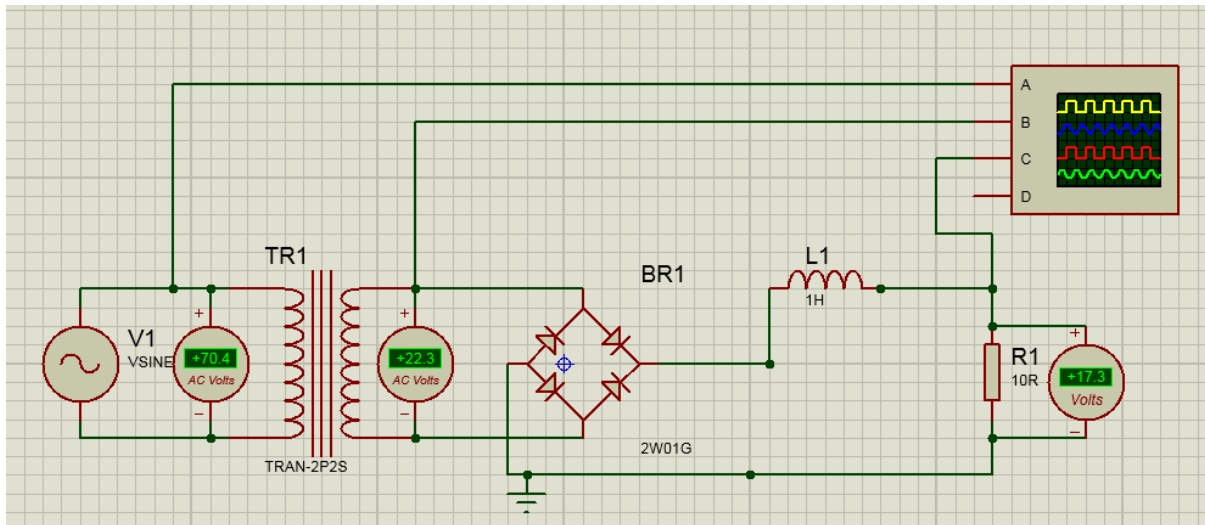


Fig 10. Circuit Diagram of Series inductor filter

Reading of Oscilloscope:

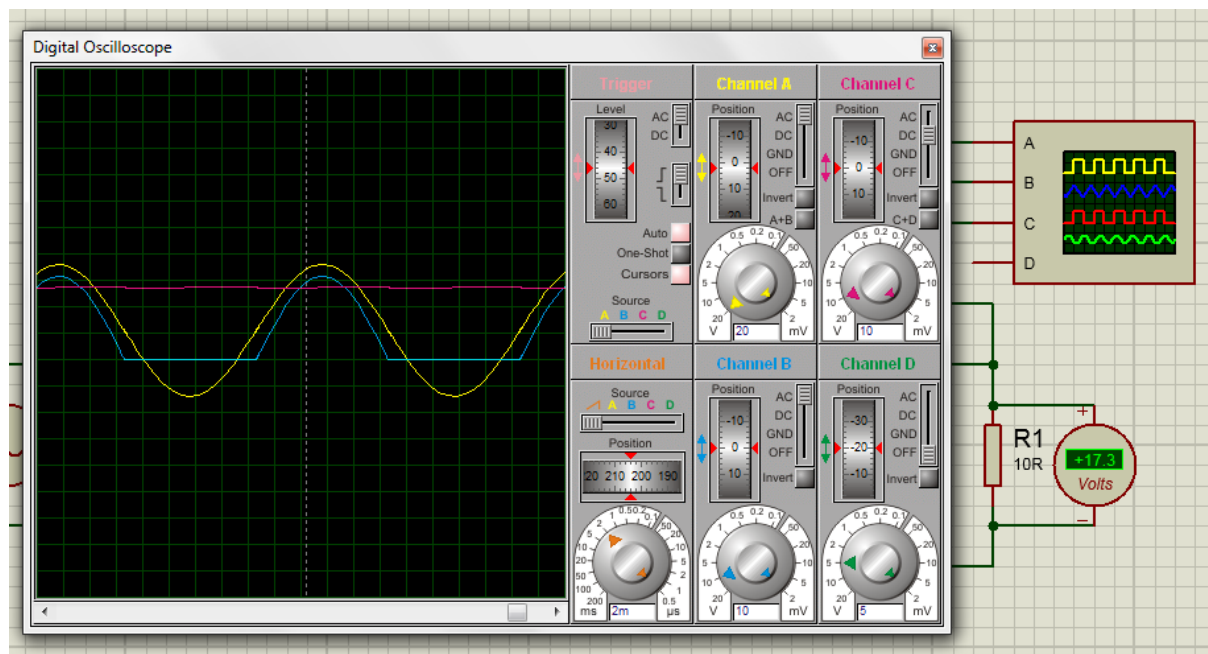


Fig 11. Oscilloscope reading of Series inductor filter

I observed the oscilloscope readings carefully. The yellow line represents the AC input signal, Blue line represents the step-down AC signal and red line represents the output DC signal after getting filtered. The DC output is not a perfect straight line showing it has little AC components in it.

Circuit Diagram of Shunt Capacitor filter :-

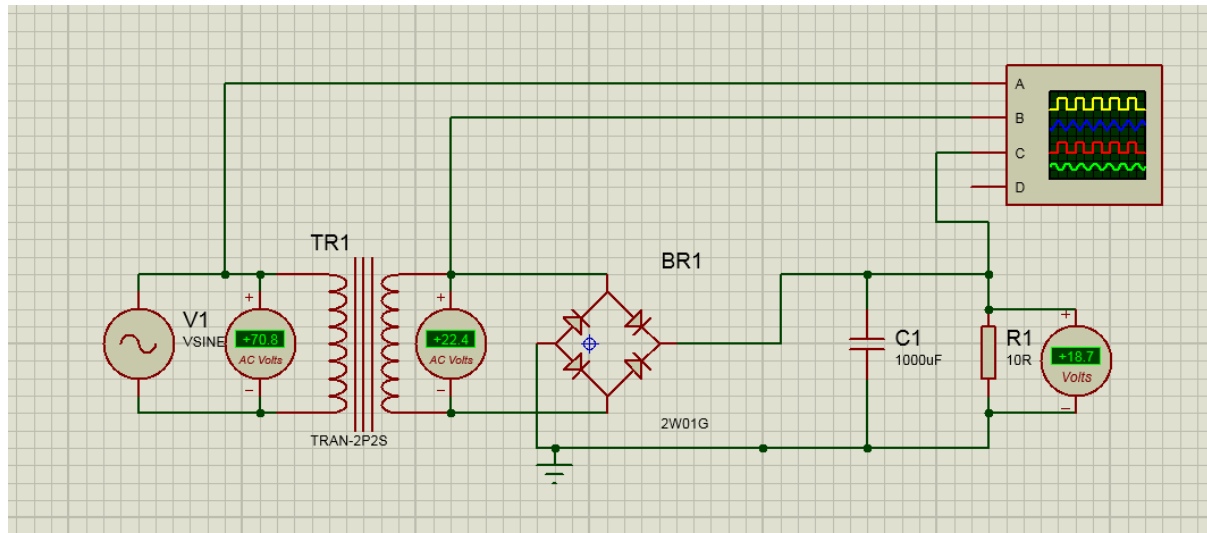


Fig 12. Circuit Diagram of Shunt Capacitor filter

Reading of Oscilloscope:

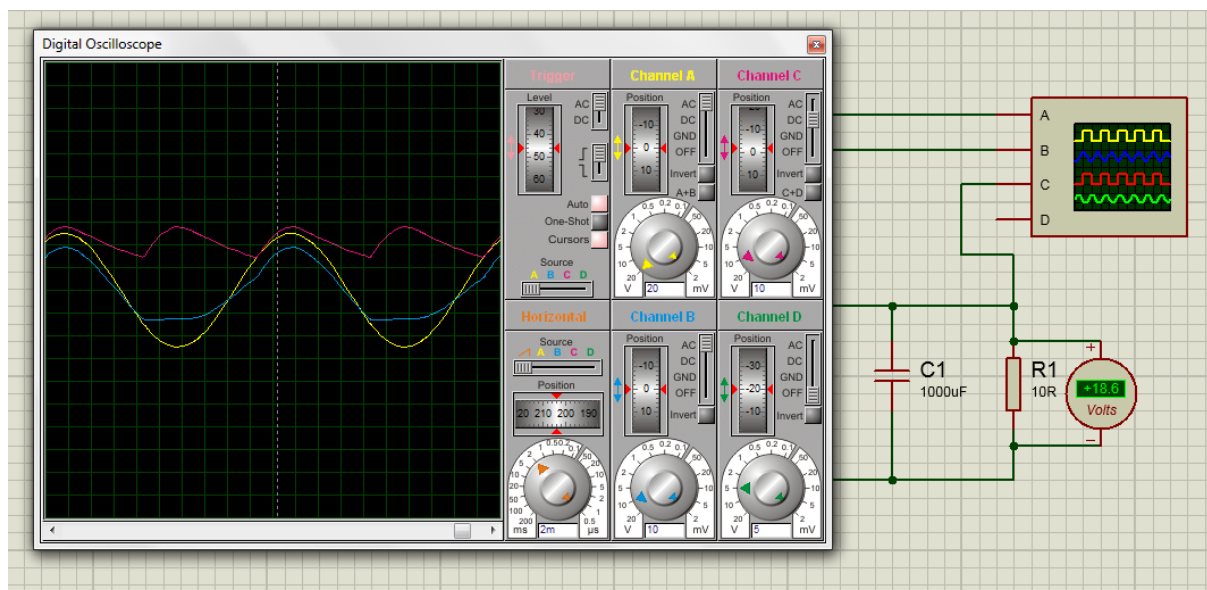


Fig 13. Oscilloscope reading of Shunt Capacitor filter

I observed the oscilloscope readings carefully. The yellow line represents the AC input signal, Blue line represents the step-down AC signal and red line represents the output DC signal after getting filtered. The DC output has ripples show it is not completely filtered the AC signal.

Circuit Diagram of LC filter :-

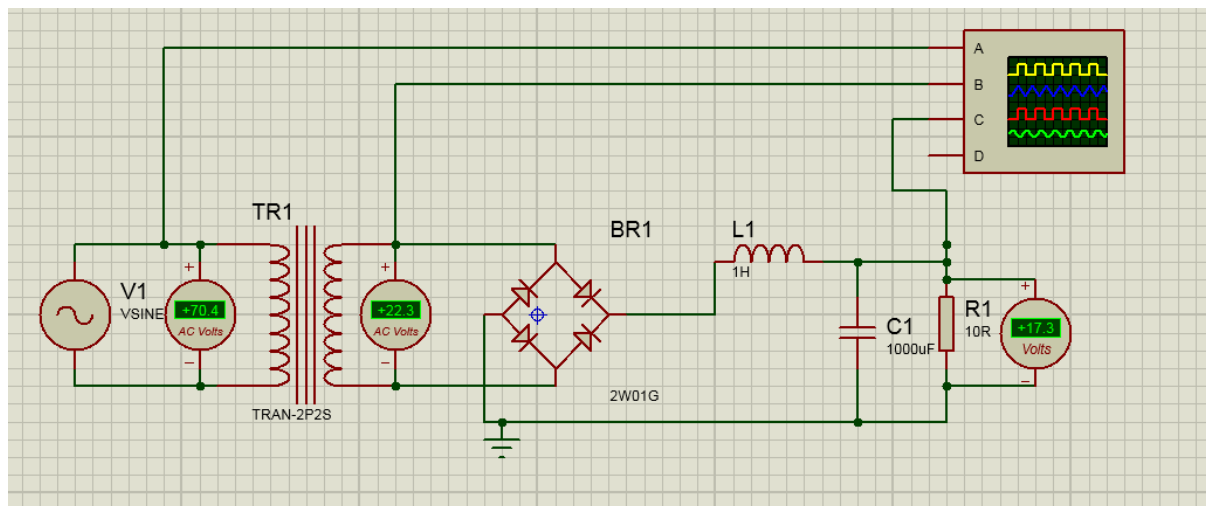


Fig 14. Circuit Diagram of LC filter

Reading of Oscilloscope:

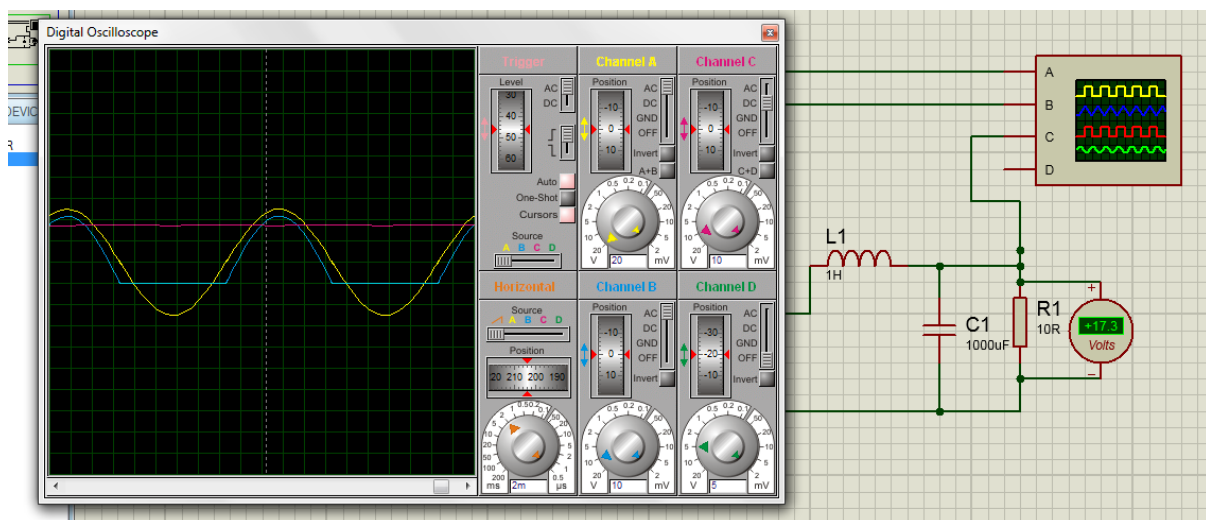


Fig 15. Oscilloscope reading of LC filter

I observed the oscilloscope readings carefully. The yellow line represents the AC input signal, Blue line represents the step-down AC signal and red line represents the output DC signal after getting filtered. The DC output is almost straight line with very little variation showing it is almost Pure DC .

Circuit Diagram of Pi filter :-

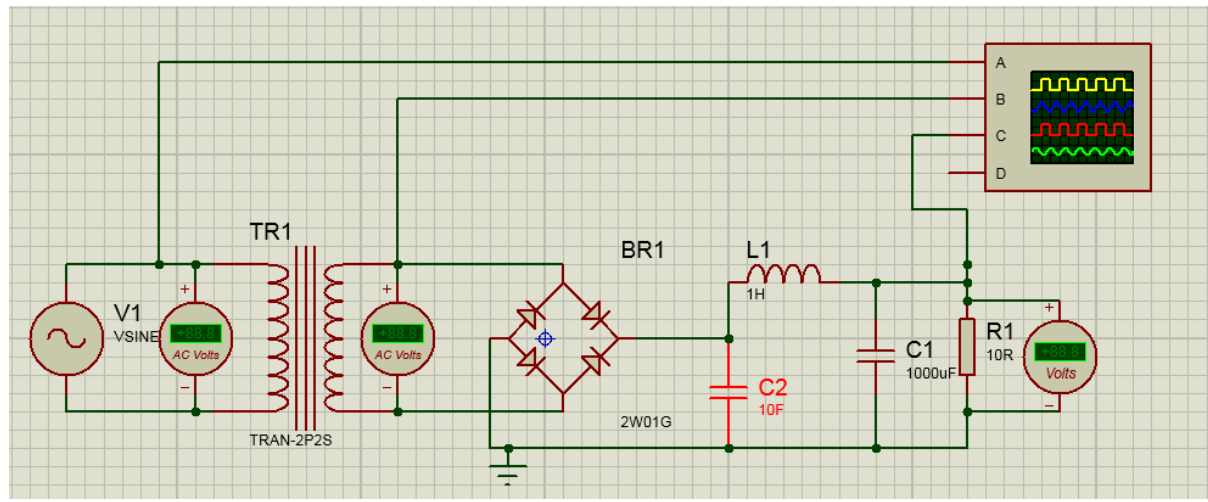


Fig 16. Circuit Diagram of Pi filter

Reading of Oscilloscope:

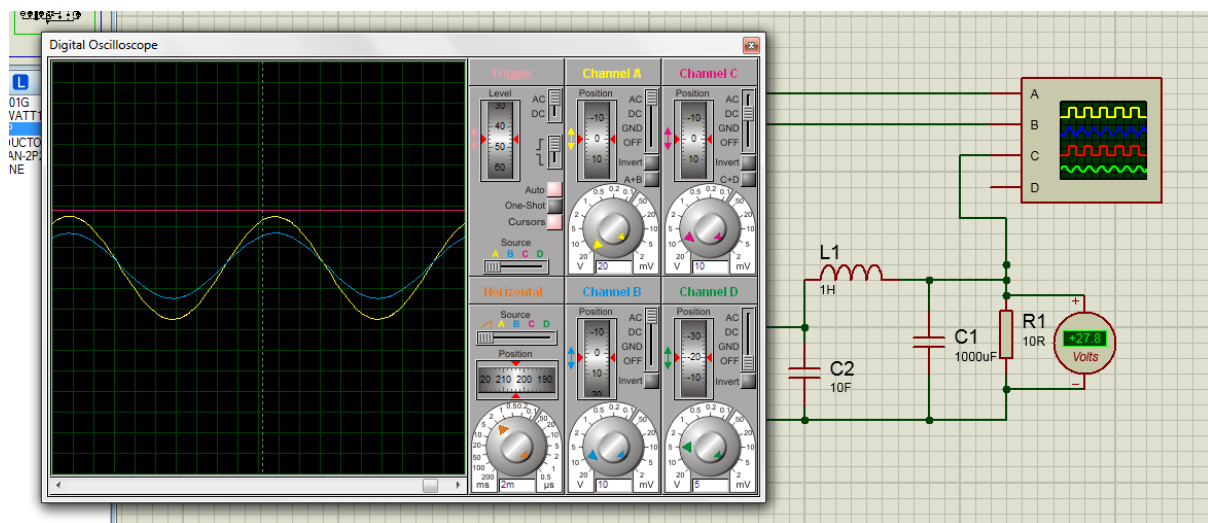


Fig 17. Oscilloscope reading of Pi filter

I observed the oscilloscope readings carefully. The yellow line represents the AC input signal, Blue line represents the step-down AC signal and red line represents the output DC signal after getting filtered. The DC output is perfect straight line showing Pure DC.

Result & Conclusion:-

After carefully watching the output of oscilloscope I can see that the output of Pi-filter is only having pure form DC other all filters were also somewhat able to convert the AC to DC

but there were little ripples left behind ,So In order to get pure DC we can prefer Pi filter over others with appropriate reading of components involved in it .

PROJECT DESCRIPTION

Design a PWM generation circuit using 555 Timer IC.

Project Objective :-

The main objective of project is to design a PWM generation circuit using 555 Timer IC in Proteus and showing output on oscilloscope. The project aims to show how resistor & capacitor value affect the turn on and turn off time of pulses.

Important Pre-requisite :-

Before starting the project it is crucial to have a good understanding of basic electronics concepts, circuit design principle and components. Also, knowledge of how to use Proteus simulation software is essential, as it will be platform for designing, simulating and analyzing the output of PWM generation circuit using 555 Timer IC .

Require Components

Sl No.	Component Name	Function
1.	Battery 9V	DC Voltage Source
2.	Capacitor	For Charge storage
3.	IC 555 timer	For Pulse generation
4.	Resistor	For Providing Resistance
5.	Oscilloscope	Measurement graph

REQUIRED INPUT PARAMETERS AND CALCULATIONS :-

Input Parameter :-

In the context of PWM generation circuit project using Proteus, the input parameter generally refers to the DC input voltage that the battery supply. Here are the key input parameters.

1. Battery

Here I have used battery of 9 volt to supply power to IC 555 timer .

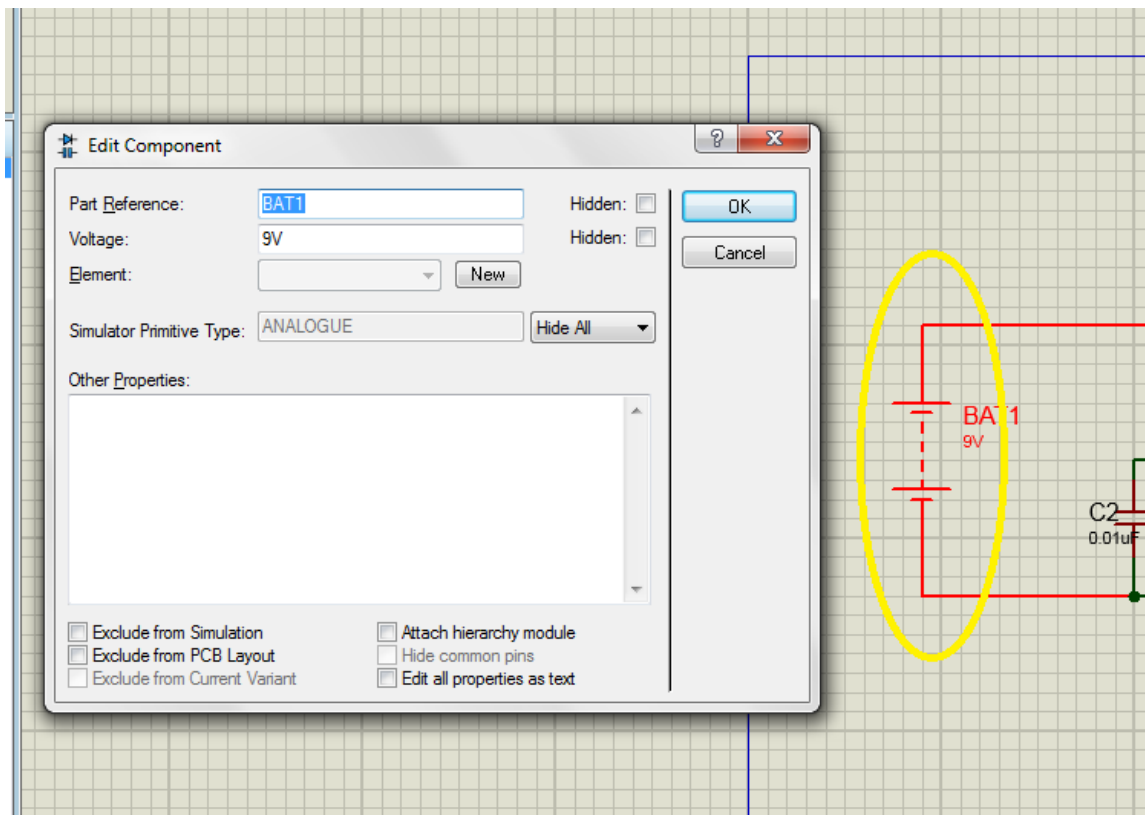


Fig 18. Battery 9V

2. Capacitor

Here I have used 2 capacitor of different charges i.e. 0.01uf and 0.1uf.

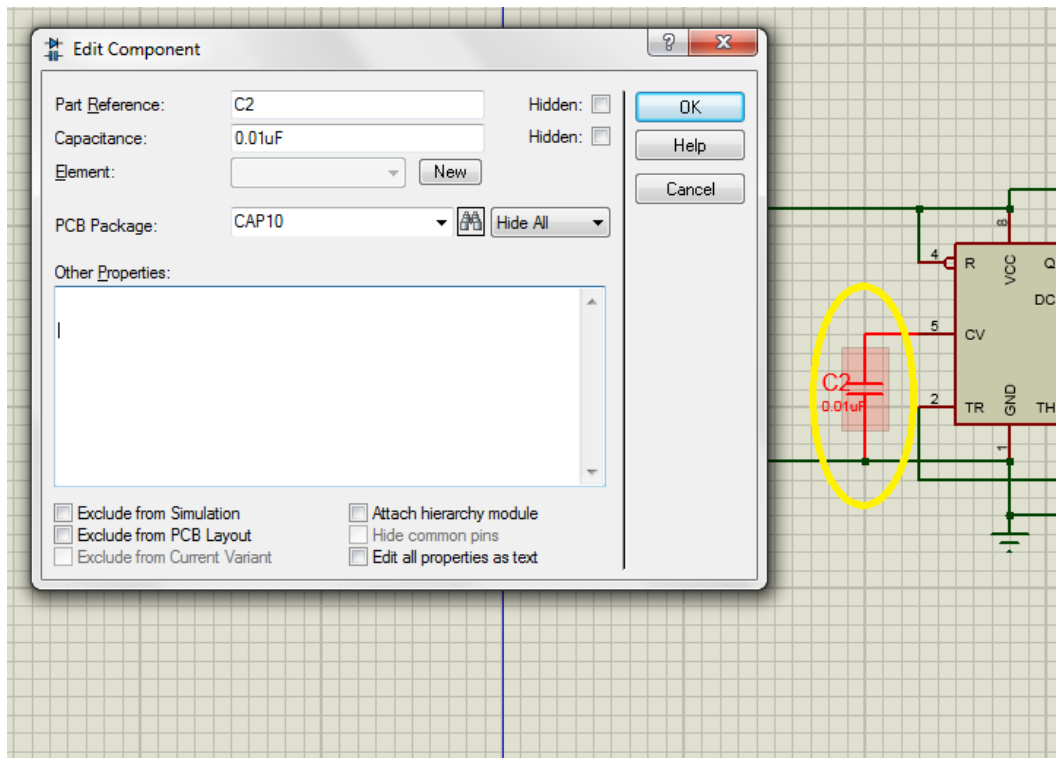


Fig 19(a). Capacitor 0.01uF

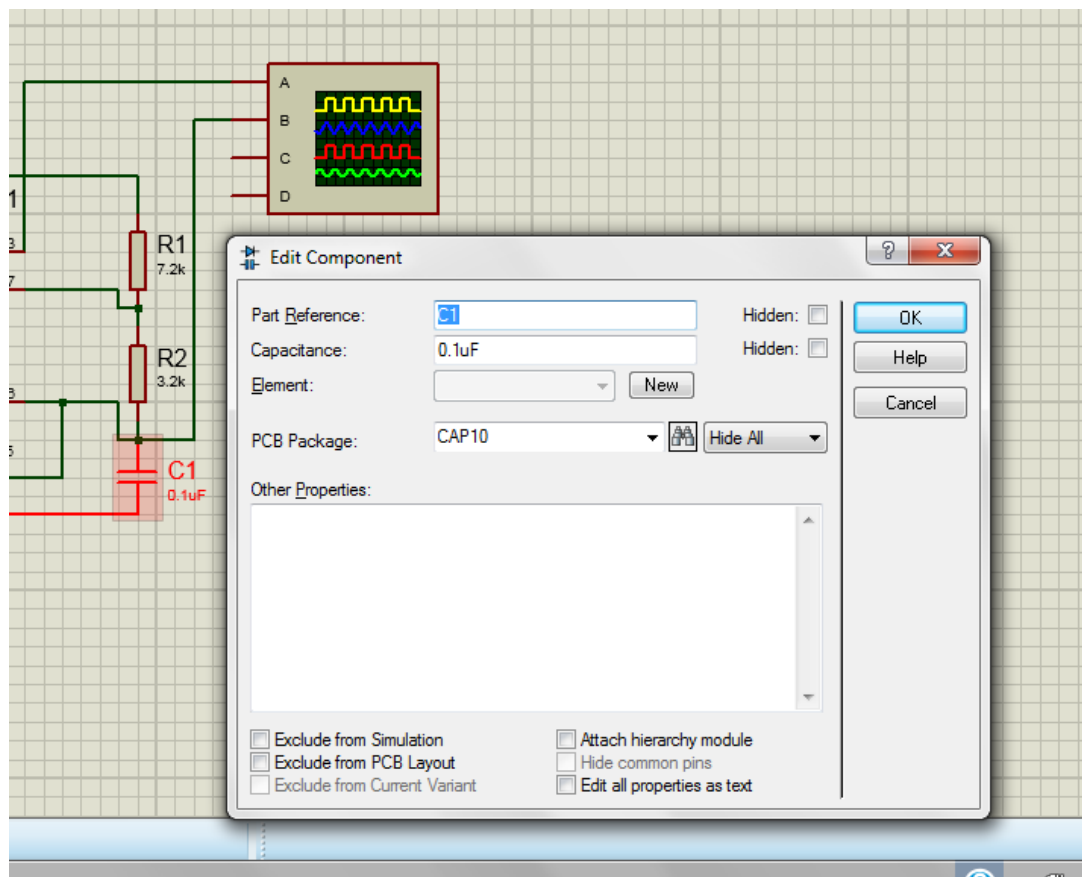


Fig 19(b). Capacitor 0.1uF

3.Resistor

Here I have used 2 resistor of resistance 7.2kohms and 3.2kohms.

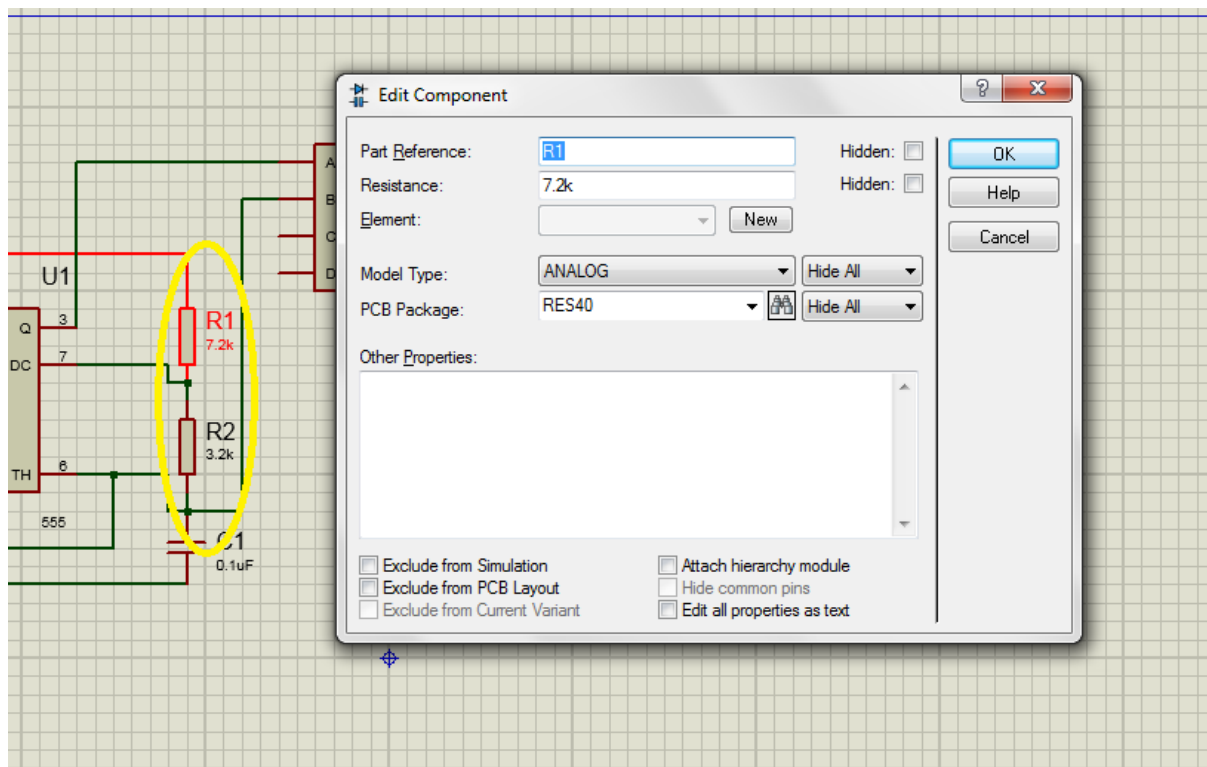


Fig 20. Resistors

3. 555 Timer IC

Here I have used 555 Timer IC to get the required pulse. These Pulses will turn On or OFF Mosfet or IGBT or JEFT.

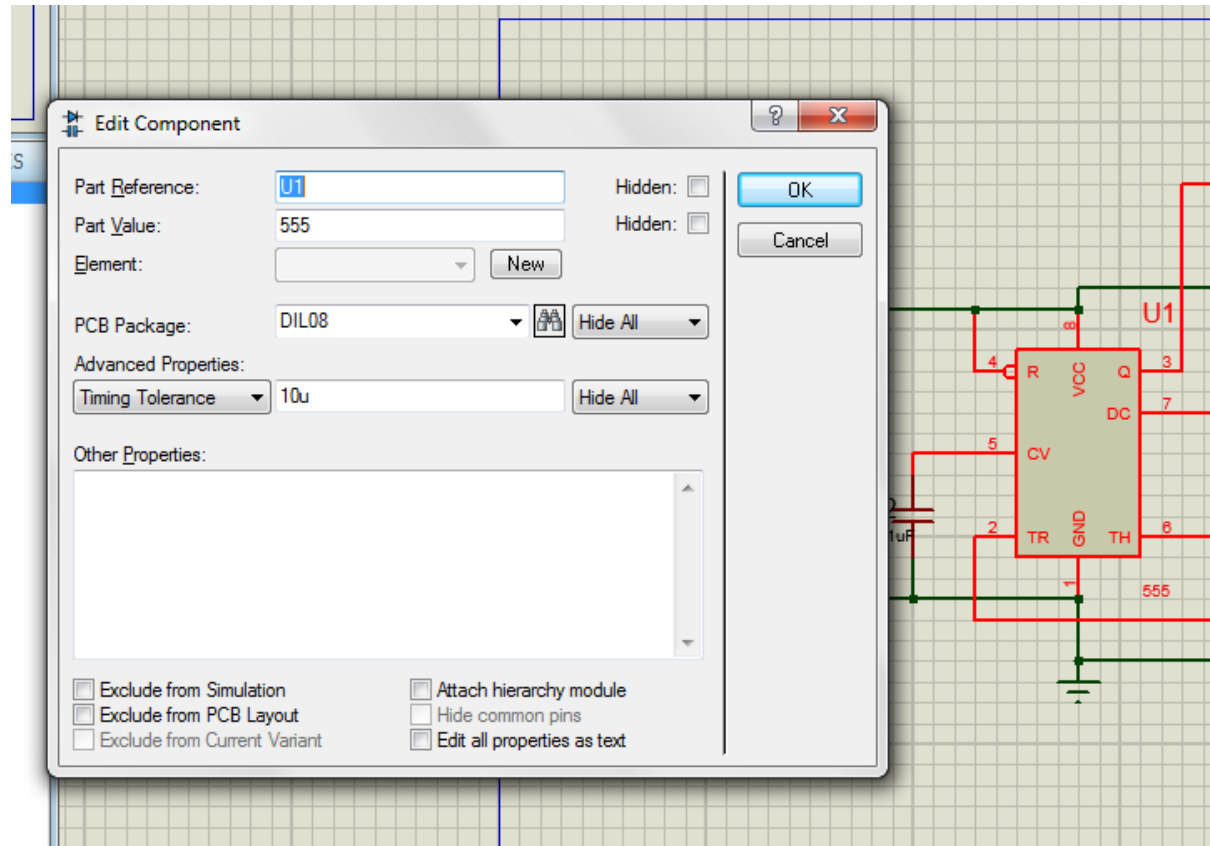


Fig 20. 555 timer IC

6. Oscilloscope

An Oscilloscope is used to visualize and analyses voltage signals by displaying their waveforms over time. It helps in measuring AC and DC signals, Frequency, amplitude.

I have used oscilloscope in my circuit with following connections:-

- **A Terminal** - Connected to LEG 3 to view Pulses.
- **B Terminal** - Connected to capacitor C1 to view charging & discharging of capacitor.
- **C Terminal** – Not Connected.
- **D Terminal** - Not Connected.

Observation and Expected Signal:

I have connected the oscilloscope to observe the circuit behaviour. Here I can see how changing the value of Resistor and capacitor also affect the Turn ON and Turn Off time of

pulses. So accordingly we can adjust the value of resistor and capacitor and adjust Turn On and Turn Off time period.

Circuit Diagram of 555 Timer IC :-

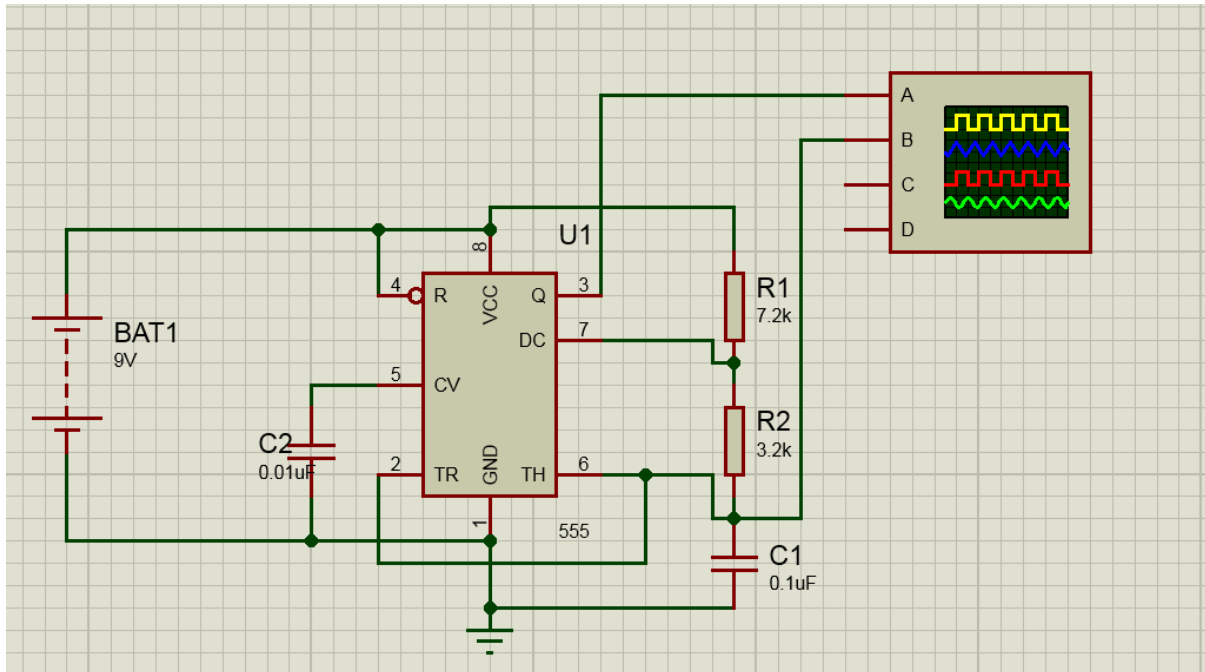


Fig 21. Circuit diagram of 555 timer IC

Reading of Oscilloscope:

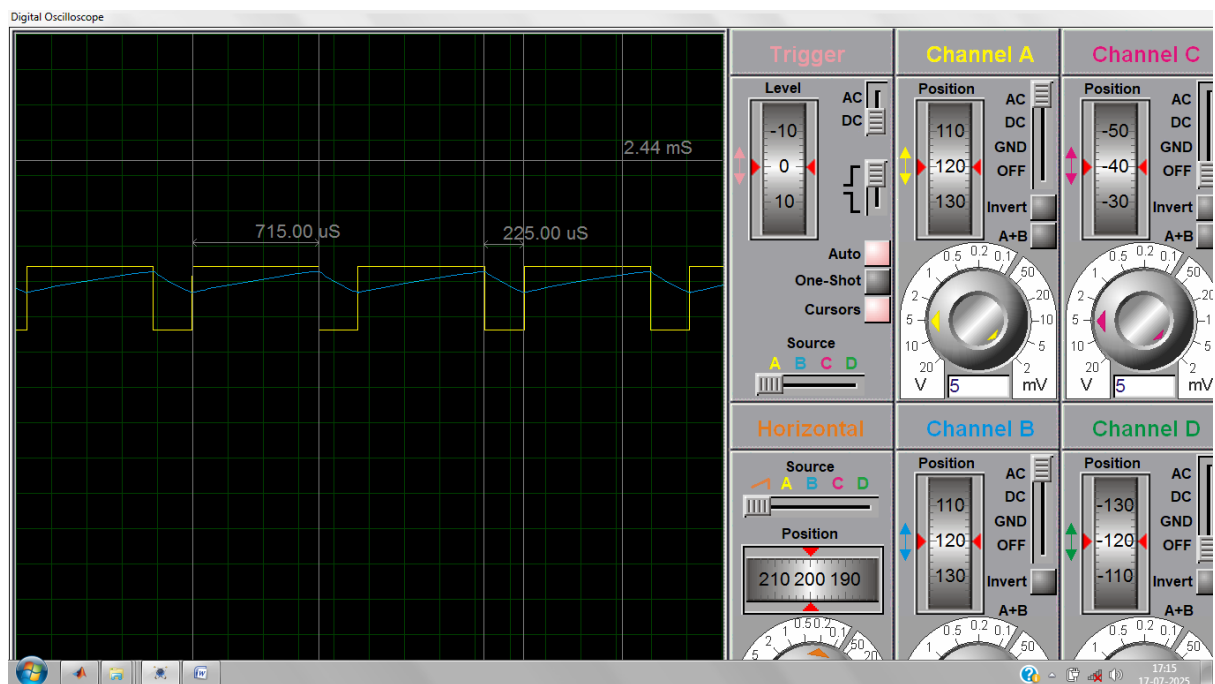


Fig 22. Oscilloscope output of 555 timer IC

I observed the oscilloscope readings carefully. The yellow line represents the pulses which we want to generate and the Turn On time is 715us and Turn Off Time is 225us. The Blue line represent charging and discharging of capacitor when the pulses is turned On the capacitor is getting charged and when pulses is turned off it getting discharged.

Circuit Diagram of 555 Timer IC (changed value of Resistor and capacitor) :-

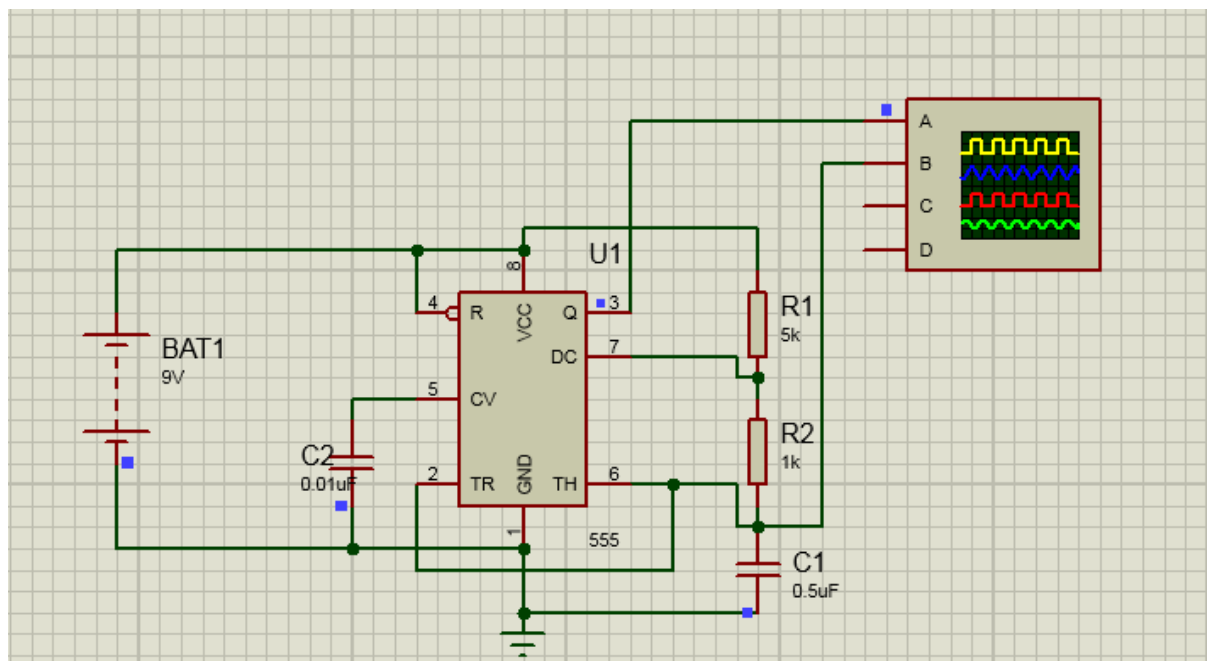


Fig 23. Circuit diagram of 555 timer IC

Reading of Oscilloscope:

I observed the oscilloscope readings carefully. The yellow line represents the pulses which we want to generate and the Turn On time is 2.40ms and Turn Off Time is 700us. The Blue line represent charging and discharging of capacitor when the pulses is turned On the capacitor is getting charged and when pulses is turned off it getting discharged.

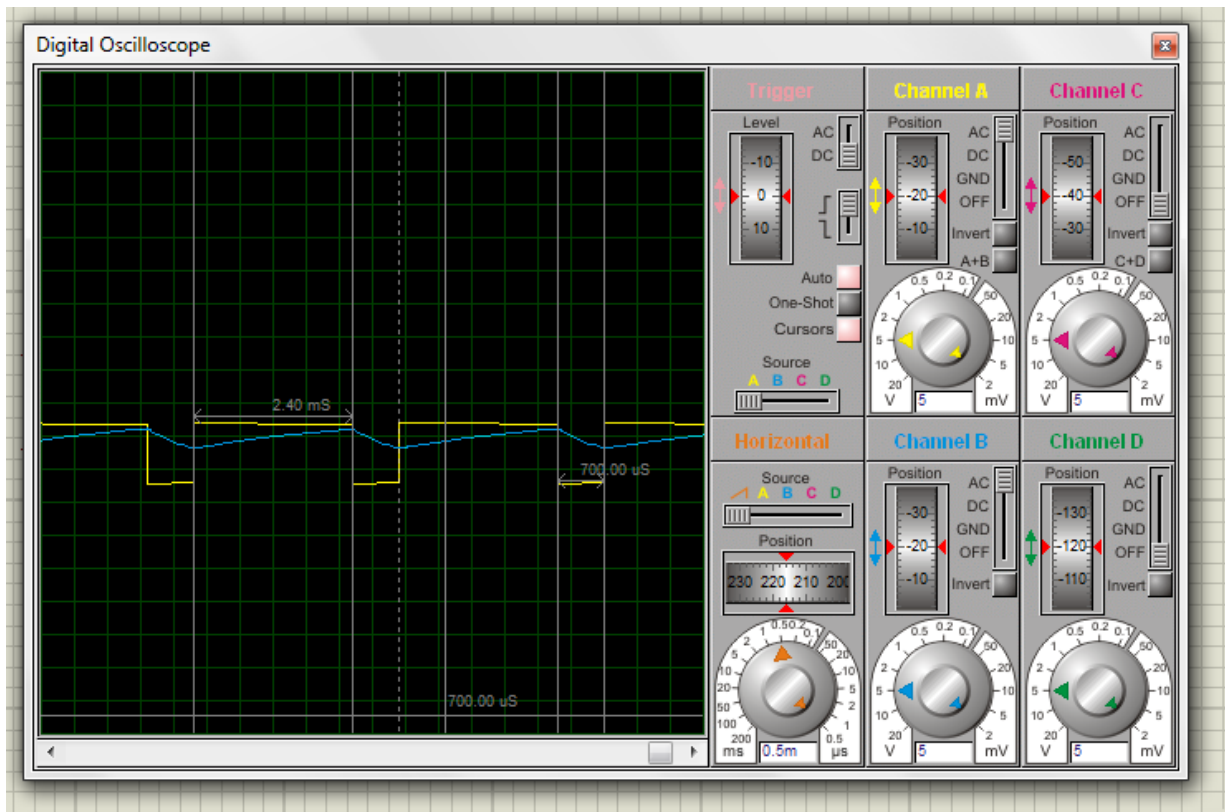


Fig 24. Oscilloscope output of 555 timer IC

Result & Conclusion:-

After carefully watching the output of oscilloscope I can see that the pulses generated by 555 timer IC can be varied by changing the value of Resistor and capacitor of the circuit .

According to our need we can adjust the value of Turn ON and Turn Off time by varying the value of Resistance and capacitance.

Converter and Inverter Simulation using Matlab

Part 01- Converter

Introduction:-

In Electric Vehicle (EV) DC-DC converter play a crucial role in regulating and managing power between components that operate at different voltage levels. These converter either step up (i.e. Boost Converter), step down (i.e. Buck Converter), or adjust in both ways (i.e. Buck Boost Converter) so that efficient power is delivered to each and every components like motor, Auxiliary systems etc.

Objective:-

- To model Buck, Boost DC-DC converters using MATLAB Simulink.
- To validate output voltage regulation.
- To observe current ripple.
- To find efficiency of DC-DC converter.

DC-DC converter:-

A DC-DC converter is a device that changes one DC voltage to another DC voltage. It act like voltage controller, it sometime increases the voltage or decreases the voltage as our need.

In EV DC-DC converter plays very important role as the main battery usually gives high voltage(around 350 Volt) and many components like light ,sensor etc need lower voltage like (around 12 Volt). The DC-DC converter reduces the battery voltage safely and efficiently so that all components works properly without getting damaged.

There are mainly three types of DC-DC converter which are used in EV:-

1. Boost Converter – Step Up
2. Buck Converter – Step Down
3. Buck-Boost Converter- Step Up-Down

1.Boost Converter:-

Boost converter is DC-DC converter that increases lower input voltage to higher output voltage. These converters are generally needed to boost voltage from battery to power the electric motor and other high voltage component.

Working:-

A Boost converter works by turning switch (like MOSFET) ON and OFF very fast using PWM(Pulse Width Modulation) signal, When switch is ON current flows from battery and stores energy in the inductor. When switch turns off the energy stored in inductor is released which adds to input voltage and passes through a diode to the output. A capacitor is used to smooth the output voltage and passes through a diode to the output. A capacitor is used to smooth the output voltage and reduce ripples. This process increases the input voltage to higher, steady output voltage.

Simulation Process:-

For simulation of the boost converter I used MATLAB –Simulink. Since it is a step up DC-DC converter, I converted 16V DC to 24V DC in the simulation.

Blocks Used

1. DC Voltage Source (16V)
2. Current Measurement – 2Pc
3. Voltage Measurement – 3Pc
4. Display – 2Pc
5. Inductor
6. MOSFET
7. Diode
8. Capacitor
9. Resistor
10. Pulse Generator
11. Power GUI
12. Scope

Working Step –

1. Input Setup:

I have used a DC Voltage source and set it to 16 Volt which act as input. A Voltage measurement block was connected in parallel to the DC voltage source to measure input voltage and display block and scope was connected to this voltage measurement block to visualize the input voltage. I have connected the current measurement block in series with battery to measure the input current.

2. Main Circuit Component:

I have added a MOSFET, diode, Inductor, capacitor, resistor, to build the buck converter circuit and I connected inductor ,diode in series with input voltage source. I connected MOSFET, Capacitor, and resistor in parallel to form the output side of the circuit.

3. Output Monitoring :

To observe the output voltage I used another voltage measurement block and connected parallel to resistor and current measurement block is connected in series with resistor. Also voltage measurement block is connected in parallel to inductor. A current measurement block in series with DC voltage source is connected to view the current output. And scope was connected to output of all these voltage measurement and current measurement block to visualize the behaviour. I have connected the display to the output of voltage measurement block at DC voltage source and at load resistor.

4. PWM Control:

A PWM generator block is used to control the MOSFET switching.

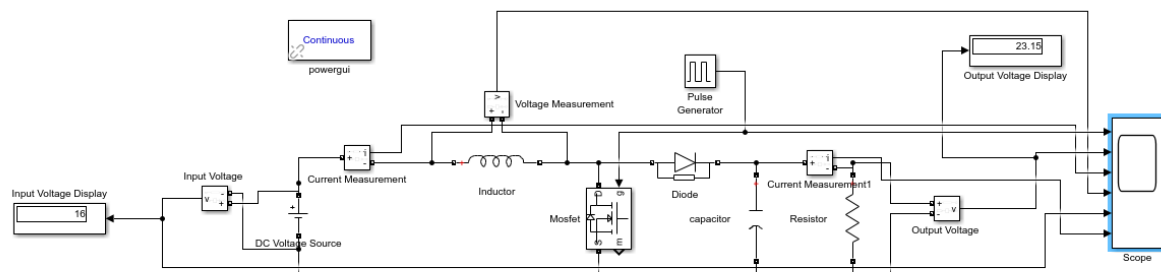


Fig. Boost Converter 16V to 24V

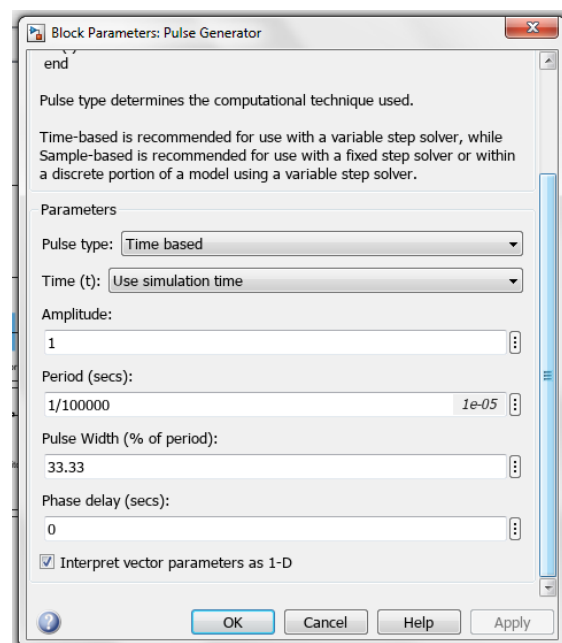


Fig. Pulse generator input parameter

Output Analysis –

I have run the simulation at 0.1 sec. I have zoomed the scope to 0.0427 to 0.0431 seconds to analyse the output clearly.

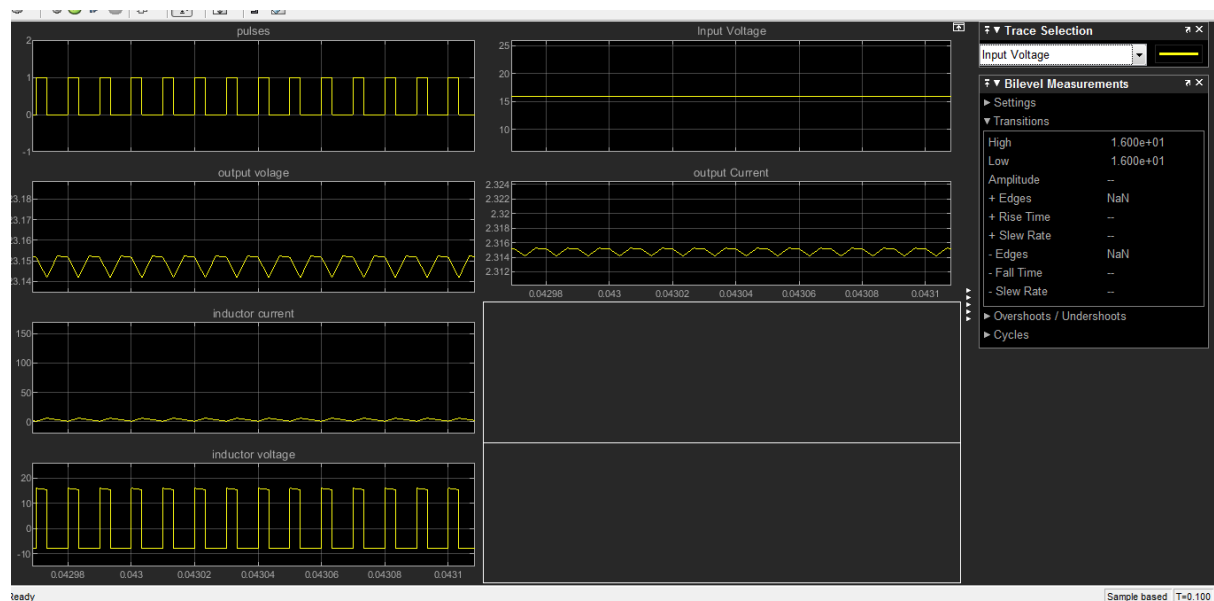


Fig. Output Analysis

In scope analysis it is clearly seen that pulse is varying from 0 to 1 which helps MOSFET in switching ON and OFF.

I have used trace selection for input voltage and can see that input voltage is constant at value 1.600e+01 Volt.

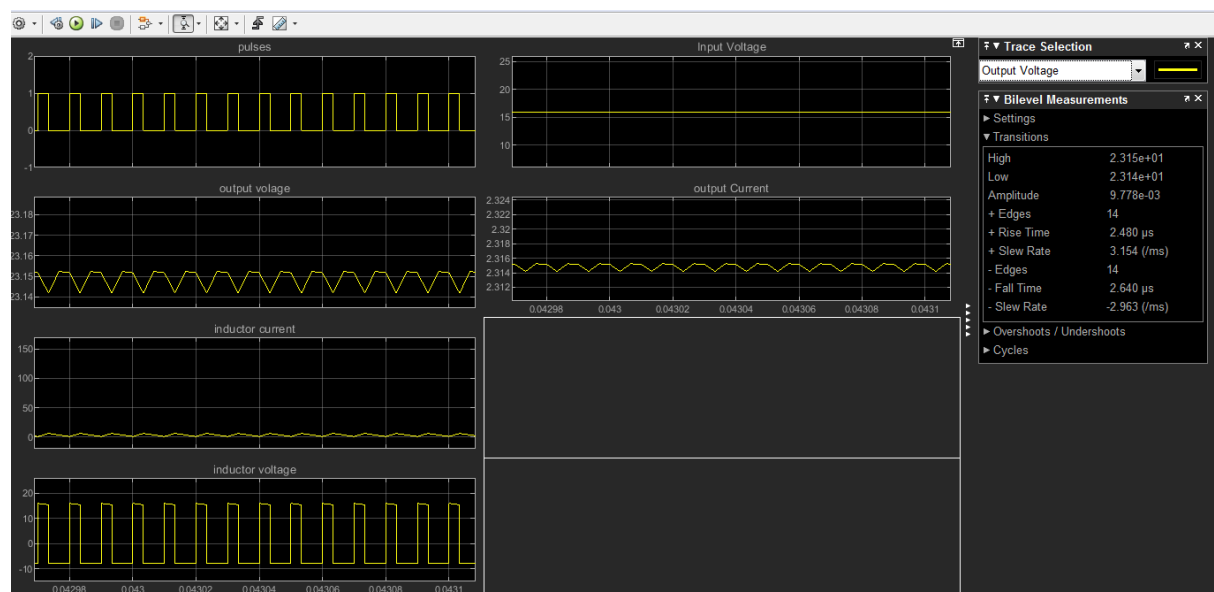


Fig. Output Analysis

I have used trace selection for Output voltage and I can see after getting boost its value has increased but the value is fluctuating between 2.315×10^1 to 2.314×10^1 volt which show that output is not perfectly constant DC.

I can see that output current is happening ripples .

I can see that inductor voltage is constantly increasing and decreasing which shows how inductor is storing and releasing energy.

Above 2 graph shows I have successfully boosted the input voltage from 16 Volt to 23.15 Volt.

2.Buck Converter:-

Buck converter is DC-DC converter that reduces higher input voltage to lower output voltage. These converter are generally needed to run the auxiliary which run at low DC voltage such as headlight, taillight etc.

Working:-

A Buck converter works by turning switch (like MOSFET) ON and OFF very fast using PWM(Pulse Width Modulation) signal, When switch is ON current flows from battery and stores energy in the inductor. The diode is reverse-biased and does not conduct. When switch turns off the inductor resists the sudden drop in current and maintain current flow by releasing stored energy. The magnetic field collapses and inductor reverses polarity to maintain current flow. The diode become forward biased and current flows through the diode to the load. The inductor discharge into load and capacitor. This process decreases the input voltage to lower, steady output voltage.

Simulation Process:-

For simulation of the Buck converter I used MATLAB –Simulink. Since it is a step down DC-DC converter, I converted 24V DC to 12V DC in the simulation.

Blocks Used

13. DC Voltage Source (24V)
14. Current Measurement – 2Pc
15. Voltage Measurement – 3Pc
16. Display – 2Pc
17. Inductor
18. MOSFET
19. Diode
20. Capacitor
21. Resistor
22. Pulse Generator
23. Power GUI
24. Scope

Working Step –

1. Input Setup:

I have used a DC Voltage source and set it to 24 Volt which act as input. A Voltage measurement block was connected in parallel to the DC voltage source to measure input voltage and display block and scope was connected to this voltage measurement block to visualize the input voltage. I have connected the current measurement block in series with MOSFET to measure the input current.

2. Main Circuit Component:

I have added a MOSFET, diode, Inductor, capacitor, resistor, to build the buck converter circuit and I connected MOSFET, inductor in series with input voltage source. I connected diode, Capacitor, and resistor in parallel to form the output side of the circuit.

3. Output Monitoring :

To observe the output voltage I used another voltage measurement block and connected parallel to resistor and current measurement block is connected in series with resistor. Also voltage measurement block is connected in parallel to inductor. And scope was connected to output of all these voltage measurement and current measurement block to visualize the behaviour. I have connected the display to the output of voltage measurement block at DC voltage source and at load resistor.

4. PWM Control:

A PWM generator block is used to control the MOSFET switching.

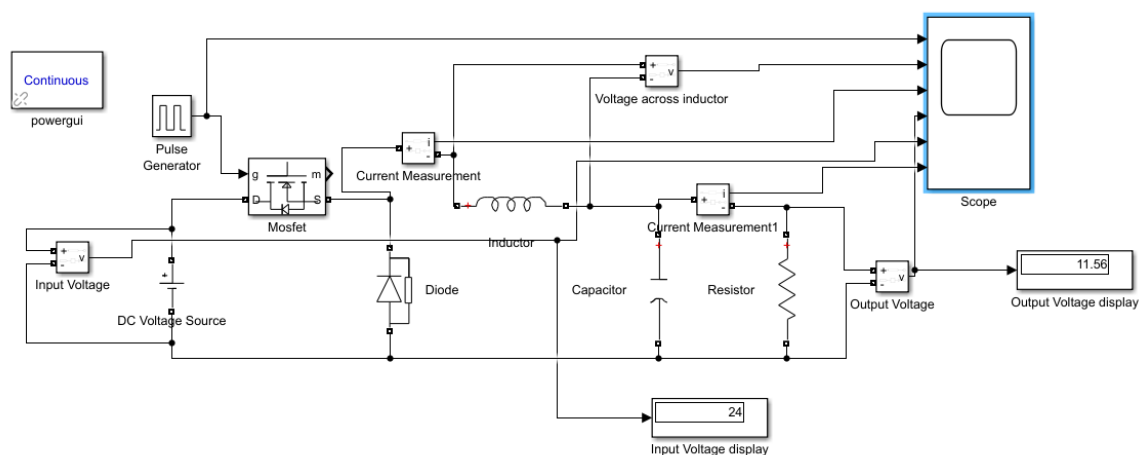


Fig. Buck Converter 24V to 12V

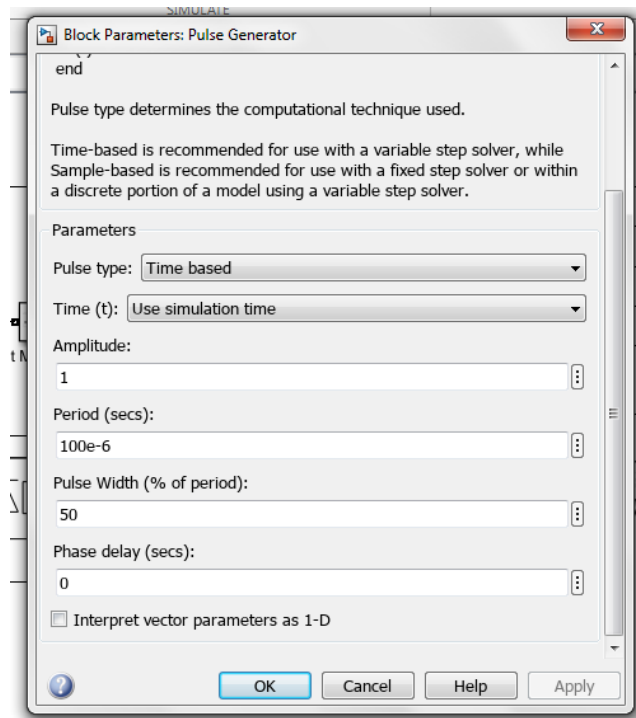


Fig. Pulse generator input parameter

Output Analysis –

I have run the simulation at 0.1 sec. I have zoomed the scope to 0.0438 to 0.0460 seconds to analyse the output clearly.

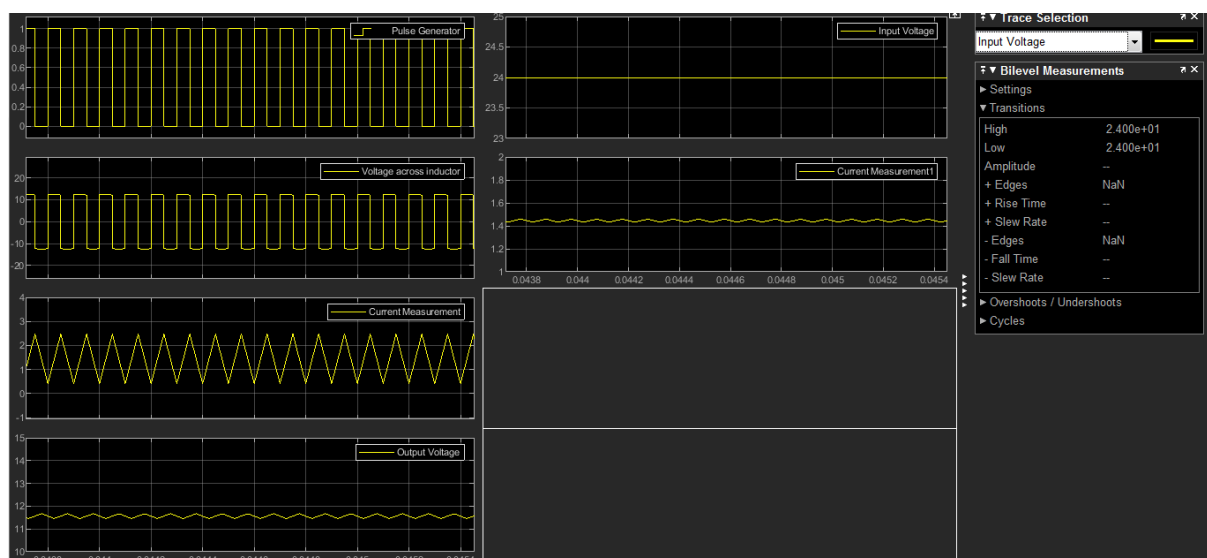


Fig. Output Analysis

In scope analysis it is clearly seen that pulse is varying from 0 to 1 which helps MOSFET in switching ON and OFF.

I have used trace selection for input voltage and can see that input voltage is constant at value 2.400e+01 Volt.

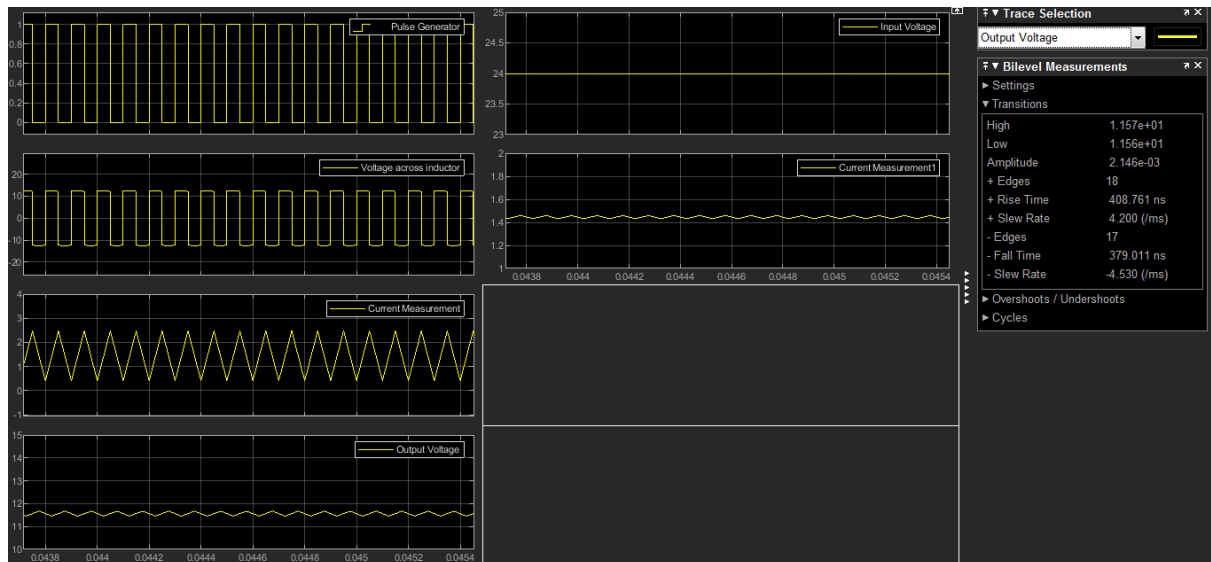


Fig. Output Analysis

I have used trace selection for Output voltage and I can see after getting step-down its value has decreased but the value is fluctuating between 1.157e+01 to 1.156e+01 Volt which show that output is not perfectly constant DC.

I can see that output current is happening ripples .

I can see that inductor voltage is constantly increasing and decreasing which shows how inductor is storing and releasing energy.

Above 2 graph shows I have successfully stepped down the the input voltage from 24 Volt to 11.57 Volt.

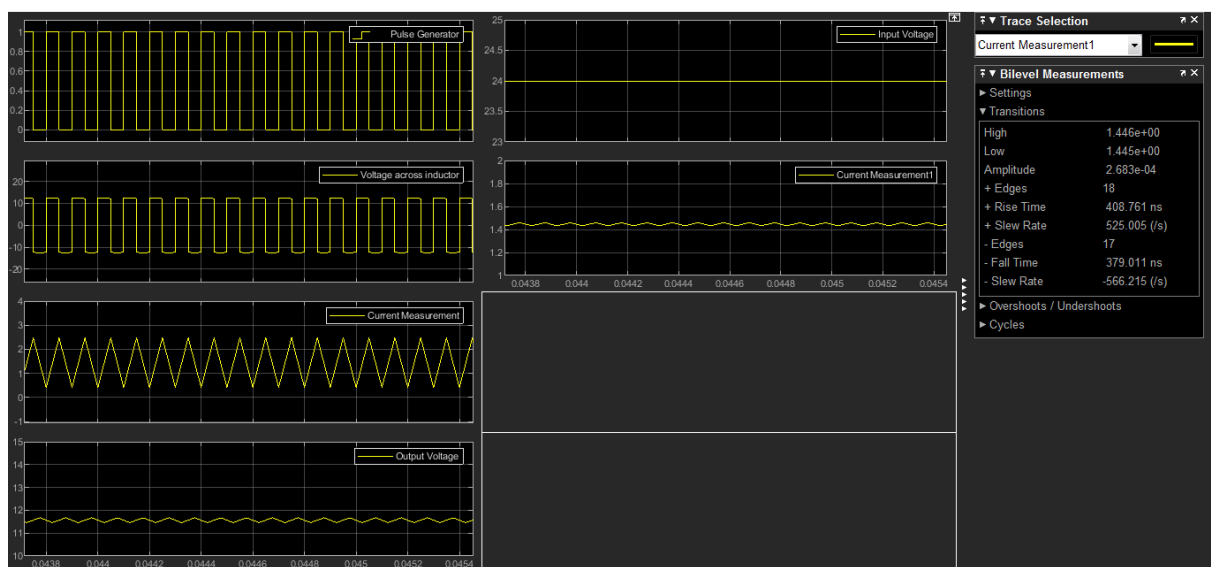


Fig. Output Analysis

I have used trace selection for current measurement block and can see current varying between 1.446e+00 to 1.445e+00 amp.

Converter and Inverter Simulation using Matlab

Part 02-Inverter

Introduction:-

In Electric Vehicle (EV) inverters play a critical role in converting the DC power from battery to AC power needed to drive the motors.

Objective:-

- To model Single-phase & Three - phase Inverter using MATLAB Simulink.
- To understand PWM technique used in Inverter.
- To find the efficiency.

Inverter:-

Inverters are electronic device which convert Direct Current (DC) to Alternating Current (AC). It is widely used in EV. It is key component in EV as it convert DC from battery into AC to power AC motor which are used in many EV's. EV stores energy in battery in form of DC but most of motor like PMSM, Induction Motor runs on AC So inverter are very necessary component in EV.

Operations:-

The process of changing DC to AC is done using the energy stored in battery or from rectifier. This is done using electronic switches that turn off and on very fast like MOSFET. They switch DC voltage very fast to make it behave like AC.

Types of Inverter:-

1. Single Phase Inverter – Half Bridge Inverter, Full Bridge Inverter
2. Three Phase Inverter – 120 degree mode, 180 degree mode

Half Bridge Inverter

It is type of inverter that convert DC to AC using 2 MOSFET and 2 Capacitor. These MOSFET Turn ON and OFF one by one. This create waveform similar to AC, It is called Half Bridge because the output voltage is half of input DC voltage. These are generally used in low power application. The 2 capacitor divide the DC voltage into equal parts which help in easy switching.

Simulation Process:-

For simulation of Single Phase Inverter I used MATLAB –Simulink .

Blocks Used -

1. DC voltage source
2. Resistor - 2No.
3. Capacitor – 2No.
4. Voltage Measurement Block – 2No.
5. MOSFET- 2No.
6. Pulse generator – 2No.
7. Scope
8. Power GUI

In simulink for simulation of single phase bridge Inverter I have used DC voltage source amplitude of 24V and connected resistor (1 ohm) in series with this DC source. Next I have used 2 capacitors (0.1F each) which are in series and parallel to DC voltage source, Next I used a 10ohm resistor as load which is resistive load to look the AC output. For switching I have used MOSFET. After that I connected 2 pulse generator blocks to control MOSFET switches. I set the time period to 0.02sec, pulse width to 50%, Amplitude to 1, phase delay to 0Sec, 0.01Sec and connected them to Mosfet s1, s2 respectively. Finally I used voltage measurement block connected parallel to load resistor and it's output is connected to scope and display to view the result.

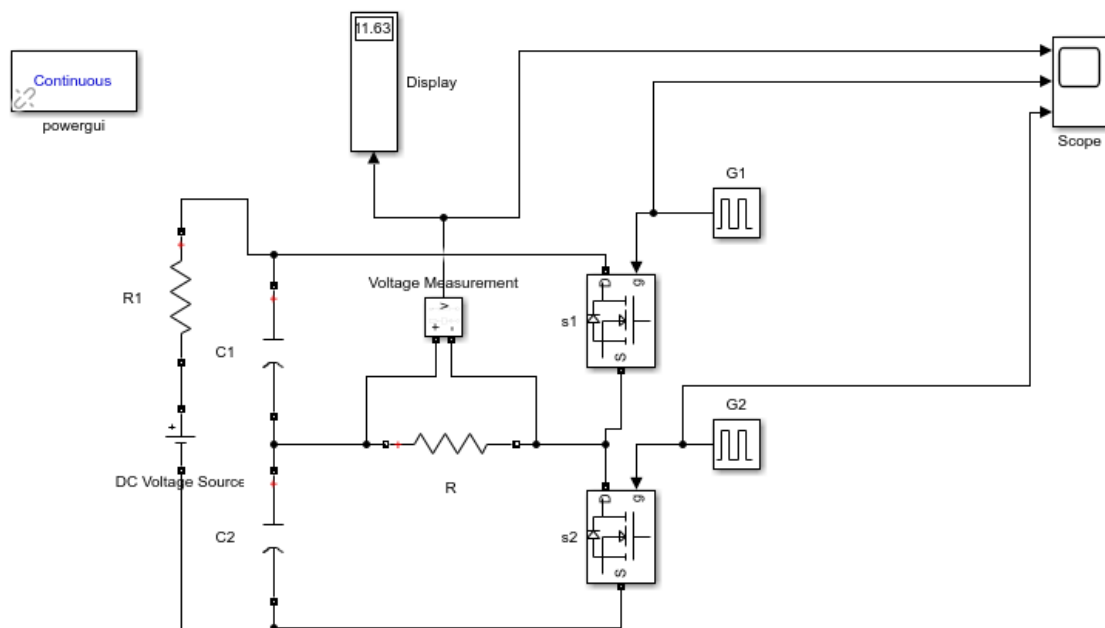


Fig : Single Phase Halfwave inverter using MOSFET.

Output Analysis:-

In simulation of single phase half bridge inverter I created a model in simulink which uses MOSFET. I have simulate single phase half bridge inverter at 0.1sec .

Scope –

- 1st row - It produce square wave as output voltage.
- 2nd row & 3rd row shows the switching of MOSFET.
- The output Voltage is varying from $1.189\text{e}+01$ to $-1.188\text{e}+01$ Volt in form of square wave.

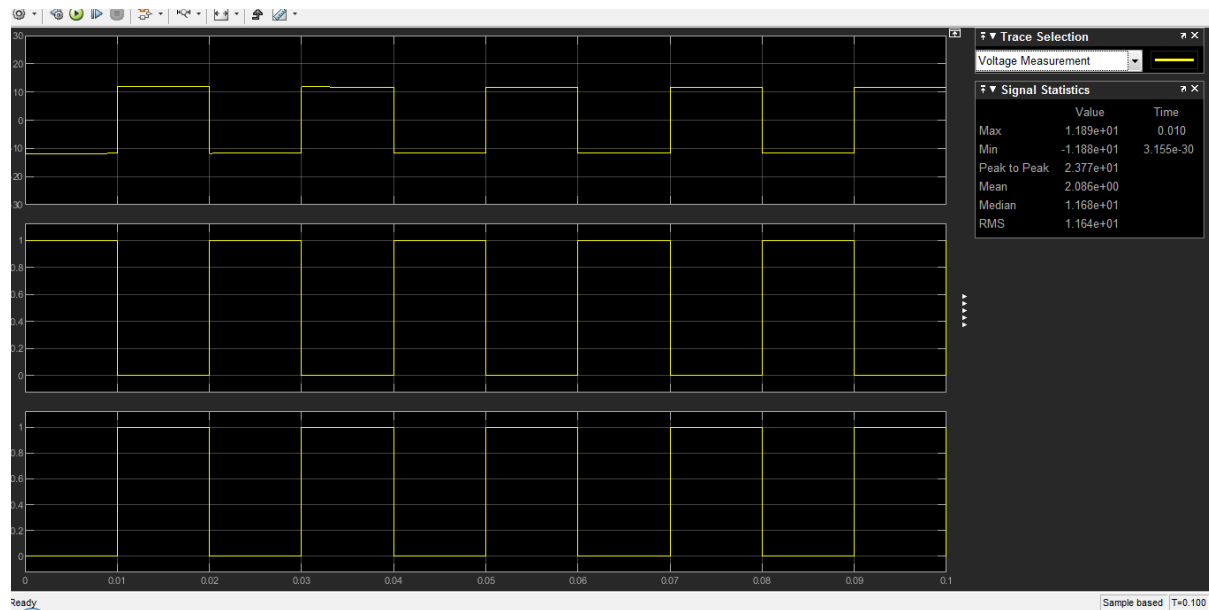


Fig :Scope view of Single Phase Halfwave inverter .

Conclusion:

Half Bridge Inverter MOSFET produce square wave.

Efficiency Calculation:

For efficiency calculation I have to know power out and power in.

Block included apart from above

- Voltage measurement -1pc
- Current measurement -2pc
- Product -2pc
- Mean – 2pc
- Divide – 1pc
- Gain -1pc
- Display – 1pc

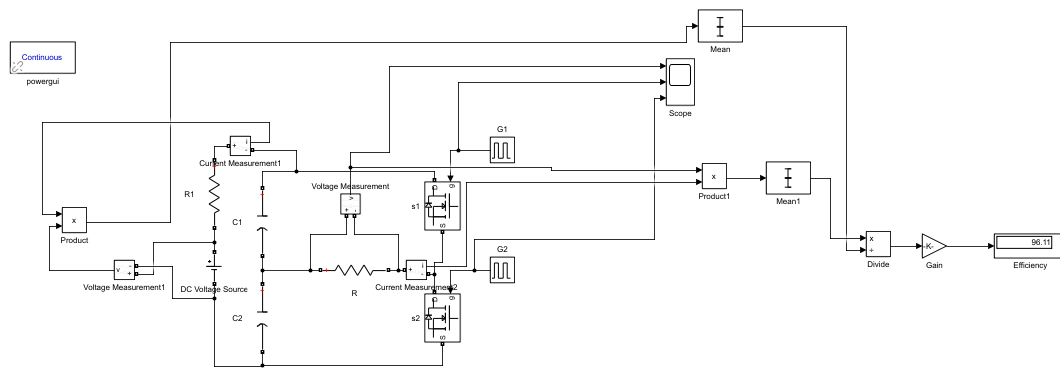


Fig : Single Phase Halfwave inverter using MOSFET with efficiency.

I have connected voltage measurement device parallel to input voltage source and in series with resistor I have connected current measurement device, I have taken product of these voltage and current and connected them to mean block to get the average power input. Similarly I connected voltage measurement device parallel to load resistor and in series I connected current measurement device and product of these is taken and connected to mean block to get average power output. This average power output and average power input is connected to divide block such that output is connected to multiply port and input is connected to divide port and then it's output is connected to gain block (gain=100) and in display output can be seen as efficiency.

Result-

In our model I calculated power out and used mean block to get average power over few cycle and similarly for power in and efficiency is coming out to be = 96.11%.

Full Bridge Inverter

A single Phase Full Bridge inverter is a circuit that convert DC into AC. It is type of inverter that convert DC to AC using 4 MOSFET and 2 Capacitor. MOSFET (1,3) and (2,4) Turn ON and OFF one by one. This create waveform similar to AC, It is called Full Bridge because the output voltage is equal to input DC voltage. These are generally used where AC output is needed from DC Supply.

Simulation Process:-

For simulation of Single Phase Inverter I used MATLAB –Simulink.

Blocks Used -

1. DC voltage source
2. Resistor - 2No.
3. Capacitor – 2No.
4. Voltage Measurement Block – 2No.
5. MOSFET- 4No.
6. Pulse generator – 2No.
7. Scope- 2 No.
8. Goto- 2No.
9. From-2No.
10. Power GUI

In simulink for simulation of single phase full bridge Inverter I have used DC voltage source amplitude of 100V and connected resistor (1 ohm) in series with this DC source. Next I have used 2 capacitors (1e-6 Feach) which are in series and parallel to DC voltage source, Next I used a 10ohm resistor as load which is resistive load to look the AC output. For switching I have used MOSFET. After that I connected 2 pulse generator blocks to control MOSFET switches. I set the time period to 0.02sec, pulse width to 50%, Amplitude to 1, phase delay to 0Sec, 0.01Sec and connected them to MOSFET(1,3) and MOSFET (2,4) respectively. Now I used 2 voltage measurement block 1st connected parallel to load resistor and 2nd connected parallel to DC voltage source and their output is connected to scope to view the result. Finally I connected goto (A) block to pulse generator 1 and goto (B) block to pulse generator 2 and I have used from (A) and from (B) block to connect them to scope to view the output of Pulse.

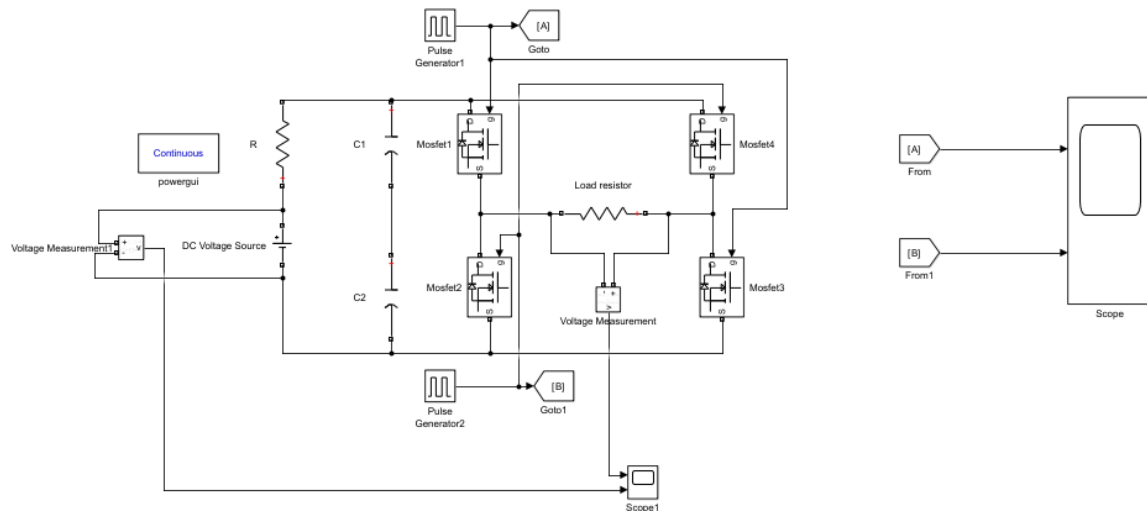


Fig : Single Phase fullwave inverter using MOSFET.

When MOSFET 1, MOSFET 3 are On current flow in one direction through load (positive half cycle)

When MOSFET 2, MOSFET 4 are On current flow in opposite direction through load (negative half cycle)

The alternating switch creates an AC waveform across the load from DC source.

Output Analysis:-

In simulation of single phase Full bridge inverter I created a model in simulink which uses MOSFET. I have simulated single phase Full bridge inverter at 0.1sec.

Scope 1 Result–

- 1st row - It produce square wave as output voltage.
- 2nd row shows the input voltage.
- Out put voltage is in form of square wave and its value is varying from 4.545×10^1 to -4.545×10^1 volt and input is constant throughout having value 1.000×10^2 volt. This shows we are getting somewhat alternating current.

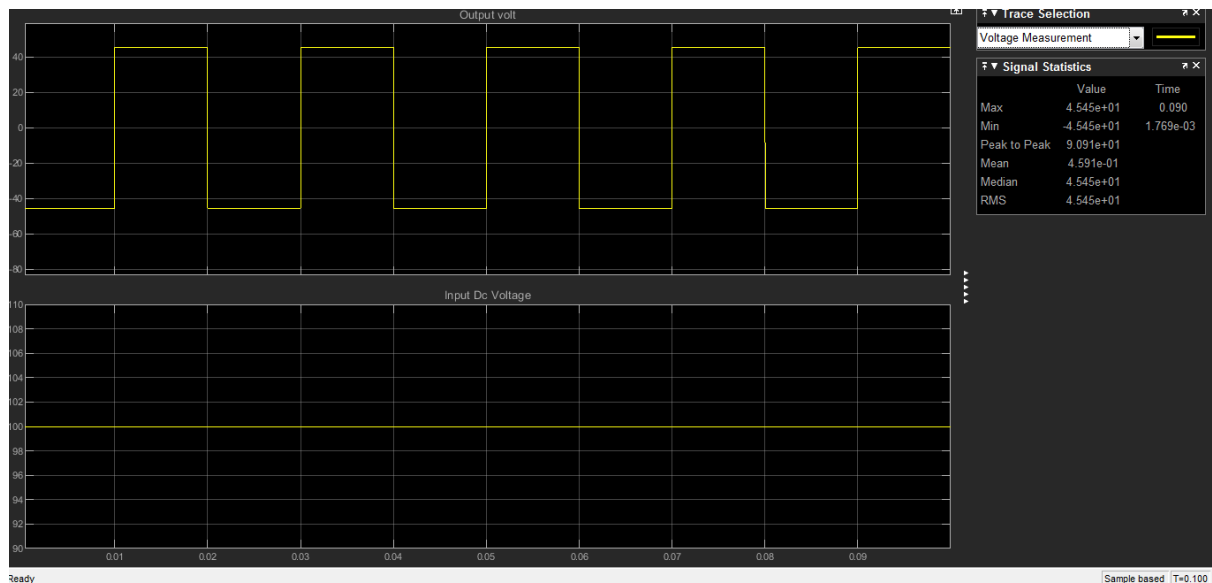


Fig : Scope view of scope1

Scope Result-

- Mosfet 1 and 3 are operated by same pulse generator and Mosfet 2 and 4 are operated by another same pulse generator and at time when 1,3 are On at that time 2,4 will be off and vice versa which help continuous flow of electricity.

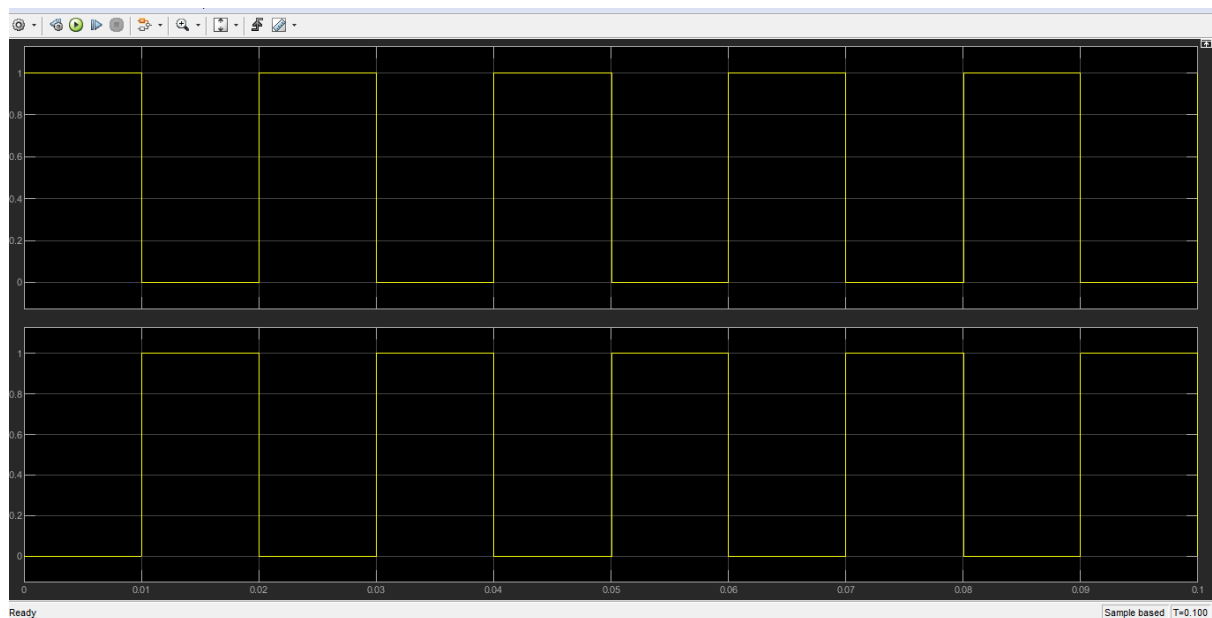


Fig : Scope View of scope.

Conclusion:

Full Bridge Inverter MOSFET produce square wave.

Efficiency Calculation:

For efficiency calculation I have to know power out and power in.

Block included apart from above

- Current measurement -2pc
- Product -2pc
- Mean – 2pc
- Divide – 1pc
- Gain -1pc
- Display – 1pc

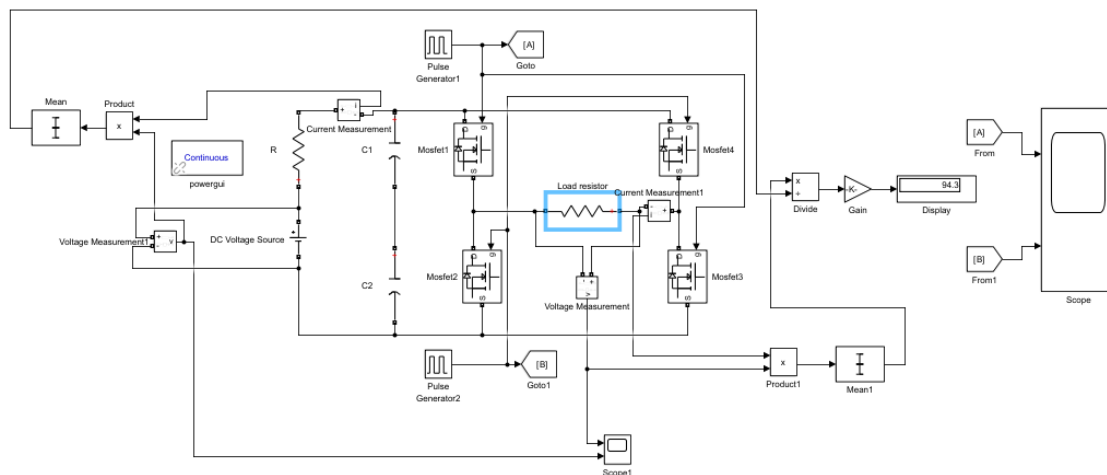


Fig : Single Phase fullwave inverter using MOSFET with efficiency.

I have connected current measurement device in series with resistor at DC voltage source and with load resistor. I have taken product of these voltage and current and connected them to mean block to get the average power input. Similarly I connected voltage measurement device parallel to load resistor and in series I connected current measurement device and product of these is taken and connected to mean block to get average power output. This average power output and average power input is connected to divide block such that output is connected to multiply port and input is connected to divide port and then it's output is connected to gain block (gain=100) and in display output can be seen as efficiency

Result-

In our model I calculated power out and used mean block to get average power over few cycle and similarly for power in and efficiency is coming out to be = 94.3%.

Three Phase Inverter

Three phase inverter is power electronics circuit that converts DC input to three phase AC output. It is widely used in EV.

It cover medium to high power applications. Main purpose of these topologies is to provide a 3 phase voltage source where amplitude, phase, frequency of voltage can be controlled.

Main advantage of 3 phase inverter are –

1. Direction of rotation of motor can be reversed by changing the output phase sequence of the inverter.
2. AC output voltage can be controlled by varying DC link Voltage.
3. Frequency of output voltage waveform depends on switching rate of switches and hence can be varied over wide range.

Type of three phase inverter

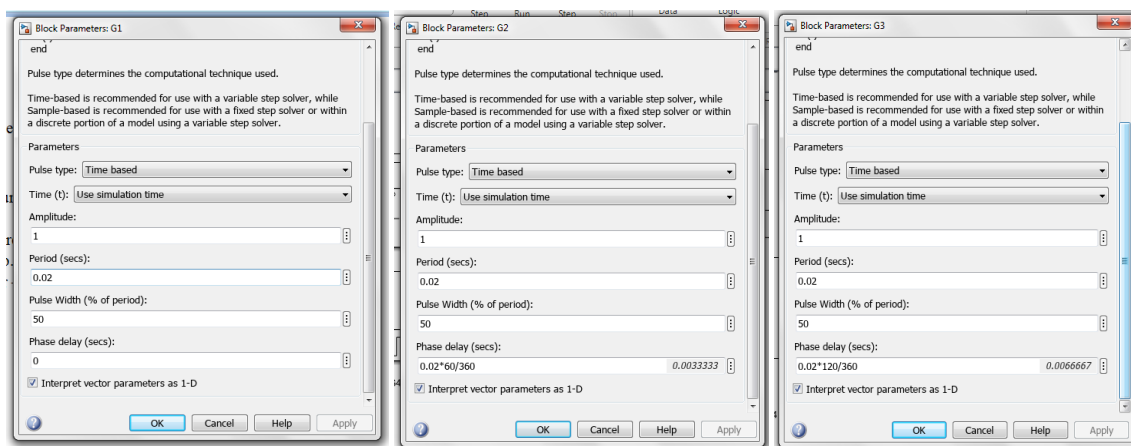
1. 120 degree
2. 180 degree

Simulation Process:-

For simulation of Three Phase Inverter I used MATLAB – Simulink.

Blocks Used -

1. DC voltage source
2. Resistor - 3No.
3. Voltage Measurement Block – 3No.
4. MOSFET- 6No.
5. Pulse generator – 6No.
6. Scope- 2 No.
7. Goto - 6No.
8. From-6No.
9. Power GUI



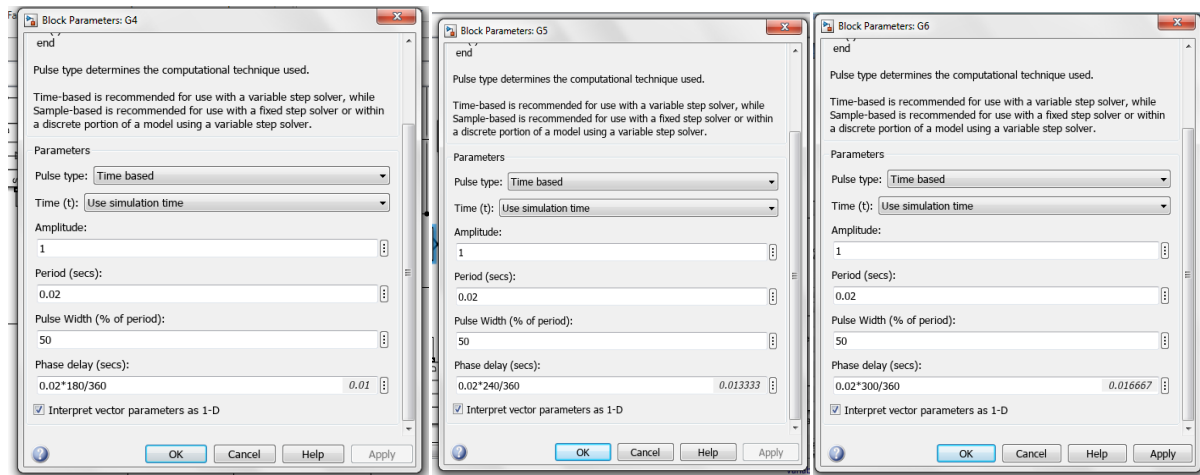


Fig :Pulse generated values for different MOSFET.

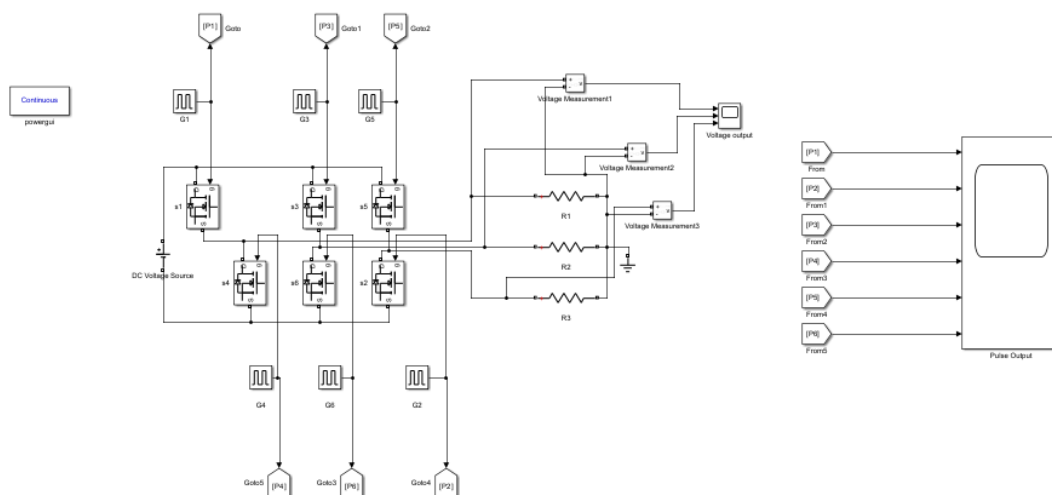


Fig : Three Phase full wave inverter using MOSFET.

In simulink for simulation of three phase full bridge Inverter I have used DC voltage source amplitude of 100V. Next I used a three 10 ohm resistor as load which is resistive load to look the AC output. For switching I have used 6 MOSFET. After that I connected 6 pulse generator blocks to control MOSFET switching. I set the parameters of pulse generator as figure just above figure of three phase full wave inverter. Now I connected 3 voltage measurement block parallel to each resistive load and their output are connected to scope named voltage output. Finally I connected goto (P1), goto (P2), goto (P3), goto (P4), goto (P5), goto (P6) block to pulse generator G1, G2, G3, G4, G5, G6 respectively, Then I

use from block that is from(P1), from(P2), from(P3), from(P4), from(P5), from(P6) and connected them to scope to view the output of Pulses.

Output Analysis:-

In simulation of three phase Full bridge inverter I created a model in simulink which uses MOSFET. I have simulated three phase Full bridge inverter at 0.1sec.

Scope Result–

- 1st, 2nd, 3rd, 4th, 5th, 6th Row - It shows the output given to gate channel of different MOSFET generated by different pulse generator.

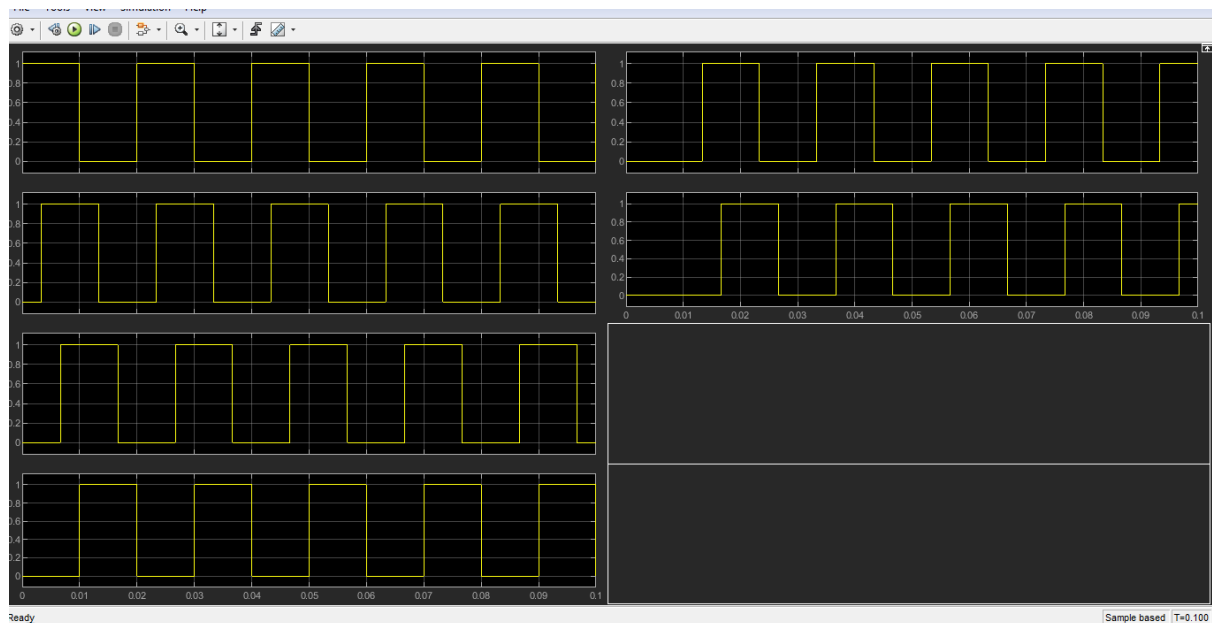


Fig : Pulse generated by different pulse generator.

- In the output of 3 phase inverter we are getting the following result at star connection
Max voltage = 3.366×10^1 volt Min = 3.234×10^1 Volt.
- 1st row graph shows the voltage for phase 1 of inverter. The yellow line goes up, then flat then down in step this square like waveform.
- 2nd row graph shows the voltage for phase 2 of inverter. It looks 1st row graph has shifted to right. It shows 120 degree phase shift which is perfect for 3 phase system.
- 3rd row graph shows the voltage for phase 3 of inverter. It is shifted even more which shows another 120 degree phase shift.

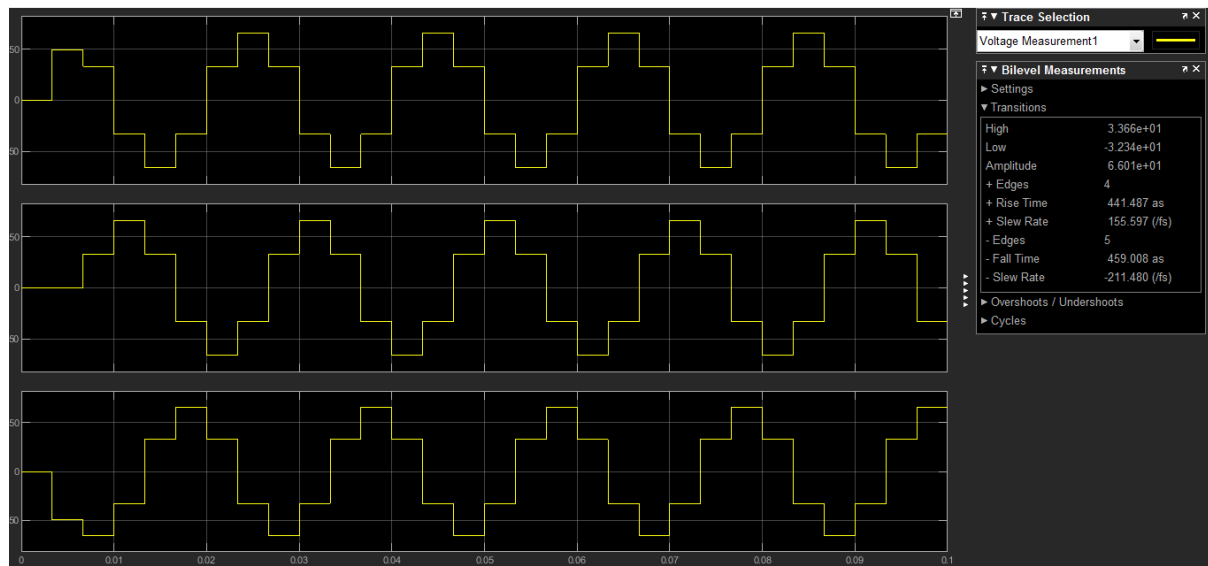


Fig : Pulse generated by different pulse generator.

Conclusion:

All 3 voltage V1, V2, V3 are equally spaced this means your inverter is giving balanced 3 phase output

Efficiency Calculation:

For efficiency calculation I have to know power out and power in.

Block included apart from above

- Voltage measurement -1pc
- Current measurement -4pc
- Product -4pc
- Mean – 2pc
- Divide – 1pc
- Gain -1pc
- Display – 1pc
- Add -1pc

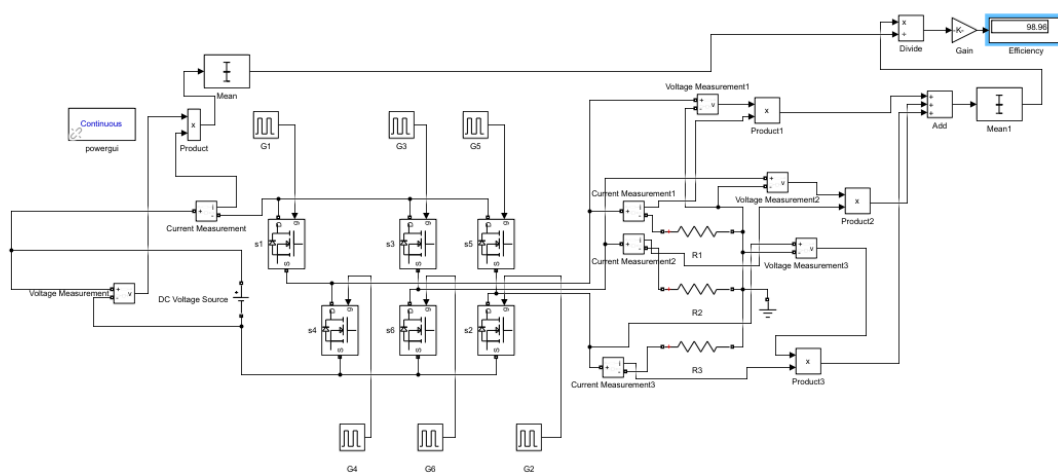


Fig : Three Phase full wave inverter using MOSFET with efficiency calculation.

I have connected voltage measurement device parallel to input voltage source and in series I have connected current measurement device, I have taken product of these voltage and current and connected them to mean block to get the average power input. Similarly I connected voltage measurement device parallel to all 3 load resistor and in series I connected current measurement device and product of these are taken like $v_1 \times i_1$, $v_2 \times i_2$ and $v_3 \times i_3$ and all are added using add block and then connected to mean block to get average power output. This average power output and average power input is connected to divide block such that output is connected to multiply port and input is connected to divide port and then it's output is connected to gain block (gain=100) and in display output can be seen as efficiency.

Result-

In our model I calculated power out and used mean block to get average power over few cycle and similarly for power in and efficiency is coming out to be = 98.96%.

PROJECT DESCRIPTION

Design a Bridge rectifier with filter and simulate it.

Project Objective :-

The main objective of project is to design Bridge Rectifier Circuit using Proteus and observe its DC output with different filter values. The project aims to achieve how filter affect the output of voltage behaviour.

Important Pre-requisite :-

Before starting the project it is crucial to have a good understanding of basic electronics concepts, circuit design principle and components. Also, knowledge of how to use Proteus simulation software is essential, as it will be platform for designing, simulating and analyzing the output of rectifier when using different filter stages.

Require Components

<u>Component List</u>			
<u>Sl No.</u>	<u>Component Name</u>	<u>Function</u>	<u>Library name</u>
1.	VSINE	AC Voltage Source	-
2.	AC Voltmeter	For Measurement	-
3.	Transformer	Voltage Conversion	2P2S
4.	Bridge Rectifier	Convert AC to DC	2W01G
5.	Capacitor	Filter	CAP
6.	DC Voltmeter	For Measurement	-
7.	Resistor	Load Resistor	-
8.	Oscilloscope	Measurement graph	-

REQUIRED INPUT PARAMETERS AND CALCULATIONS :-

Input Parameter :-

In the context of Rectifier circuit project using Proteus, the input parameter generally refer to the characteristics of the AC input voltage waveform that the rectifier circuit will be designed to handle. Here are the key input parameters.

1.Source Voltage :-

In the Proteus software as a source voltage we will take the value of the peak voltage.

$$\text{Peak voltage} = \text{Nominal voltage} \times \text{sq root}(2)$$

Peak voltage is the maximum instantaneous voltage reached by AC cycle. Nominal voltage is average voltage of AC supply.

$$\begin{aligned}
 \text{Peak voltage} &= \text{Nominal voltage} \times \text{sq root}(2) \\
 &= 70.71\text{V} \times \text{sq root}(2) \\
 &= 100\text{V}
 \end{aligned}$$

$$\text{Frequency} = 50\text{Hz}$$

Here I used VSINE component in Proteus to represent input voltage. I set amplitude to 100V and frequency to 50Hz.

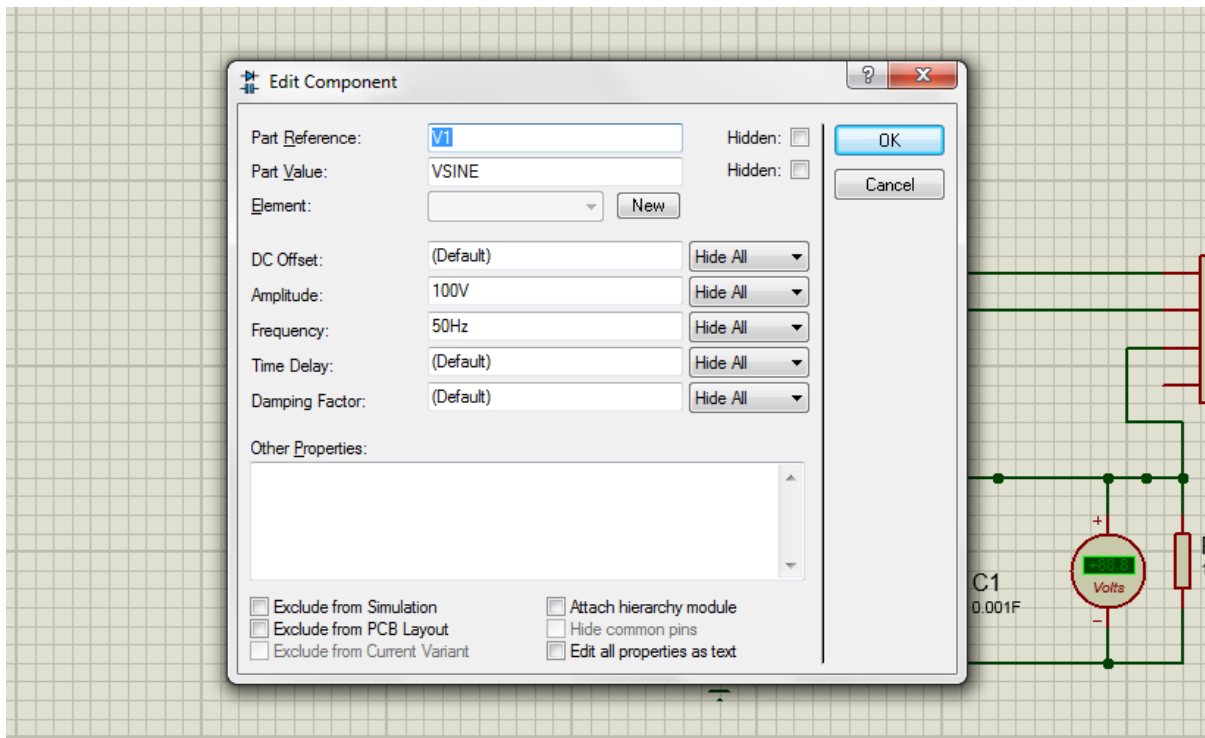


Fig . Input Sinewave Parameter

2.AC voltmeter :-

I used an AC Voltmeter to measure the output voltage of VSINE source. To get the accurate reading I connected voltmeter in parallel with VSINE source. This allows voltmeter directly to read voltage of source without affecting the circuit functionality. Voltmeter are always connected in parallel to the source as it has high internal resistance.

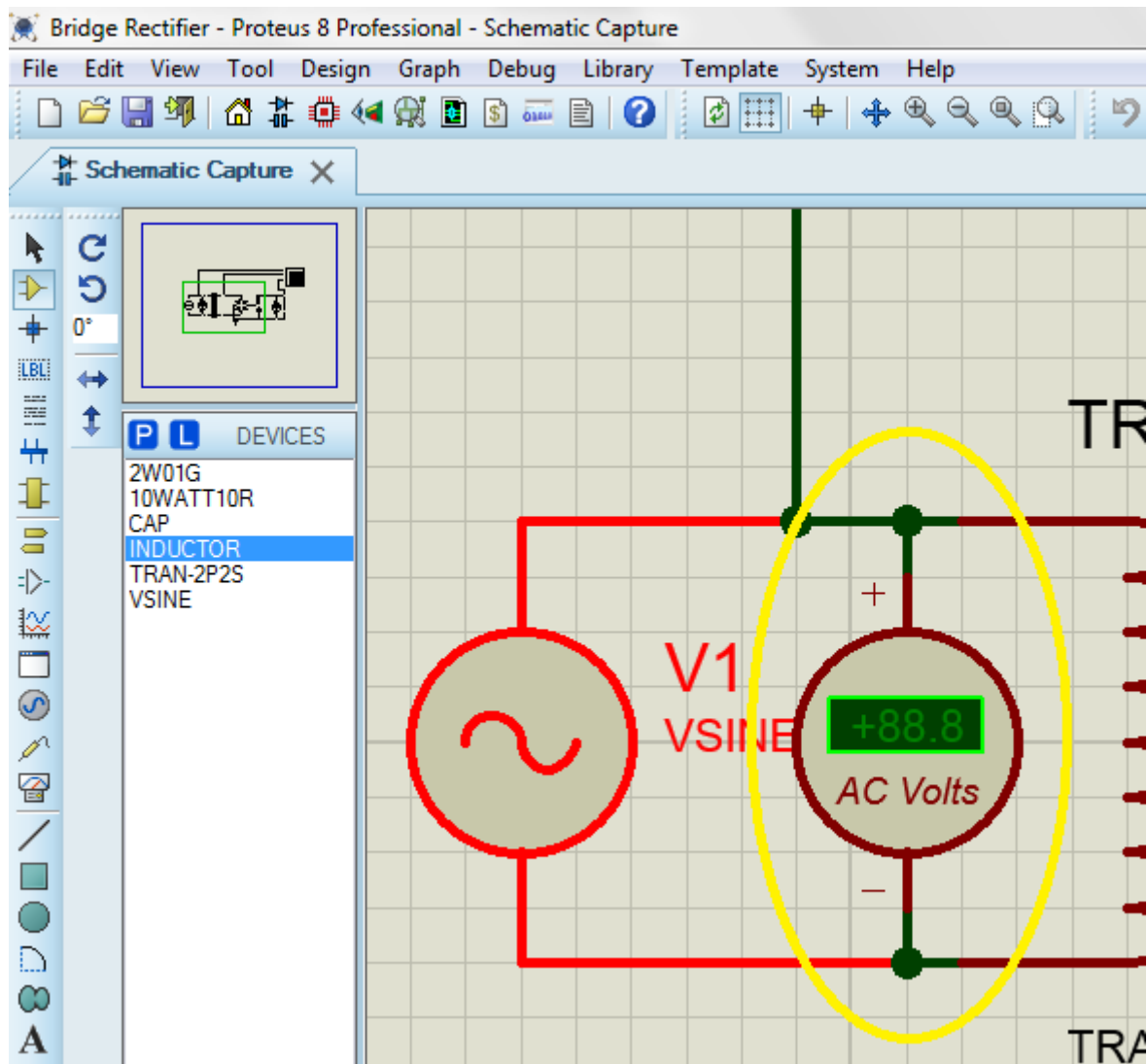


Fig . Voltmeter

3. Transformer :-

Here we have used step down transformer to step-down the sine voltage to required level. It step down transformer incoming voltage from the supply mains and isolates the output circuit from line supply.

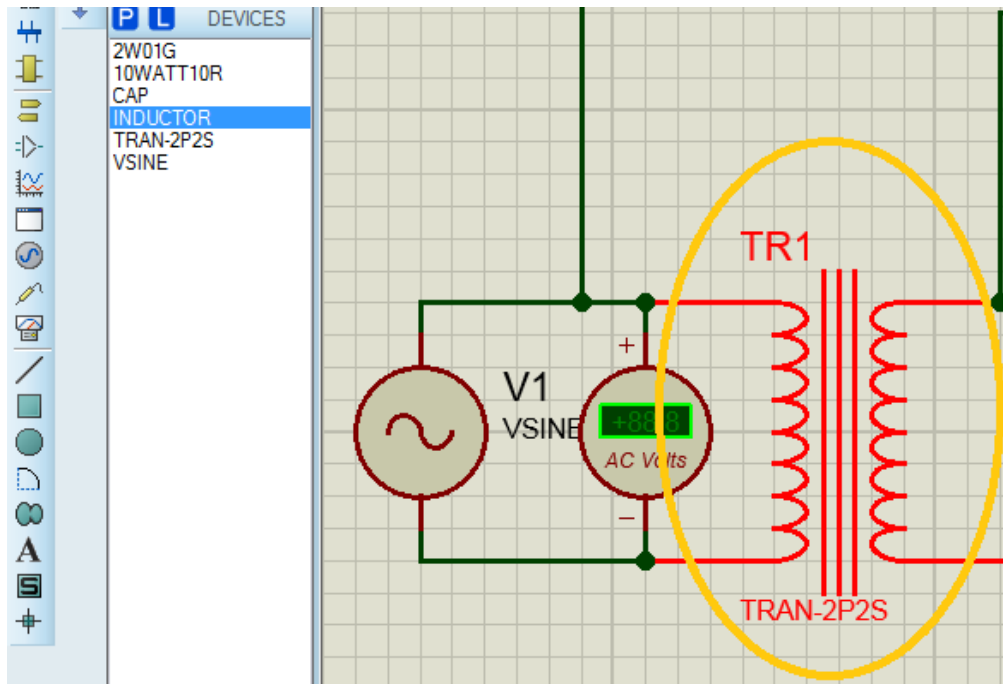


Fig . Transformer

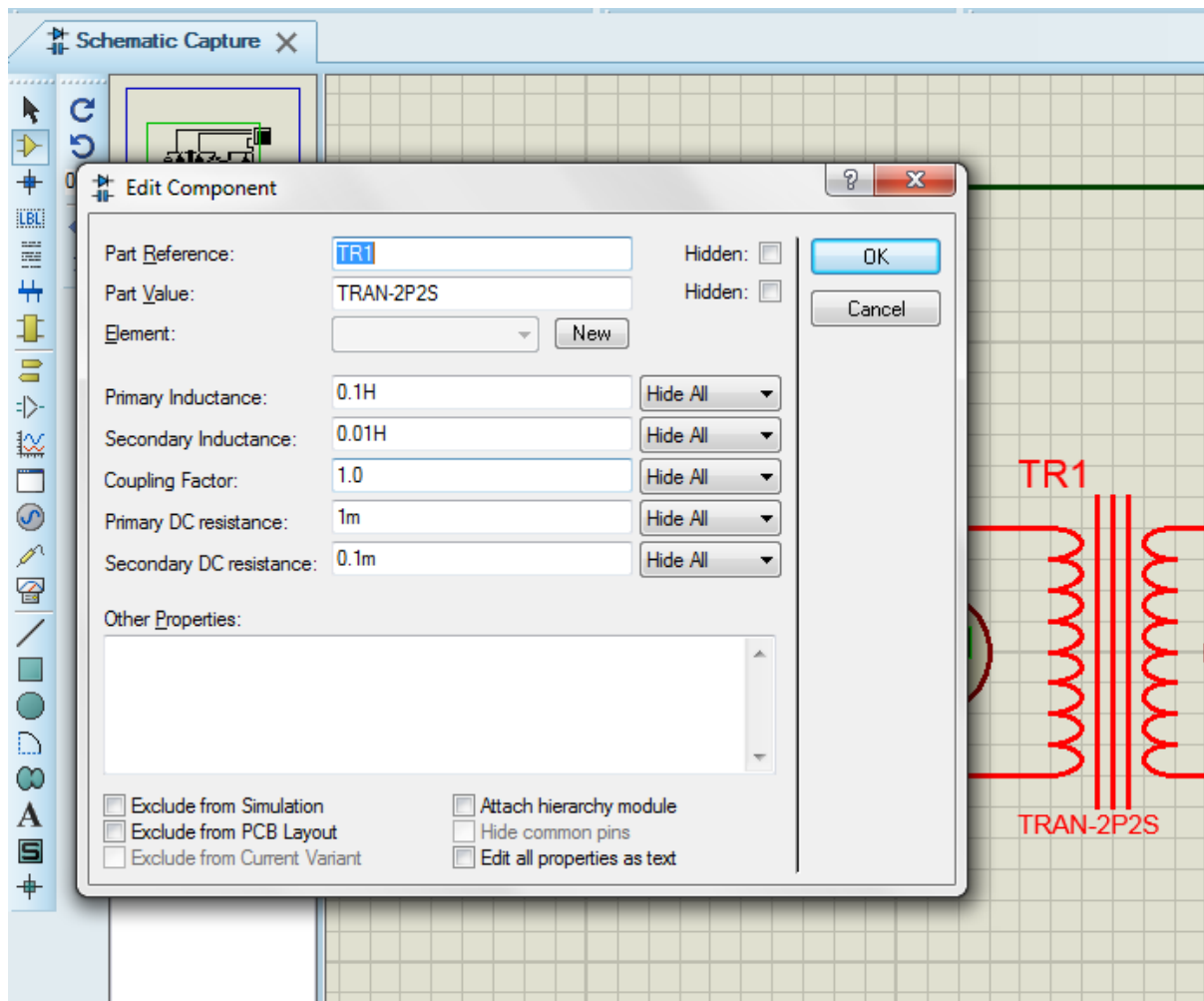


Fig . Input parameter for transormer

4. RECTIFIER

Rectifier are electronic Device which help to convert AC to DC . Basically there are two type of rectifier –

5. Half wave rectifier
6. Full Wave rectifier- Bridge rectifier , Center tapped rectifier

Here I have used full wave bridge rectifier which is used to convert AC to DC. It has total 4 terminals. 2 terminals are connected to voltage source and remaining 2 is connected to load Resistor.

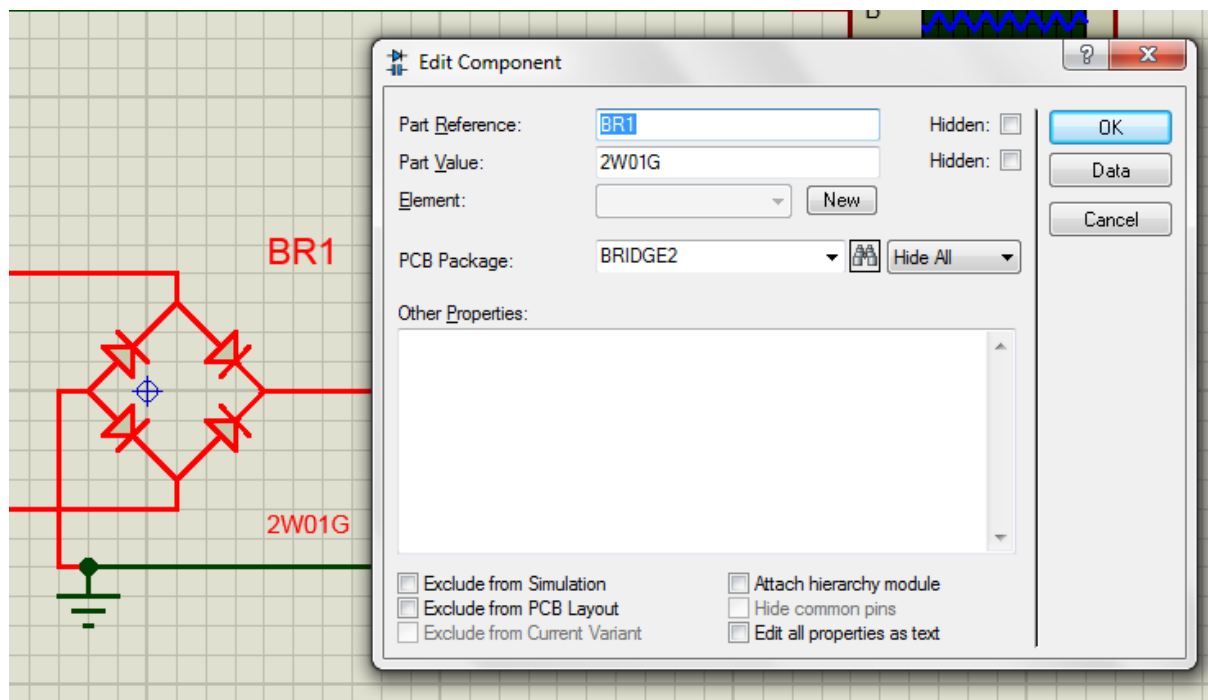


Fig Full Wave Bridge Rectifier

5. Filter

Here I have used capacitor in parallel with the load to get Pure DC.

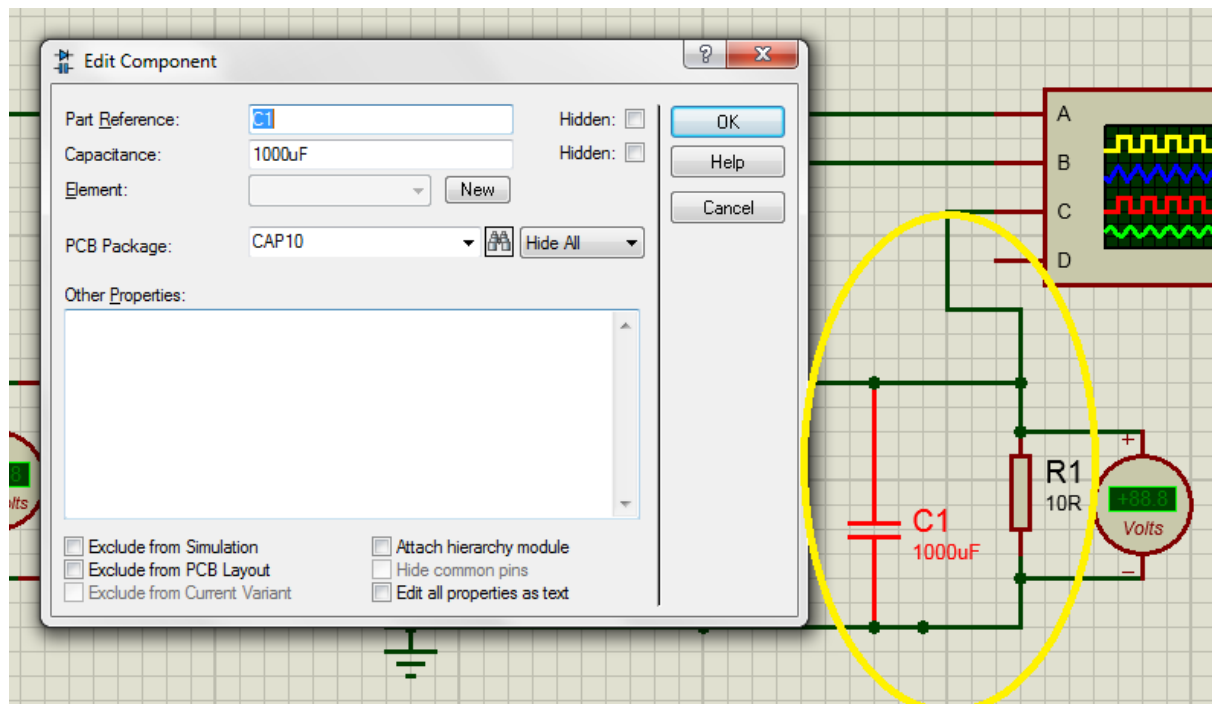


Fig . Shunt capacitor Filter

6. DC Voltmeter

Here I used DC voltmeter to measure the output of DC voltage across load resistor, I connected DC Voltmeter in parallel to load resistor. This allows voltmeter to measure the voltage of circuit without affecting the circuit as voltmeter has high internal resistance.

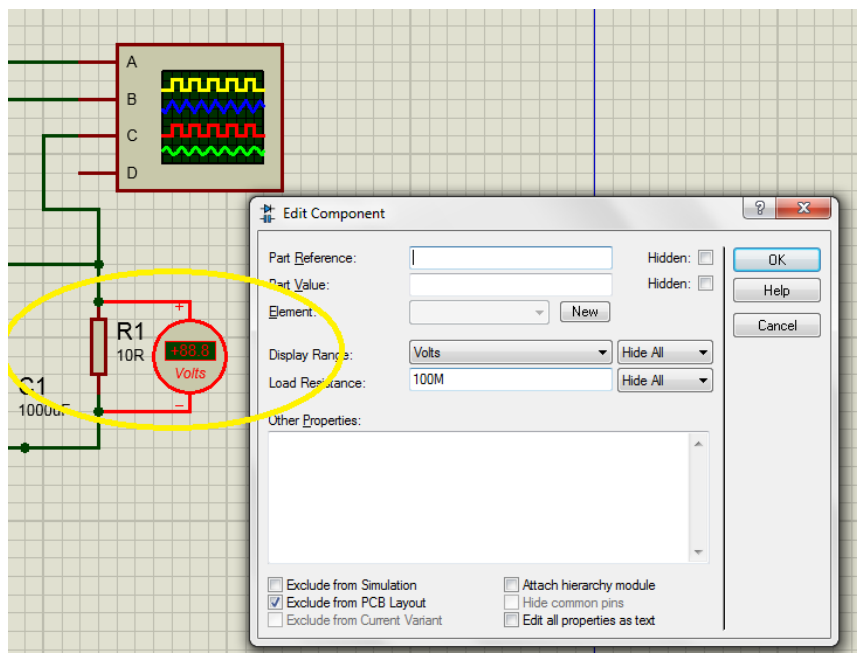


Fig. DC Voltmeter

6. Oscilloscope

An Oscilloscope is used to visualize and analyses voltage signals by displaying their waveforms over time. It helps in measuring AC and DC signals, Frequency, amplitude.

I have used oscilloscope in my circuit with following connections:-

- **A Terminal** - Connected to the primary winding of input AC Signal.
- **B Terminal** - Connected to the secondary winding of transformer.
- **C Terminal** – Connected to load resistor.
- **D Terminal** - Not connected.

Circuit Diagram of Shunt Capacitor filter :-

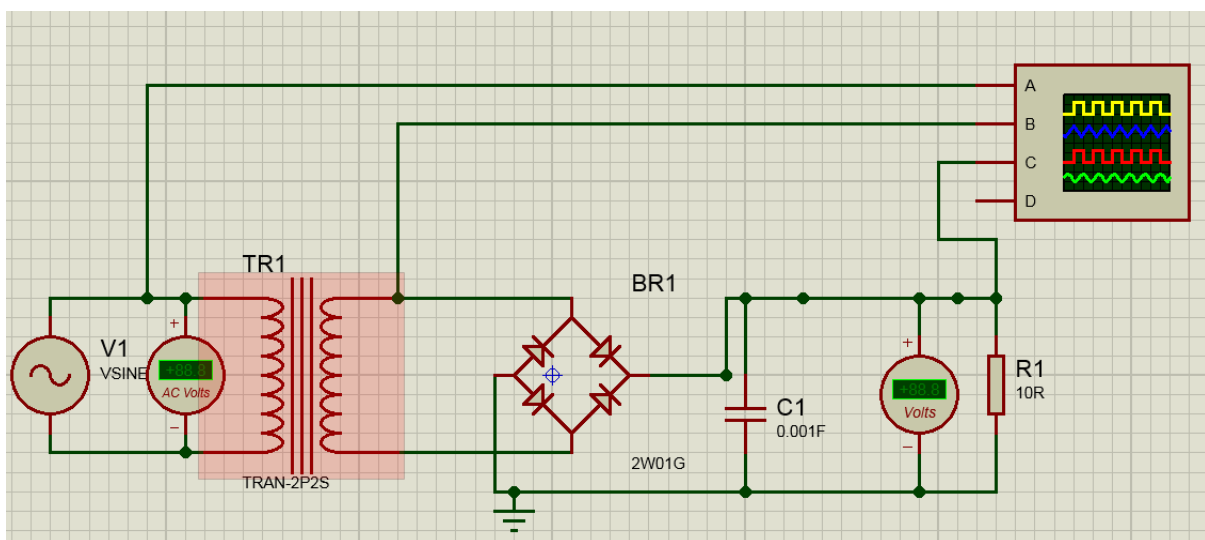


Fig. Circuit Diagram of Shunt Capacitor filter (0.001f)

Reading of Oscilloscope:

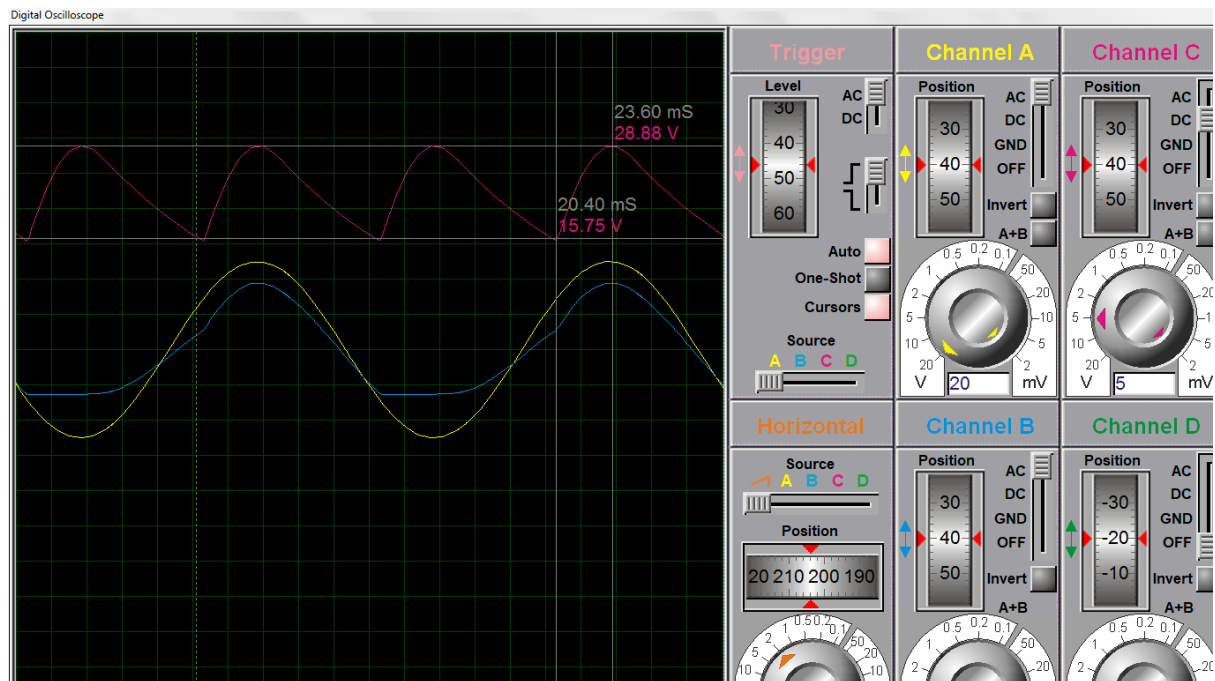


Fig. Oscilloscope reading of Shunt Capacitor filter (0.001F)

I observed the oscilloscope readings carefully. The yellow line represents the AC input signal, Blue line represents the step-down AC signal and red line represents the output DC signal after getting filtered. The DC output has ripples show it is not completely filtered the AC signal. When we have used capacitor of low capacitance (0.001f) – In oscilloscope we can see that the output DC has ripples showing impure form of DC having AC component also with max value of 28.88V & min value of 15.75V. This shows that appropriate rating of filter are very necessary to remove ripples from DC.

Circuit Diagram of Shunt Capacitor filter :-

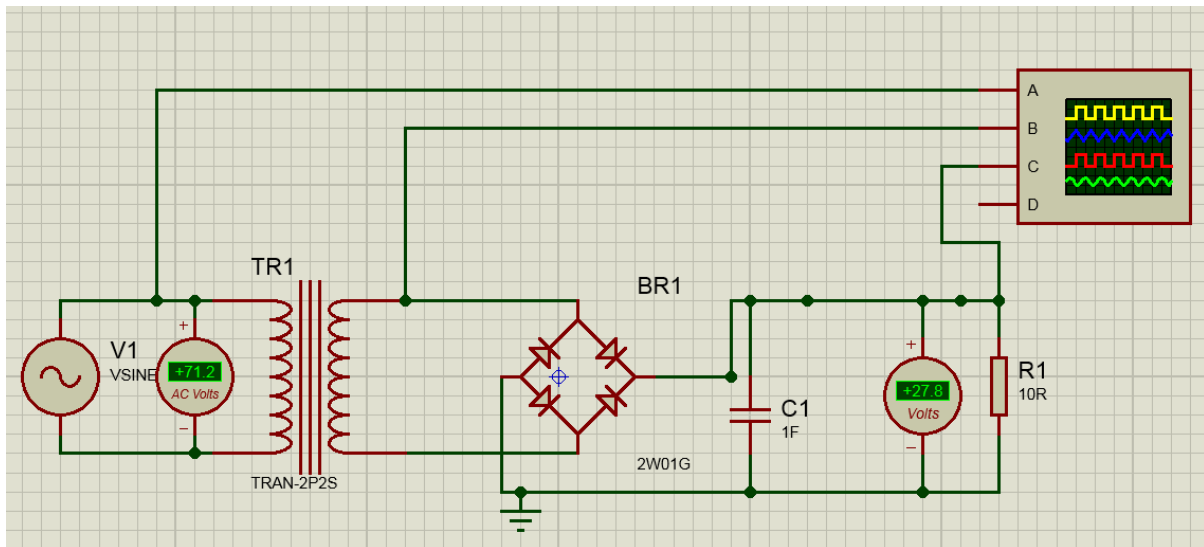


Fig. Circuit Diagram of Shunt Capacitor filter (1f)

Reading of Oscilloscope:

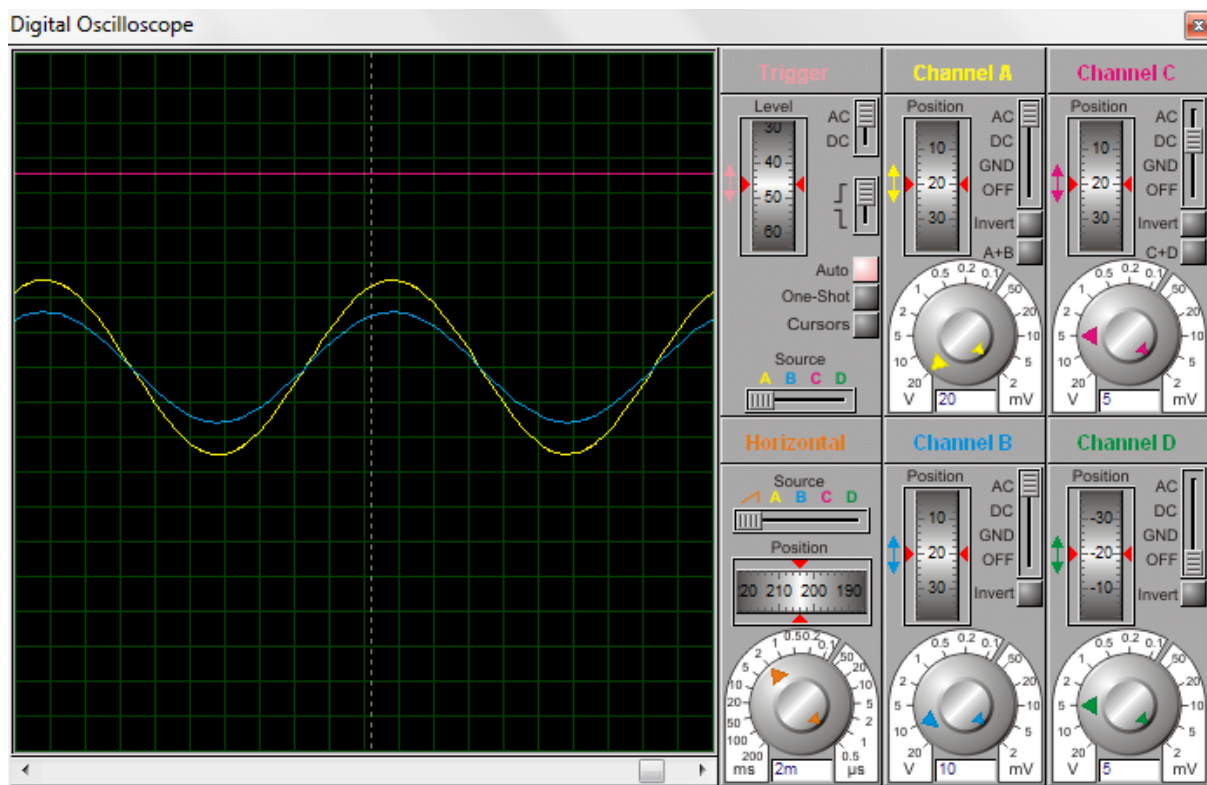


Fig. Oscilloscope Reading of Shunt Capacitor filter (1f)

I observed the oscilloscope readings carefully. The yellow line represents the AC input signal, Blue line represents the step-down Ac signal and red line represents the output DC

signal after getting filtered. The DC output is perfect straight line showing it's completely filtered the AC signal.

Result & Conclusion:-

After carefully watching the output of oscilloscope, I can see that the output of shunt capacitor filter having 1farad value is only having pure form of DC and shunt capacitor filters having 0.001f was also somewhat able to convert the AC to DC but there were little ripples left behind, So In order to get pure DC we should use appropriate value of filters so our purpose get solved.

PROJECT DESCRIPTION

Design a PWM controlled MOSFET based converter with filter and simulate it with and without load integration.

Project Objective :-

The main objective of project is to design PWM controlled MOSFET based Converter using Proteus and observe its DC output behaviour with LED load Integration. The project aims to achieve how load integration affects the output of voltage behaviour.

Important Pre-requisite :-

Before starting the project it is crucial to have a good understanding of basic electronics concepts, circuit design principle and components. Also, knowledge of how to use Proteus simulation software is essential, as it will be platform for designing, simulating and analyzing the output of rectifier when using different filter stages.

Require Components :-

<u>Component List</u>			
<u>Sl No.</u>	<u>Component Name</u>	<u>Function</u>	<u>Library name</u>
1.	Battery 24V	DC Voltage Source	CELL
2.	Capacitor	Store Energy	CAP
3.	NMOSFET	Switching	1RF740
4.	Diode	Allow current in one direction	UF4001
5.	Inductor 330uH	Store Energy	IND - AIR
6.	Resistor	Load Resistor	MINRES 1K
7.	Oscilloscope	Measurement graph	-
8.	DC voltmeter	Read Output Voltage	-
9.	LED	Load	LED AQUA
10.	Pulse Generator	Generate Pulse for switching	Q1(G)

Input Parameter :-

MOSFET

The main function of MOSFET is switching based on PWM pulses, it will make the MOSFET turn On and Off as per input of PWM Pulses it will make it to act as switch.

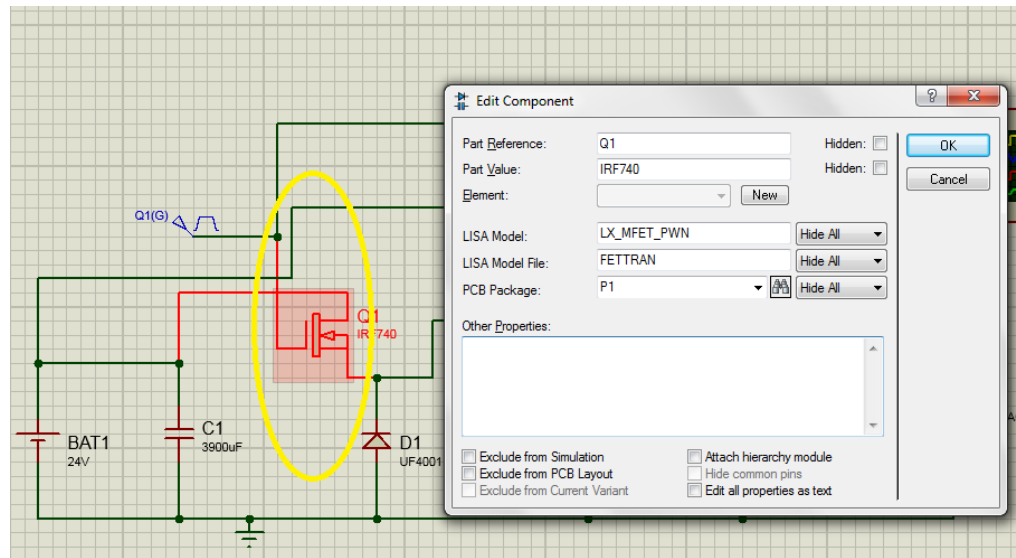


Fig. MOSFET input parameter

Inductor

Inductor is energy storing element. It store energy in a magnetic field when electric current flow through it thereby opposing changes in the current.

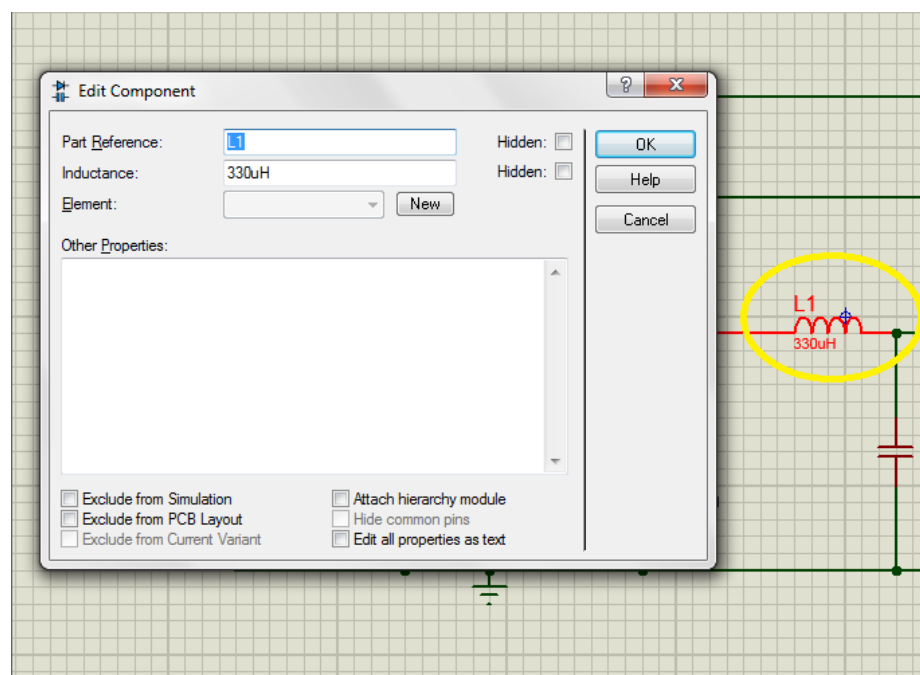


Fig. Inductor input parameter

Pulse Generator

It helps in switching MOSFET ON and OFF based on parameter feeded.

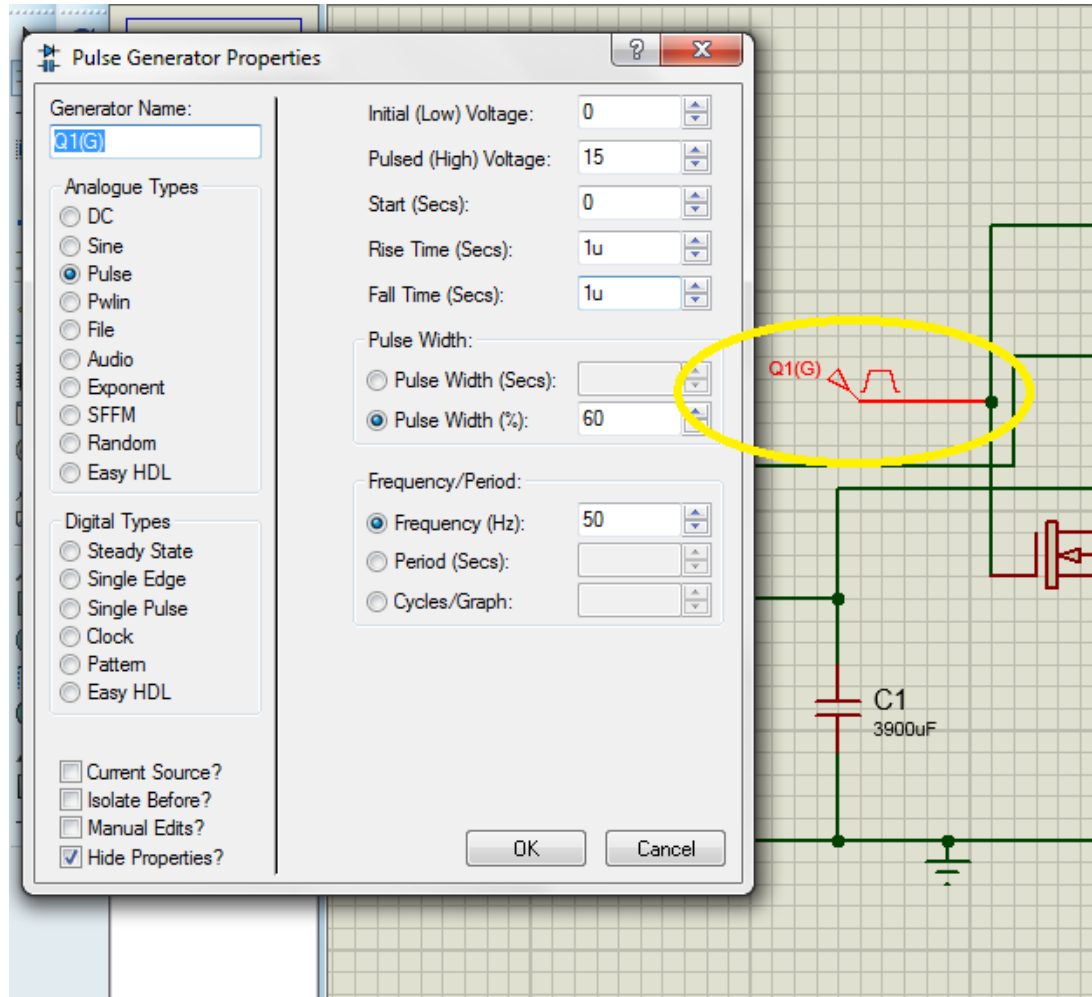


Fig.Pulse generator input parameter

Oscilloscope

An Oscilloscope is used to visualize and analyses voltage signals by displaying their waveforms over time. It helps in measuring AC and DC signals, Frequency, amplitude.

I have used oscilloscope in my circuit with following connections:-

- **A Terminal** - Connected to the Pulse generator.
- **B Terminal** - Connected to the load resistor.
- **C Terminal** – Connected to LED.
- **D Terminal** – Connected to Battery terminal.

Circuit Diagram of PWM controlled MOSFET based Converter :

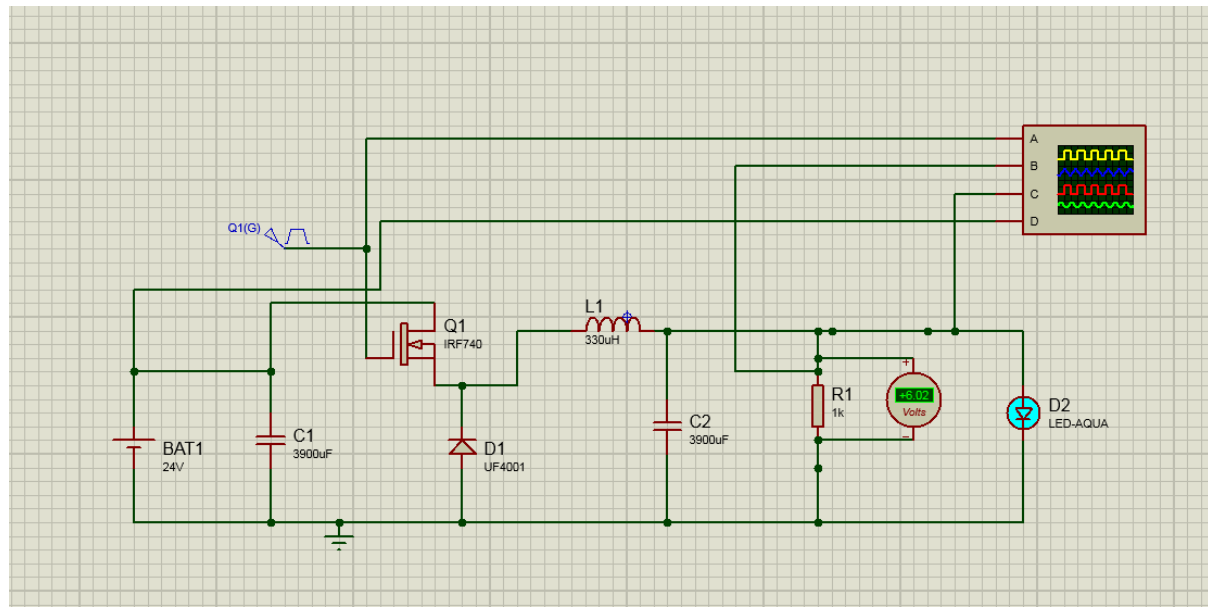


Fig.: Circuit diagram of PWM controlled MOSFET based Converter.

Reading of Oscilloscope:

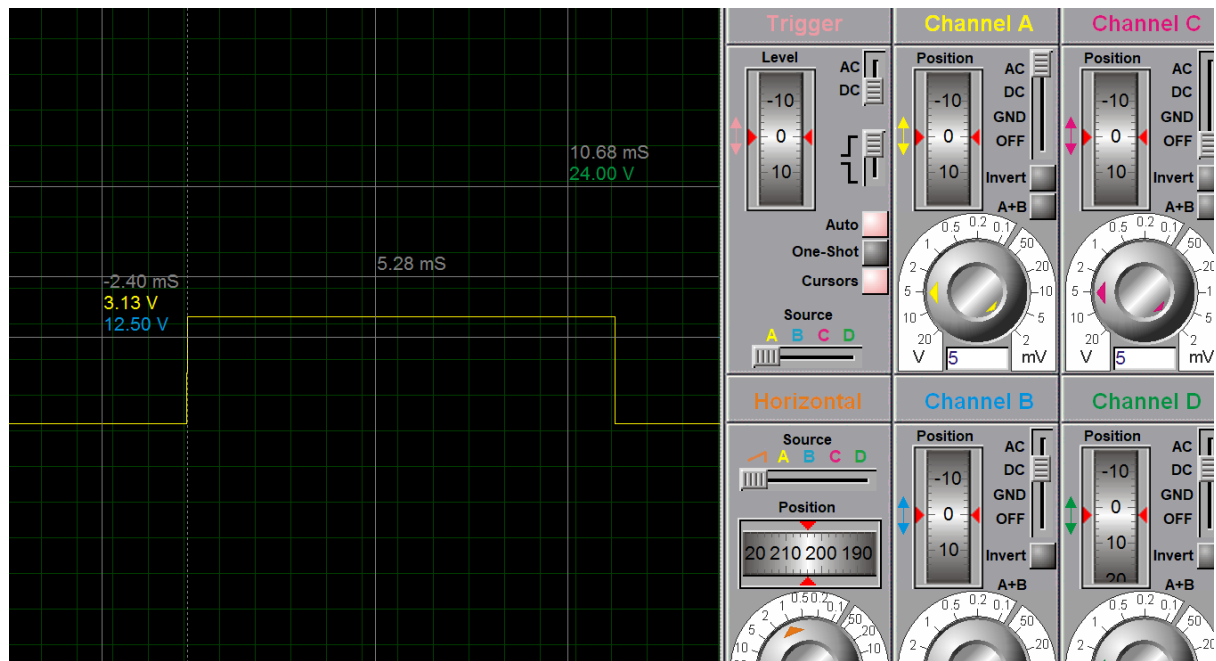


Fig . Oscilloscope Reading

I observed the oscilloscope reading carefully. I use here cursors to measure the values. The transparent horizontal line above which 24.00V in green is written (was green earlier after keeping the cursor it becomes white) and it is input voltage.

I observed that voltage is stepped down to 12.5 V (horizontal white line above which 12.5V in blue is written)

Reading of Oscilloscope after integration of LED:

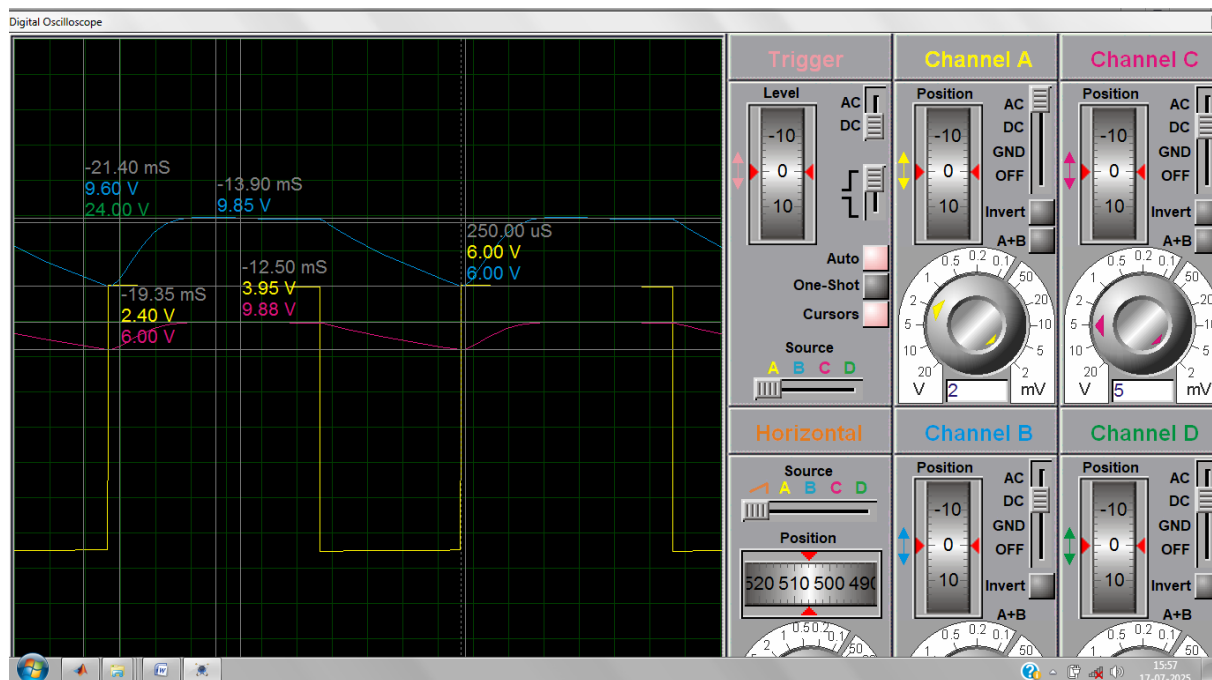


Fig . Oscilloscope Reading

I observed the oscilloscope reading carefully. I use here cursors to measure the values. After LED load integration I can see that input is still constant that is 24V (which is viewed in green) but output Dc voltage is varying from 9.85V to 6 V (which is viewed in blue).

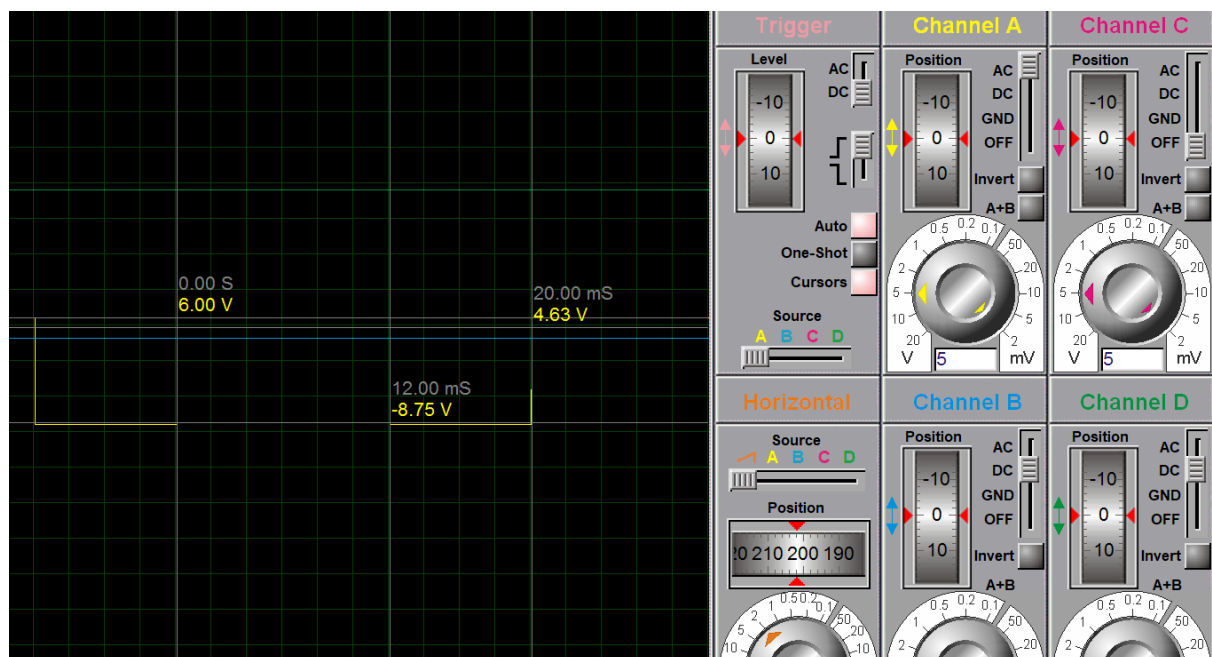


Fig . Oscilloscope Reading

After carefully observing the oscilloscope Switching Performance can be seen. I use cursor to check the timing of switching For period of 12 mS the switch is ON and for period of (20-12ms=)8mS the switch is off.

So, $\text{Period}(T) = T_{\text{on}} + T_{\text{off}} = 12 + 8 = 20\text{mS}$

And Duty ratio $= (T_{\text{on}}/T) = 12/20 = 0.6$

Result and Conclusion –

After carefully watching the output of oscilloscope, I can see that the output after and before LED integration varies before integration output was constant to 12.5V and after integration it is varying from 9.85V to 6 V which shows power is drawn to glow the LED.

PROJECT DESCRIPTION

PCB Design of Buck Converter in KiCAD.

Project Objective :-

The main objective of project is to design PCB of Buck Converter in KiCAD. The project aims to achieve PCB design.

Important Pre-requisite :-

Before starting the project it is crucial to have a good understanding of basic electronics concepts, circuit design principle and components. Also, knowledge of how to use KiCAD for PCB Design.

Require Components :-

Component List			
Sl No.	Component Name	Qty	Value
1.	Screw Terminal 01x02	2	-
2.	Capacitor	2	220uf, 470uf
3.	Resistor	1	10k
4.	MOSFET	1	IRLZ44N
5.	Diode	1	-
6.	Inductor	1	33uH
7.	Ground	1	-

Procedure -

Firstly we will select the component and then we will assign the footprint, then we connect the components, Assign the value to the component. As below-

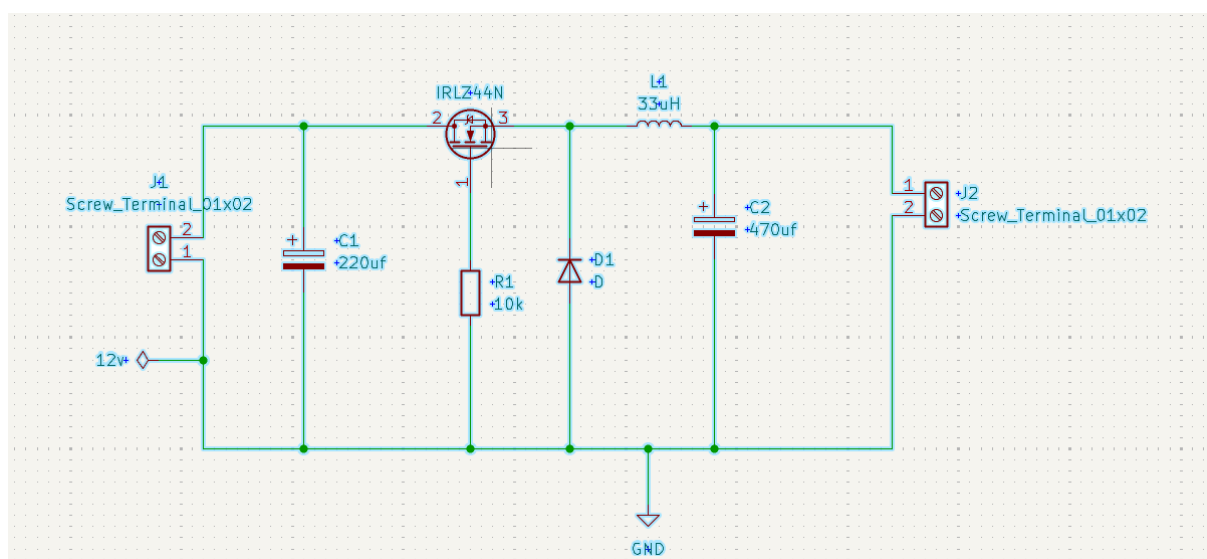


Fig . Schematic diagram

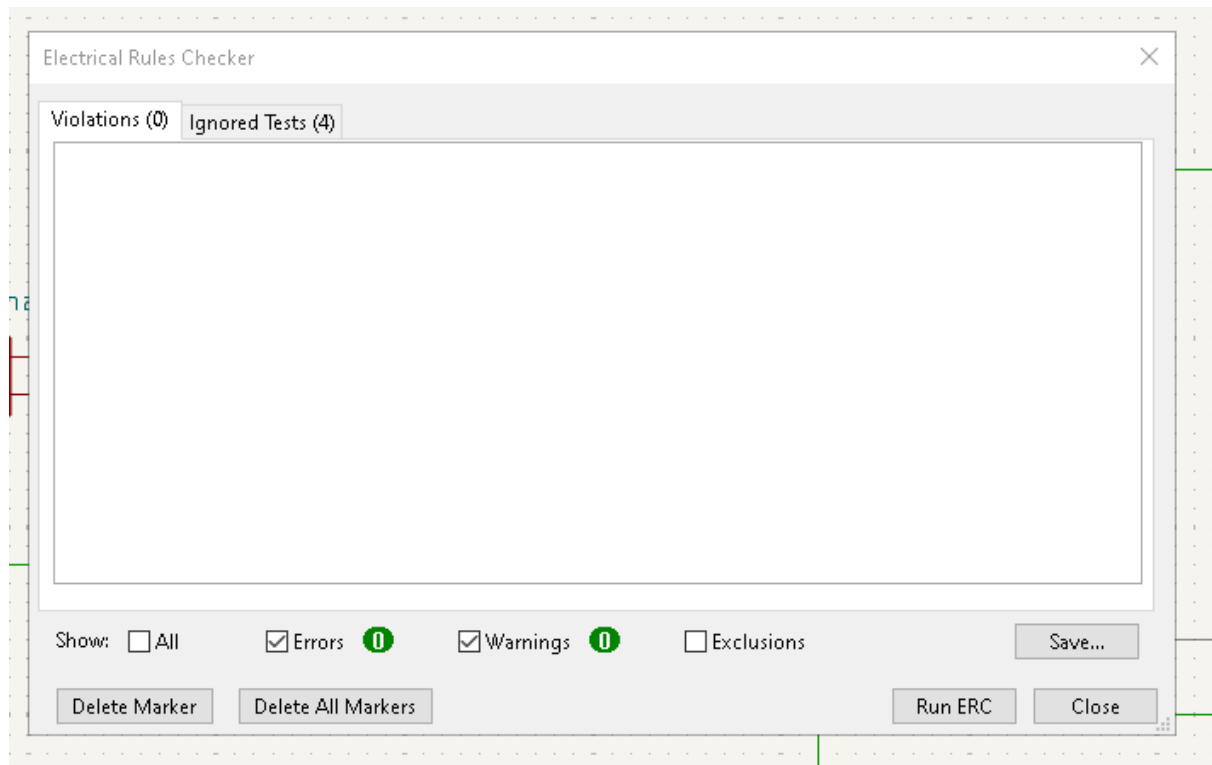


Fig . ERC showing 0 Error & 0 Warning

I checked the connection is proper or not by Electrical rule checker, If there is any error we can see that after running ERC.

After this in tools I have selected option Update PCB from Schematic. After this I had arrange the components so that they can be connected easily. Now I had given track to them as below.

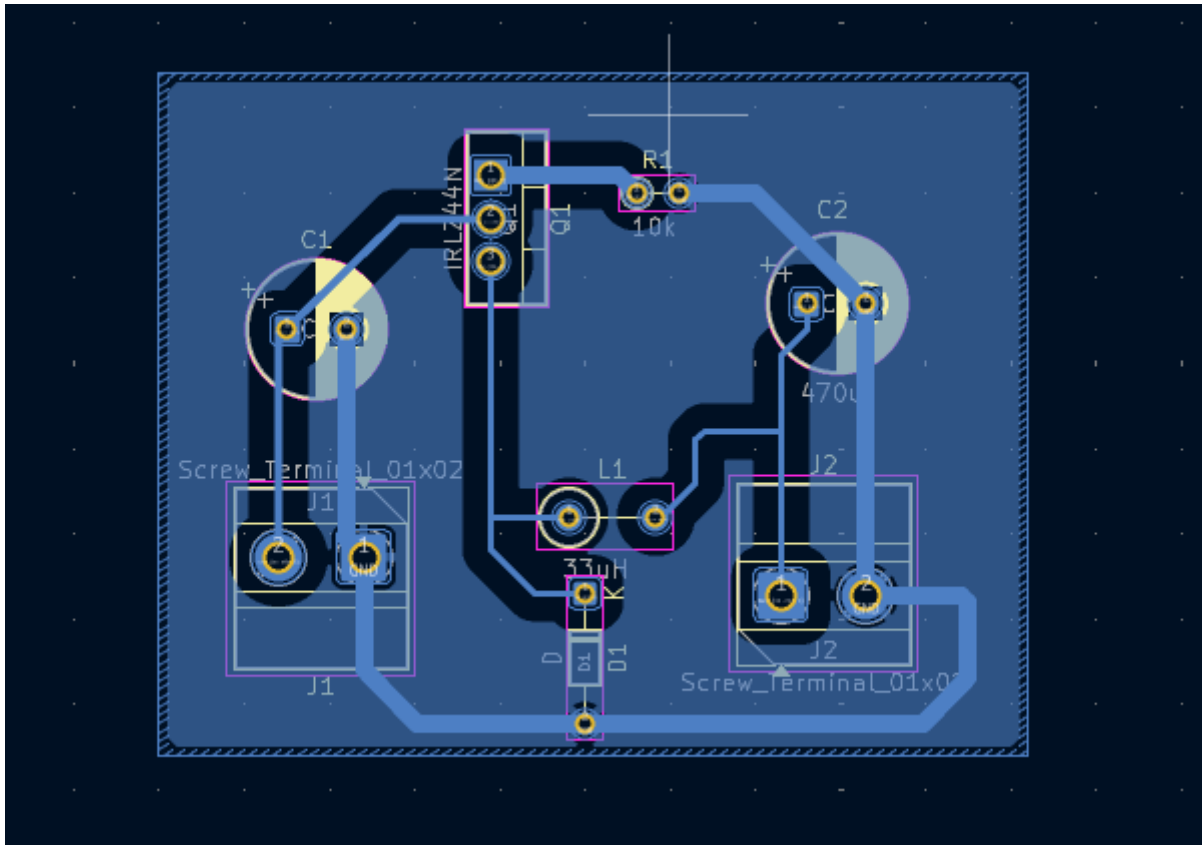


Fig . Circuit diagram after giving proper track width

Here I had given track width, ground plane, thermal vias. After providing proper inputs. I had run Design rule Checker to check error in PCB.

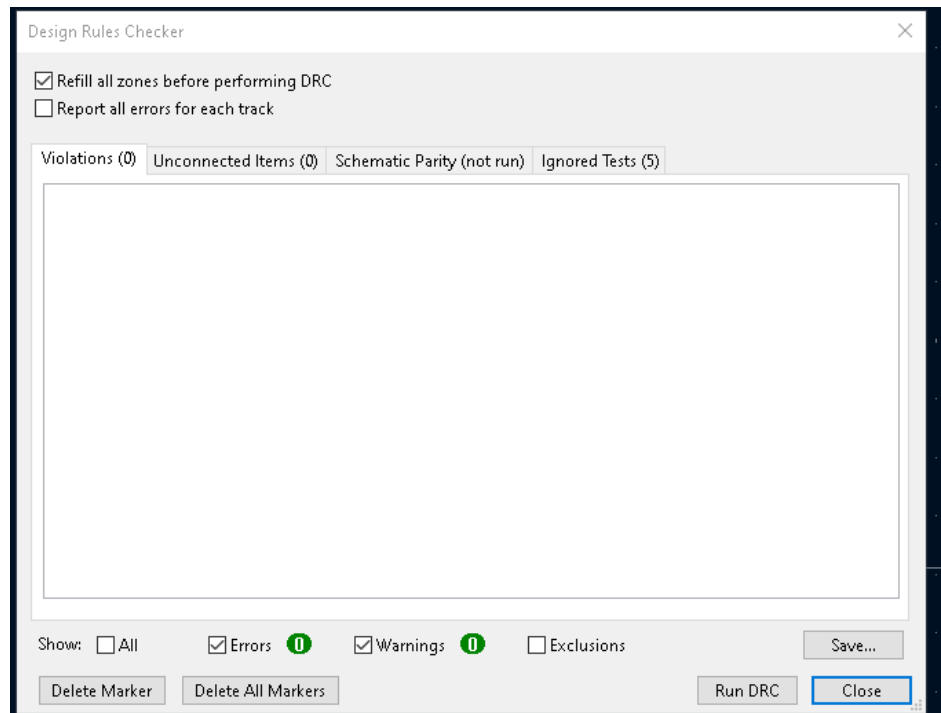


Fig . DRC showing 0 Error & Warning.

In view I selected 3D to view the PCB in 3D.

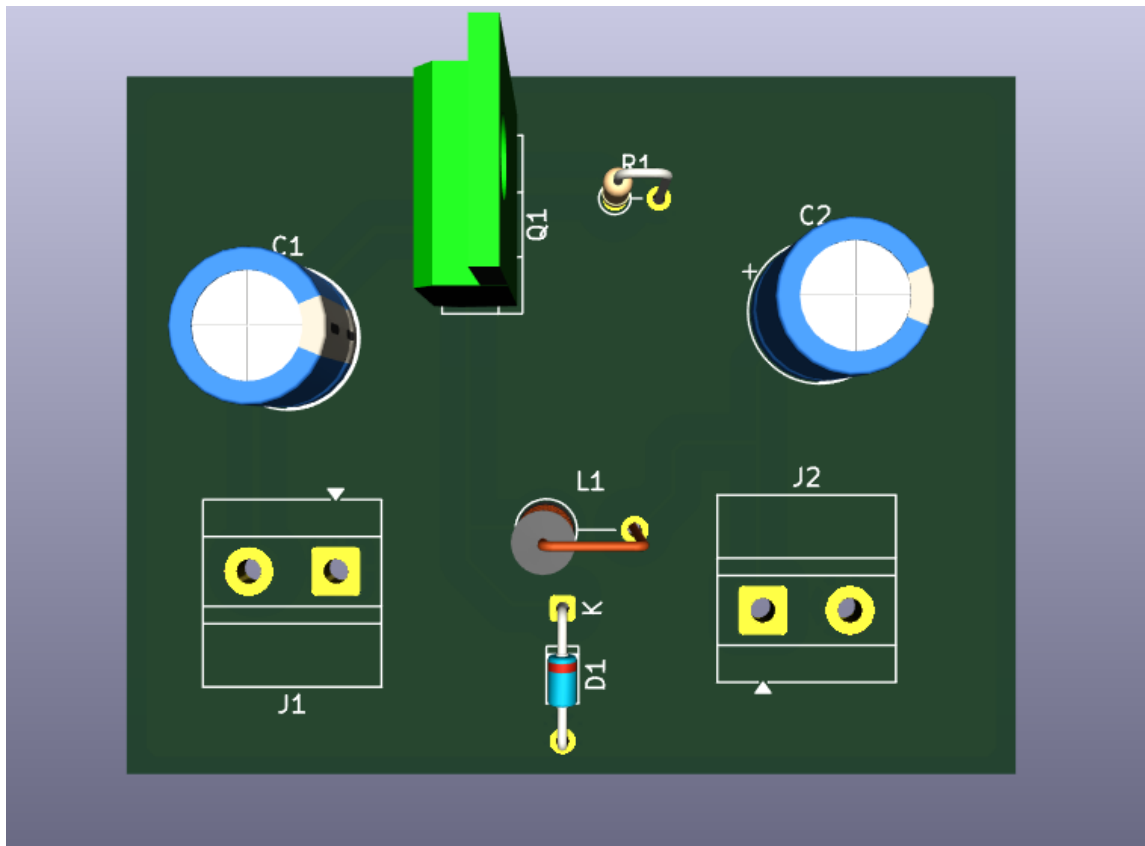


Fig.3D view of PCB

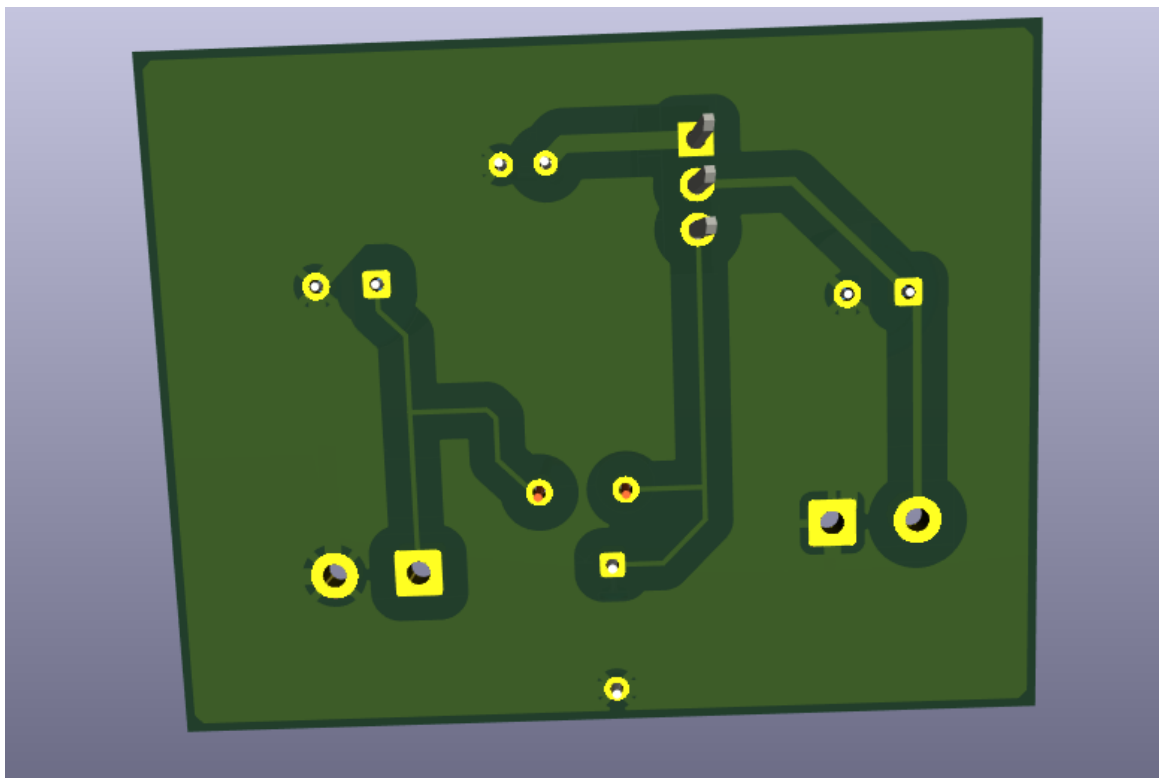


Fig.3D view of PCB

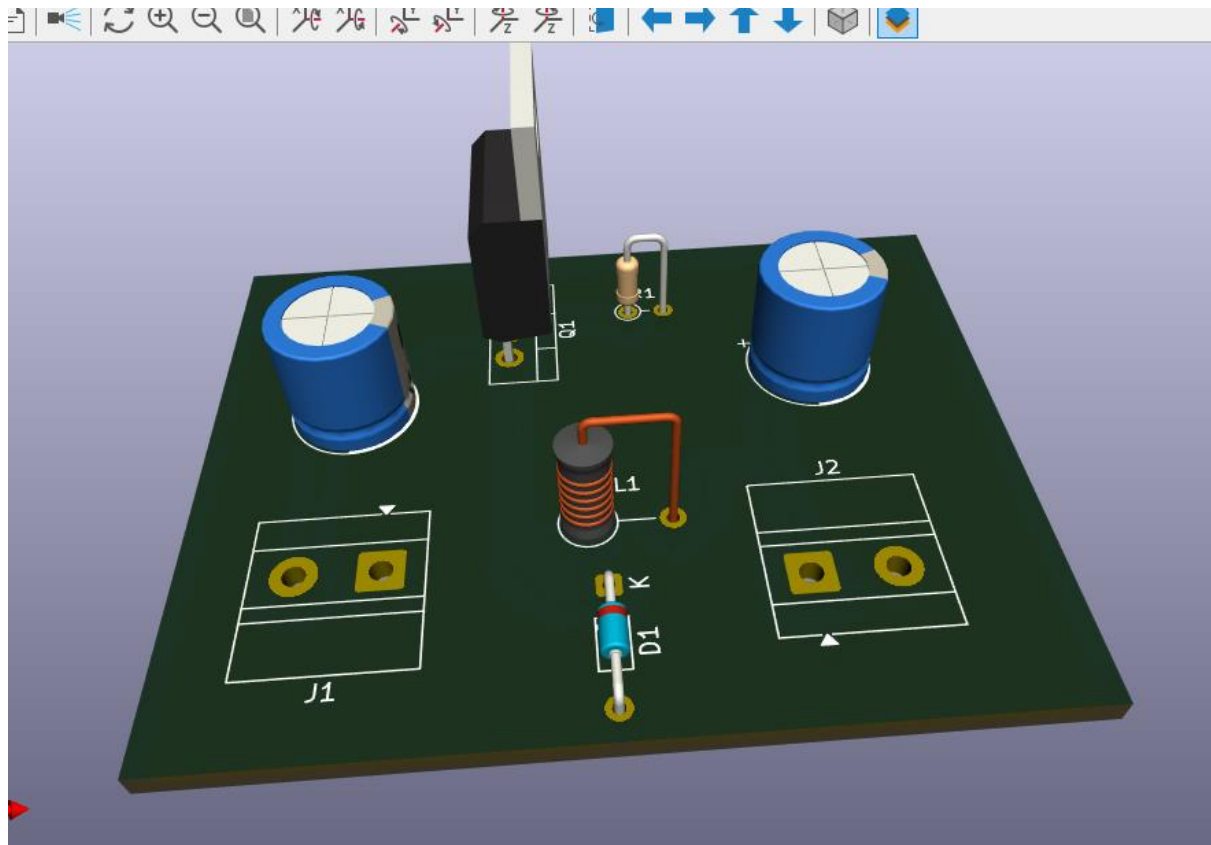


Fig.3D view of PCB

Results and Conclusion:

After designing PCB in KiCAD I can see that I can use this design to produce same PCB in industry and I can also check for errors before implementing the design to actual product.

